

GRAS Determination for the Use of L-Ergothioneine in Selected Conventional Food

Prepared for:

Gene III Biotechnology Co., Ltd.

East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu District,
Nanjing, Jiangsu 210023, China



Prepared by:

Anchor Center for Certification,
No. 1295 Chuan Qiao Road, Building 2, Suite 302,
Pudong, Shanghai, China 201206

7 May 2025

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FOREWORD

Gene III Biotechnology Co., Ltd. (referred to as “Gene III”), founded in 2021 and based in Jiangsu, China, is a bio-innovation company specializing in biosynthesis and intelligent manufacturing. The company integrates research and development, industrial production, and commercialization to provide cost-effective biosynthetic solutions. This notice pertains to L-Ergothioneine, which is produced via fermentation of genetically modified *Escherichia coli* K-12 MG1655.

Gene III has engaged the Anchor Center for Certification (referred to as “ACC”) as its agent for the submission of this GRAS notification to the U.S. FDA. ACC primarily focused on composite safety information, i.e., scientific procedures with corroboration from the history of use. The safety/toxicity studies of L-Ergothioneine, along with its compositional details, specifications, and method of preparation, were reviewed. Additionally, a search of the scientific and regulatory literature was conducted through April 14, 2025, with particular attention to reports on adverse effects and studies relevant to safety conclusions. The composite safety/toxicity studies, in concert with dietary exposure information, ultimately provide the scientific foundation for this GRAS conclusion.

Part 1. SIGNED STATEMENTS AND CERTIFICATION

1.1. Submission of GRAS Notice

Gene III Biotechnology Co., Ltd. is hereby submitting a GRAS notice in accordance with subpart E of CFR 21 part 170.

1.2. Name And Address of The Sponsor

Gene III Biotechnology Co., Ltd.
East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu District,
Nanjing, Jiangsu 210023, China

As the Responsible Party, Gene III accepts the responsibility for the GRAS conclusion that has been made for our L-Ergothioneine as described in the subject safety evaluation; consequently, the L-Ergothioneine preparation having an acceptable composition that meets the conditions described herein is not subject to the premarket approval requirements for food ingredients.

1.3. Common or Usual Name

The common name of the ingredient to be used on food labeling is “L-Ergothioneine,” “Ergothioneine,” or “EGT.” Gene III Biotechnology Co., Ltd. intends to market the product under the names “L-Ergothioneine,” “Ergothioneine,” or “EGT.”

1.4. Trade Secret or Confidential Information

This notification does not contain any trade secret or confidential information.

1.5. Intended Use

Gene III’s L-Ergothioneine is intended to be added into a variety of food applications, as detailed in Part 3, at levels determined by this safety review and produced in accordance with current Good Manufacturing Practices (cGMP). The proposed uses do not include meat and poultry products or infant formulas.

1.6. Basis of Exclusion from the Requirement for Premarket Approval under Subpart E of 21 CFR 170.30¹

Gene III has determined that our L-Ergothioneine meets the specifications described herein and is GRAS as per 21 CFR 170.30. The GRAS determination is based on scientific procedures as described in the following sections. The evaluation accurately reflects the intended conditions of use for the designated L-Ergothioneine.

¹ See 21 CFR 170.30. Accessible at: <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-170/subpart-B/section-170.30>
(Accessed: April 1, 2025)
Anchor Center for Certification (ACC)

1.7. Premarket Approval

The notified substance is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetics (FFD&C) Act based on our conclusion that the substance is GRAS under the conditions of intended use.

1.8. Availability of Information

The data and information that serve as the basis for this GRAS determination will be available for review and copying at reasonable times at the office of ACC*:

Contact:

Anchor Center for Certification (ACC)

No. 1295 Chuan Qiao Road, Building 2, Suite 302

Pudong, Shanghai, China 201206

Email: stella.si@acclab.cn

or be sent to the FDA upon request.

*ACC is the regulatory agent of Gene III Biotechnology Co., Ltd. for GRAS Notification for the Use of L-Ergothioneine in Selected Conventional Food.

1.9. Freedom of Information Act (FOIA)

Parts 2 through 7 of this notification do not contain data or information that is exempt from disclosure under the FOIA.

1.10. Information Included in the GRAS Notification

To the best of our knowledge, the information contained in this GRAS notification is complete, representative, and balanced. It contains both favorable and unfavorable information, known to the Sponsor and ACC, and pertinent to the evaluation of the safety and GRAS status of L-Ergothioneine.



Ju Hongying

Senior Reliability Manager

Gene III Biotechnology Co., Ltd.

East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu District,

Nanjing, Jiangsu 210023, China

Date: May 7, 2025



Agent for Gene III Biotechnology Co., Ltd.

Stella Si

Managing Director

Anchor Center for Certification (ACC)

No. 1295 Chuan Qiao Road, Building 2, Suite 302

Pudong, Shanghai, China 201206

Date: May 7, 2025

Part 2. IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT

2.1. Chemical Identity of the Ingredient

L-Ergothioneine is a metabolite of histidine, specifically a thiolhistidine derivative, that occurs naturally in a number of common foods, including mushrooms, offal, cereals, and some varieties of black and red beans (*Phaseolus vulgaris*). It is a product of plant origin that is synthesized by fungi and mycobacteria and captured in the plant roots. It was first isolated from rye ergot (*Claviceps purpurea*) in 1909 (EFSA, 2016). The structure of L-Ergothioneine is shown in **Figure 1**.

Common or Usual Name	L-Ergothioneine
IUPAC Name	(2S)-3-(2-sulfanylidene-1,3-dihydroimidazol-4-yl)-2-(trimethylazaniumyl)propanoate
Synonyms	Ergothioneine; 2-Thiol-L-histidine-betaine; Thioneine; L (+)-Ergothioneine
CAS Number	497-30-3
EC Number	207-843-5
Molecular Formula	C ₉ H ₁₅ N ₃ O ₂ S
Molecular Mass	229.30 Da

Referred to PubChem - <https://pubchem.ncbi.nlm.nih.gov/compound/Ergothioneine> (Accessed March 7, 2025).

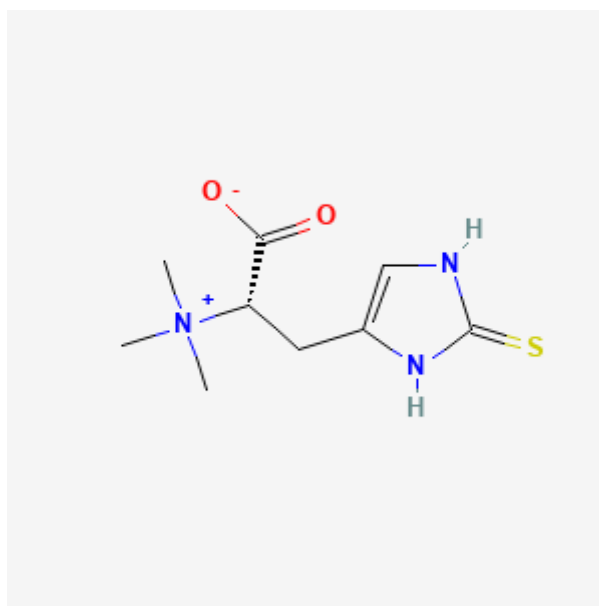


Figure 1. Chemical structure of L-Ergothioneine².

2.2. Production Organism

Gene III produces L-Ergothioneine through fermentation using a genetically modified *Escherichia coli* K-12 MG1655 strain engineered to express three biosynthetic enzymes, EgtD, Egt1, and EgtE. These

² Referred to PubChem - <https://pubchem.ncbi.nlm.nih.gov/compound/5351619> (Accessed March 7, 2025)

enzymes catalyze the stepwise conversion of histidine to L-Ergothioneine through a multi-step biosynthetic pathway, as illustrated in **Figure 2**.

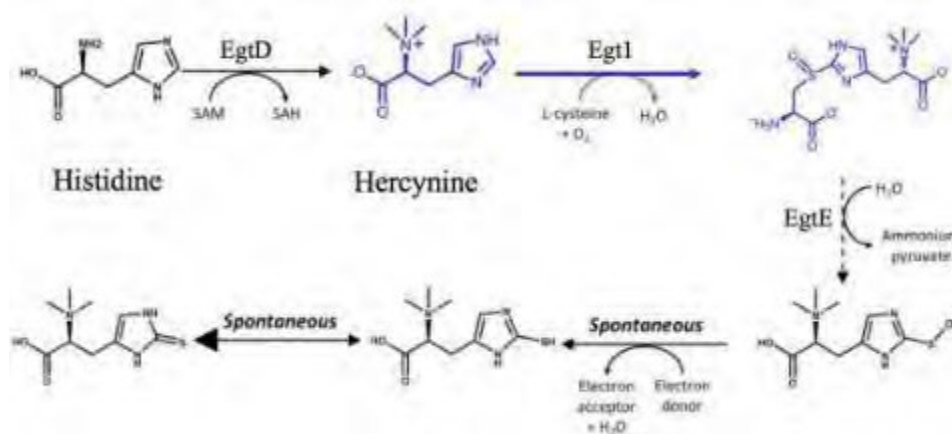


Figure 2. Biosynthesis of L-Ergothioneine.

2.2.1. Host Strain

The host strain is *E. coli* K-12 MG1655, with its development history detailed in Browning et al. (2023). *E. coli* K-12 MG1655 has been used in the manufacturing of L-Ergothioneine, as documented in GRAS Notice (GRN) 000734, which received a “no questions” letter from the FDA. The strain is classified by the American Type Culture Collection (ATCC) as a Bio-Safety Level 1³ organism (ATCC 47076 and 700926, for frozen and freeze-dried forms, respectively) and has also been used as the production organism in GRNs 000749, 000897, 000951, and 001075, all of which received “no questions” letters from the FDA.

2.2.2. Construction of the Production Strain

Three genes were optimized and expressed in *E. coli* K-12 MG1655 using plasmid pMUT1, as described by Kan et al. (2021). This plasmid, available in the National Center for Biotechnology Information (NCBI) database under accession number A84793, functions as a vector for expressing the *egtI* gene from *Neurospora crassa* (ATCC 24698) and the *egtD* and *egtE* genes from *Mycobacterium smegmatis* MC2 155 (ATCC 700084).

The *egtI* gene encodes the ergothioneine biosynthesis enzyme (EgtI), which catalyzes the S-adenosyl-L-methionine (SAM)-dependent triple methylation of histidine’s alpha-amino group to form hercynine, followed by conjugation with cysteine and oxygen to form hercynylcysteine sulfoxide. The *egtD* gene encodes L-histidine N(alpha)-methyltransferase (EgtD), which also catalyzes the SAM-dependent methylation of histidine to form hercynine. The *egtE* gene encodes hercynylcysteine sulfoxide lyase (EgtE), a pyridoxal-phosphate-dependent enzyme that catalyzes the conversion of hercynylcysteine

³ Bio-safety Level 1 is defined as “Well-characterized agents not known to consistently cause disease in immunocompetent adult humans and present minimal potential hazard to laboratory personnel and the environment.” (https://www.cdc.gov/labs/pdf/SF_19_308133-A_BMBL6_00-BOOK-WEB-final-3.pdf)

sulfoxide to ergothioneine. A summary of the inserted genes and their respective enzymes is provided in **Table 1** below.

Table 1. Details of Inserted Genes

Gene	Enzyme	Function	GenBank Accession No.	Origin
<i>egtI</i>	EgtI	Methylation of histidine to hercynine, followed by conjugation with cysteine and oxygen to form hercynylcysteine sulfoxide	XP_956324.3	<i>N. crassa</i> (ATCC 24698)
<i>egtD</i>	EgtD	Methylation of histidine to form hercynine	WP_011731156.1	<i>M. smegmatis</i> MC ² 155 (ATCC 700084)
<i>egtE</i>	EgtE	Conversion of hercynylcysteine sulfoxide to ergothioneine	WP_011731155.1	<i>M. smegmatis</i> MC ² 155 (ATCC 700084)

The resulting plasmid, containing the inserted genes along with its intrinsic elements, was introduced into *E. coli* K-12 MG1655 competent cells via chemical transformation. The gene expression system is based on the bacterial quorum sensing system as described in Soma et al. (2021), which enables the bacteria to coordinate the expression of EgtD, EgtI, and EgtE depending on their population density.

Successful transformation was verified through Polymerase Chain Reaction (PCR) amplification, agarose gel electrophoresis (AGE), and sequencing. PCR and AGE confirmed the presence of target gene fragments at the expected sizes (**Figure 3**), while sequencing validated their alignment with the intended gene sequences. Additionally, Gene III confirmed that the antibiotic resistance genes, virulence factors, and toxin genes detected in the genetically modified *E. coli* K-12 MG1655 are intrinsic to *E. coli* K-12 MG1655 (Appendix 1). The production strain has been deposited in the China Center for Type Culture Collection (CCTCC) under deposit number M2023894.

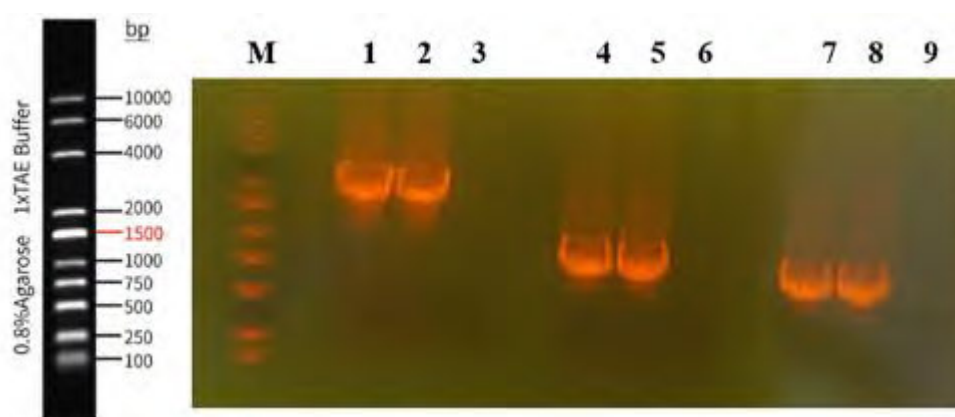


Figure 3. Verification of successful transformation using AGE.

M: DNA marker; 1: *egtI* from genetically modified *E. coli* K-12 MG1655; 2: *egtI* from positive control; 3: *egtI* from negative control; 4: *egtE* from genetically modified *E. coli* K-12 MG1655; 5: *egtE* from positive control; 6: *egtE* from negative control; 7: *egtD* from genetically modified *E. coli* K-12 MG1655; 8: *egtD* from positive control; 9: *egtD* from negative control.

2.2.3. Stability of Production Strain

The stability of genetically modified *E. coli* K-12 MG1655 was evaluated over five consecutive subcultures (~210 generations) based on growth rate, genetic stability, and ergothioneine production.

Growth rate was assessed using turbimetry, with consistent growth curves observed across generations, indicating stable growth.

Genetic stability was examined through PCR and sequencing analysis. PCR amplification confirmed the presence of *egtI*, *egtD*, and *egtE* across generations with consistent band sizes. Sequencing results further verified that the target gene sequences remained unchanged, demonstrating genetic stability.

Ergothioneine production was evaluated in shake-flask fermentation. Synthesis was detected in all tested generations, with stable yields observed, confirming consistent production capability.

2.3. Manufacturing Process

2.3.1. Raw Materials and Processing Aids

Gene III maintains that all raw materials and processing aids used in the production of L-Ergothioneine are food-grade and suitable for their intended use. An attestation statement is provided in Appendix 2. **Table 2** summarizes these materials. All raw materials and processing aids comply with the Food Allergen Labeling and Consumer Protection Act of 2004 (FALCPA), are not derived from major allergens, and would not introduce major allergens into the final product.

Table 2. Raw Materials and Processing Aids

Materials	Function	Regulatory Status
Yeast peptone	Fermentation medium ^a	21 CFR 184.1553
Yeast Extract	Fermentation medium	21 CFR 184.1983
Sodium chloride	Reaction component	21 CFR 182.1
Potassium dihydrogen phosphate	Reaction component	JECFA Monograph 1 (2006)
Ammonium sulfate	Reaction component	21 CFR 184.1143
Dextrose	Fermentation medium	21 CFR 184.1857
Magnesium sulfate	Reaction component	21 CFR 184.1443
Defoamer	Fermentation aid	21 CFR 173.340
Ammonia water	pH regulation and elution solvent	21 CFR 184.1139
L-methionine	Biosynthesis precursor	21 CFR 172.320
L-histidine	Biosynthesis precursor	21 CFR 172.320
L-cysteine	Biosynthesis precursor	21 CFR 172.320
Activated carbon	Decolorization	JECFA Monographs 10 (2010)
Ethanol	Crystallization	21 CFR 184.1293
Polyethylene bag	Packaging	Pharmaceutical-grade
Aluminum composite bag (PET/AL/PE)	Packaging	Pharmaceutical-grade

^a Yeast peptone is used in seed culture fermentation only.

PET: Polyethylene terephthalate; Al: Aluminum; PE: Polyethylene

2.3.2. Production of L-Ergothioneine

Gene III's L-Ergothioneine is manufactured in compliance with cGMP, with relevant documentation provided in Appendix 3. Production is conducted in a sterile, contained environment. In-process controls include media sterilization, macromolecular filtration, and resin column filtration. The manufacturing process is summarized in **Figure 4** and further described in the following sections.

The production water meets the national drinking water standard GB5749-2022, consistent with those set by the United States Environmental Protection Agency (U.S. EPA), except for per- and poly-fluoroalkyl substances (PFAS). The purification process includes sequential steps such as filtration, sand filtration, activated carbon filtration, softening, reverse osmosis with electrodeionization, and ultraviolet (UV) sterilization. Both activated carbon filtration and reverse osmosis effectively remove PFAS (U.S. EPA, 2018). Along with the national standard, this purification process ensures the production water complies with U.S. EPA drinking water standards.

Solid waste is periodically collected and disposed of by a third-party waste management service. Wastewater undergoes initial on-site treatment in accordance with national discharge standards before being transferred to a third-party treatment facility.

2.3.2.1. Inoculation and Seed Culture

The glycerol stocks of the modified *E. coli* strain are stored at $-70 \pm 5^{\circ}\text{C}$ and thawed at room temperature prior to use. The seed culture medium, composed of yeast peptone, yeast extract, and sodium chloride, is sterilized at $120\text{--}123^{\circ}\text{C}$ for 25–30 minutes before inoculation. The inoculated seed culture (1% stock) is incubated at $37 \pm 2^{\circ}\text{C}$ for 15–17 hours with shaking at 220 ± 20 rpm. Once the culture transitions from clear to turbid, it proceeds to the fermentation stage.

2.3.2.2. Fermentation

A basal culture medium is prepared using yeast extract, potassium dihydrogen phosphate, ammonium sulfate, dextrose, magnesium sulfate, defoamer, and potable water. The pH is adjusted to 6.5–7.3 with ammonia water, and the medium is sterilized in a fermentation tank at $119\text{--}123^{\circ}\text{C}$ for 30 minutes. Supplement 1, composed of dextrose, magnesium sulfate, and potable water, is sterilized under pressure at $118\text{--}121^{\circ}\text{C}$ for 20 minutes in an auxiliary tank. Supplement 2, containing L-methionine, L-histidine, L-cysteine, and potable water, undergoes high-pressure sterilization at $118\text{--}121^{\circ}\text{C}$ for 20 minutes in a separate auxiliary tank.

The seed culture is inoculated into the fermentation tank containing the basal culture medium at 0.04%–1% of the initial fermentation volume. During fermentation, Supplements 1 and 2 are added stepwise in multiple batches over time. The fermentation temperature is controlled at $33\text{--}37^{\circ}\text{C}$ for the first 8–10 hours and then maintained at $20\text{--}24^{\circ}\text{C}$ until the end of the process. The pH of the medium is maintained at 6.8–6.9 throughout the fermentation process. Ventilation is maintained at 600–1200

m³/h, and the agitator speed is controlled at 80–165 rpm. Fermentation is carried out for 90–96 hours and is terminated by sterilization at 95°C for 25–30 minutes once the L-Ergothioneine concentration reaches ≥ 6000 mg/L.

2.3.2.3. Filtration

Bacterial cells are removed by plate-and-frame pressure filtration, followed by removal of macromolecular impurities using an ultrafiltration membrane. The collected supernatant is preliminarily concentrated by nanofiltration before prior to purification.

2.3.2.4. Purification

The supernatant undergoes decolorization and impurity removal using macroporous adsorption resin. The resulting flow-through is subjected to cation exchange chromatography, where it is washed with primary reverse osmosis water and eluted with 0.3–0.5% ammonia water.

2.3.2.5. Concentration

The eluent (purity >95%) is transferred to a concentration unit to be concentrated, where the temperature is controlled at $\leq 60^{\circ}\text{C}$ and the vacuum pressure at ≤ -0.08 MPa.

2.3.2.6. Decolorization

The concentrated solution is decolorized and purified by removing insoluble substances using activated carbon. Activated carbon (0.2%–0.5%) is added and stirred for 30–35 minutes.

2.3.2.7. Filtration, Crystallization, and Centrifugation

The solution is filtered using a plate-and-frame system and transferred to a crystallization tank. Ethanol is added, and the mixture is stirred for 0.5–1.5 hours. Crystals are collected using a tray centrifuge operated at high speed.

2.3.2.8. Drying, Sieving, and Iron Removal

The obtained crystals are placed in a vacuum drying oven, where the vacuum pressure is maintained at ≤ -0.08 MPa and the hot water temperature at 60–65°C. After drying for 6–8 hours, the product is sieved through a 40-mesh oscillating sieve and undergoes iron removal using a high-intensity magnetic separator, yielding the final L-Ergothioneine product.

2.3.2.9. Packaging

The final L-ergothioneine powder is packed in vacuum-heat-sealed polyethylene bags. The sealed product is then packed and sealed into a light-shielding aluminum composite bag and stored at room temperature.

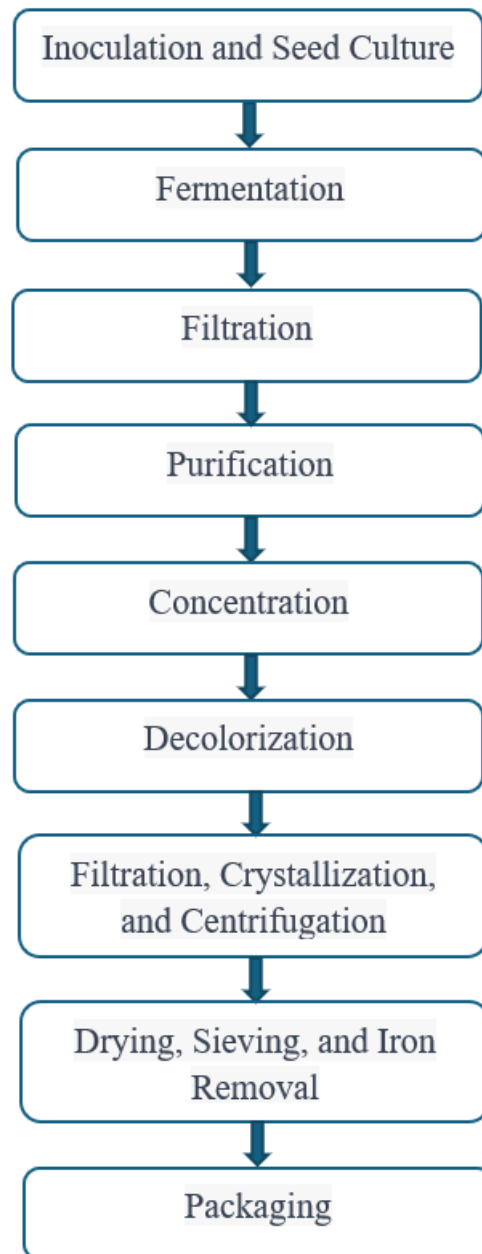


Figure 4. Production flow chart of Gene III's L-Ergothioneine.

2.4. Product Specification

The product specifications for Gene III's L-Ergothioneine are summarized in **Table 3**. They are consistent with the specifications established for L-Ergothioneine outlined in GRN 000734 and EFSA's scientific opinion (EFSA, 2016). Given the similarity in specifications, Gene III's L-Ergothioneine is therefore substantially equivalent to the L-ergothioneine described in GRN 000734 and the EFSA evaluation. Gene III confirms that all analytical methods used for specification testing are internationally recognized or internally validated.

Analyses of five non-consecutive, commercially representative batches of L-Ergothioneine, summarized in **Table 4**, demonstrate compliance with these specifications and batch-to-batch consistency. Certificates of analysis (CoAs) are provided in Appendix 4.

Table 3. Specifications for Gene III's L-Ergothioneine

Parameters	Gene III's L-Ergothioneine Specifications		Comparative L-Ergothioneine Specifications	
	Specification	Method	Blue California ^b (GRN 000734)	Tetrahedron (EFSA, 2016)
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	Visual	White powder	White powder with <0.8% foreign matter
Identification	HPLC chromatogram matches the standard sample	HPLC ^a	NS	NS
Optical Rotation	$[\alpha]_D \geq (+)122^\circ$ (C=1, H ₂ O)	Polarimetry ^a	NS	$[\alpha]_D \geq (+)122^\circ$ (C=1, H ₂ O)
Assay of L-Ergothioneine (%)	≥99.5	HPLC ^a	≥98	≥99.5
Loss on Drying (%)	≤0.5	JECFA Vol. 4	≤5	≤0.5
Total Residual Solvents (Ethanol) (ppm)	≤1000	GC ^a	<1000	<1000
Heavy Metals*				
Lead (mg/kg)	≤0.5	AOAC 2015.01	≤1.0	≤3
Arsenic (mg/kg)	≤0.5	AOAC 2015.01	≤1.0	NS
Mercury (mg/kg)	≤0.1	AOAC 2015.01	≤1.0	≤0.1
Cadmium (mg/kg)	≤0.1	AOAC 2015.01	≤1.0	≤1
Microbiological Criteria				

Parameters	Gene III's L-Ergothioneine Specifications		Comparative L-Ergothioneine Specifications	
	Specification	Method	Blue California ^b (GRN 000734)	Tetrahedron (EFSA, 2016)
Total Aerobic Bacteria Count (CFU/g)	≤1000	AOAC 966.23	≤5000	≤1000
Molds and Yeasts (CFU/g)	≤50	ISO 21527-2	<100	≤100
<i>Escherichia coli</i>	Negative/g	ISO 7251:2005	Negative	Absent in 1g
<i>Salmonella</i>	Negative/25g	ISO 65791:2017	Negative	NS
<i>Staphylococcus aureus</i>	Negative/25g	GB 4789.10	NS	NS

^a Internal method.

^b The specifications of Blue California are referred from Table 1, Page 12 of 214, GRN 000734.

* Not a routine testing item for batch release.

NS: Not specified

Table 4. Batch Analysis of L-Ergothioneine

Parameters	Specifications	Representative Lots				
		EGT 20230720	EGT 20240207	EGT 20250107	EGT 20250115	EGT 20250117
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform	Conform	Conform	Conform	Conform
Optical Rotation	$[\alpha]_D \geq (+)122^\circ$ (C=1, H ₂ O)	+127.0°	+126.0°	+127.0°	+128.0°	+128.0°
Assay of L-Ergothioneine (%)	≥ 99.5	99.9	99.9	99.9	99.9	99.9
Loss on Drying (%)	≤ 0.5	0.04	0.05	0.04	0.14	0.14
Total Residual Solvents (Ethanol) (ppm)	≤ 1000	32.5	197	410	422	440
Heavy Metals*						
Lead (mg/kg)	≤ 0.5	<0.05	<0.05	<0.05	<0.05	<0.05
Arsenic (mg/kg)	≤ 0.5	<0.05	<0.05	<0.05	<0.05	<0.05
Mercury (mg/kg)	≤ 0.1	<0.01	<0.01	<0.01	<0.01	<0.01
Cadmium (mg/kg)	≤ 0.1	<0.05	<0.05	<0.05	<0.05	<0.05
Microbiological Limits						
Total Aerobic Bacteria Count (CFU/g)	≤ 1000	<10	<10	<10	<10	<10

Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative

* Not a routine testing item for batch release.

2.5. Physical or Technical Effect

L-Ergothioneine will be added as an ingredient in conventional foods.

2.6. Stability

2.6.1. General Stability Data for L-Ergothioneine

Based on a six-month accelerated stability study ($40\pm 2^{\circ}\text{C}$, $75\pm 5\%$ relative humidity) and a 12-month intermediate stability study ($30\pm 2^{\circ}\text{C}$, $65\pm 5\%$ relative humidity), Tetrahedron reported that its L-Ergothioneine material is expected to remain stable for one year (EFSA, 2016).

In GRN 000734, Blue California reported the initial results of a planned 36-month shelf-stability study for its L-Ergothioneine. Samples stored under ambient conditions and tested at 6 months and 12 months showed no changes (FDA, 2018b).

2.6.2. Stability Data of Gene III's L-Ergothioneine

Gene III conducted two stability studies on L-Ergothioneine under different conditions. All samples were vacuum-heat sealed in polyethylene bags and then packed in light-shielding aluminum composite bags. Samples were taken from individual sealed bags at each time point.

The long-term stability study is conducted at $25\pm 2^{\circ}\text{C}$ for 24 months, with sampling at 3, 6, 9, 12, 18, and 24 months. The currently available results (**Table 5**) indicate that product quality remained stable under the specified conditions for up to 18 months. Further evaluations are ongoing, with updates to be provided as additional data becomes available.

The accelerated stability study is conducted at $40\pm 2^{\circ}\text{C}$ and $75\pm 5\%$ relative humidity for 6 months, with sampling at 1, 2, 3, and 6 months. Results from the accelerated stability study (**Table 6**) demonstrate that L-Ergothioneine remained stable throughout the study period.

The accelerated stability data support a two-year shelf life for L-Ergothioneine when stored at room temperature in its intended commercial packaging, which is vacuum-heat sealed polyethylene bags enclosed within light-shielding aluminum composite bags.

Table 5. Long-term Stability Data

Parameter	Specification	EGT20230720						
		0 Month	3 Months	6 Months	9 Months	12 Months	18 Months	24 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	--
Assay of L-Ergothioneine (%)	≥99.5	99.9	100	100	99.9	100	99.9	--
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10	<10	--
Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10	<10	--
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
Conclusion	NS	Stable	Stable	Stable	Stable	Stable	Stable	--
Parameter	Specification	EGT20230904						
		0 Month	3 Months	6 Months	9 Months	12 Months	18 Months	24 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	--
Assay of L-Ergothioneine (%)	≥99.5	99.9	99.9	99.9	100	100	99.9	--
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10	<10	--

Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10	<10	--
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
Conclusion	NS	Stable	Stable	Stable	Stable	Stable	Stable	--
Parameter	Specification	EGT20231001						
		0 Month	3 Months	6 Months	9 Months	12 Months	18 Months	24 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	--
Assay of L-Ergothioneine (%)	≥99.5	100	99.9	100	100	100	100	--
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10	<10	--
Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10	<10	--
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative	Negative	--
Conclusion	NS	Stable	Stable	Stable	Stable	Stable	Stable	--

"--" indicates that the batch is under stability study conditions but was not yet due for testing.

Table 6. Accelerated Stability Data

Parameter	Specification	EGT20230720				
		0 Month	1 Month	2 Months	3 Months	6 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Assay of L-Ergothioneine (%)	≥99.5	99.9	99.9	100	99.9	100
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10
Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
Conclusion	NS	Stable	Stable	Stable	Stable	Stable
Parameter	Specification	EGT20230904				
		0 Month	1 Month	2 Months	3 Months	6 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Assay of L-Ergothioneine (%)	≥99.5	99.9	99.9	100	100	100
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10
Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative

<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
Conclusion	NS	Stable	Stable	Stable	Stable	Stable
Parameter	Specification	EGT20231001				
		0 Month	1 Month	2 Months	3 Months	6 Months
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Assay of L-Ergothioneine (%)	≥99.5	100	99.9	100	100	100
Total Aerobic Bacteria Count (CFU/g)	≤1000	<10	<10	<10	<10	<10
Molds and Yeasts (CFU/g)	≤50	<10	<10	<10	<10	<10
<i>Escherichia coli</i>	Negative/g	Negative	Negative	Negative	Negative	Negative
<i>Salmonella</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative	Negative	Negative	Negative	Negative
Conclusion	NS	Stable	Stable	Stable	Stable	Stable

Part 3. DIETARY EXPOSURE

3.1. Background Exposure

As part of their evaluation of synthetic L-Ergothioneine manufactured by Tetrahedron, the European Food Safety Authority (EFSA) Panel reviewed a published intake assessment by Ramirez-Martinez et al. (2016) that discussed the estimated daily intakes of ergothioneine in the European population. The EFSA Panel noted that the mean chronic dietary intake of ergothioneine in several European countries ranged from 0.051 to 0.255 mg/kg body weight/day and from 0.306 to 0.409 mg/kg bw/day for adult and adolescent consumers, respectively.

Ramirez-Martinez et al. (2016) also estimated the dietary intake of ergothioneine in US consumers using probabilistic estimates considering the consumption of the main dietary sources of ergothioneine (i.e., mushrooms, beans, meat organs, and oatmeal). In the US, the mean chronic background dietary intake of ergothioneine was estimated to be 0.153 mg/kg bw/day and 0.299 mg/kg bw/day for adult and adolescent consumers, respectively. As compared with European populations, US consumers were estimated to have a lower overall consumption of ergothioneine in their diet.

The L-Ergothioneine reviewed by EFSA had an intended use level of up to 5 mg per serving in a limited number of conventional foods, including alcohol-free beverages, cereal bars, milk, fresh dairy products, and chocolates, and dietary supplements at 30 mg per day for adults and 20 mg per day for children (EFSA, 2016).

In GRN 000734, submitted by Blue California, the estimated 90th percentile daily intake (calculated as mean intake $\times$ 2) of ergothioneine from the intended uses was reported to be 0.48 mg/kg bw/day for adults, 1.55 mg/kg bw/day for children aged 3 to 10 years, 0.8 mg/kg bw/day for children aged 10 to 14 years, and 0.56 mg/kg bw/day for children aged 14 to 18 years. Combining the GRN 000734 proposed intake estimate with the background intake reported by Ramirez-Martinez et al. (2016) results in the highest cumulative intake estimates of 0.633 mg/kg bw/day for adults and 1.85 mg/kg bw/day for children aged 3 to 10 years.

3.2. Estimate of Dietary Exposure to the Substance

Gene III's L-Ergothioneine is intended for use as an ingredient in beverages (coffee, tea, fruit drinks and ades, carbonated soft drinks, sports and energy drinks, nutritional beverages), protein and nutritional powders, bakery products (nutrition bars, cereal bars, sweet baked goods), and candy. It is not intended for use in infant formulas or foods regulated by the USDA. The proposed products and inclusion levels are presented in **Table 7**.

Table 7. Intended Food Categories for Gene III's L-Ergothioneine

Food Categories	Amount per Serving ^a (mg)	Use Level (mg/ 100g)
Beverages		
Coffee	60	16.7
Tea	60	16.7

Carbonated Soft Drinks	60	16.7
Sport and Energy Drinks	30	8.3
Fruit drinks and ades	30	12.5
Nutritional Beverages	60	25
Protein and nutritional powders	60	150
Bakery and other baked goods		
Nutrition bars and cereal bars	30	75
Sweet bakery (Cakes, cookies and pastries)	30	34.6–85.7
Candy		
Gelatin (surrogate for gummy candy)	150	120
Candy containing chocolate	30	100
Candy not containing chocolate	30	100

^a Serving size determined from Reference Amount Customarily Consumed (RACC) - 21 CFR § 101.12. Assumption that 1 g = 1 ml for RACC.

The estimated dietary exposures for L-Ergothioneine were calculated using the Creme Food Safety® model (<https://www.cremeglobal.com>). The model includes food consumption data included in the NHANES 2017 – 2020 pre-pandemic survey (CDC, 2022). Output calculation types include daily average intakes, acute exposures, as well as population statistics such as mean, percentiles, standard errors, and confidence intervals. Results are output for “Consumers Only” (i.e., consumers of the intended food categories). Results of the exposure assessment are given in absolute terms (mg/day) as well as relative to the consumer’s body weight (mg/kg bw/day) at the mean and 90th percentile. The results are shown in **Table 8**, and a cumulative intake estimate is also provided (**Table 9**).

The per unit of body weight exposure is calculated on a subject level using the body weight recorded by the NHANES survey data. The estimated intake per kg of body weight is calculated per individual survey respondent and then averaged to provide the results in **Table 8**. The results for “Consumers Only” are shown in **Table 8** and a detailed report, including *per capita* exposure estimates, is provided in Appendix 5. The intended use for Gene III’s L-Ergothioneine does not include infant formula or foods for young children; therefore, intake estimates for 0–23.9-month-olds are not included.

Table 8. Summary of Consumer Only Intake from the Intended Use of Gene III’s L-Ergothioneine Based on NHANES 2017-2020 Survey

Population Group	N	% Population	Group Mean Body Weight (kg) ^a	Consumer Only Intake (mg/day)		Consumer Only Intake (mg/kg bw/day)	
				Mean ± SE	90th Percentile ± SE	Mean ± SE	90th Percentile ± SE
Children, Females and Males (2-5 years)	729	91%	17.3	49.8 ± 1.5	95.5 ± 3.8	2.87 ± 0.08	5.77 ± 0.35
Children, Females and Males (6-11)	1156	96%	32.9	75.0 ± 1.4	141.5 ± 3.5	2.26 ± 0.05	4.54 ± 0.12

years)							
Teenage, Female (12-18 years)	626	96%	57.8	83.1 ± 2.2	149.4 ± 5.2	1.43 ± 0.05	2.72 ± 0.13
Teenage, Male (12-18 years)	637	96%	62.4	105.3 ± 3.5	207.8 ± 8.5	1.67 ± 0.06	3.14 ± 0.11
Adults, Female (19-110 years)	3459	97%	73.8	147.8 ± 1.8	272.8 ± 4.1	2.00 ± 0.03	3.66 ± 0.08
Adults, Male (19-110 years)	3161	97%	87.3	182.8 ± 2.3	338.1 ± 8.5	2.09 ± 0.03	3.90 ± 0.10
Males and Females (2-110 years)	9768	97%	69.4	145.4 ± 1.2	285.0 ± 3.3	2.05 ± 0.02	3.92 ± 0.06

^a Mean body weight calculated by dividing the mean intake in mg/day by the mean intake in mg/kg bw/day. This is used for calculation of cumulative intake, as noted in Table 9.

The estimated mean and 90th percentile intake of L-Ergothioneine for the overall population is 145.4 mg/day (2.05 mg/kg bw/day) and 285.0 mg/day (3.92 mg/kg bw/day), respectively. In the population with the highest consumption on an absolute basis (adult males), the mean and 90th percentile estimated dietary intakes at the proposed maximum intended use levels of Gene III's L- Ergothioneine is 182.8 mg/day and 338.1 mg/day, respectively. The highest intake on a body weight basis is from children 2 – 5 years, with the mean and 90th percentile intake of 2.87 mg/kg bw/day and 5.77 mg/kg bw/day respectively.

Table 9. Summary of Cumulative Intake from Diet and from the Intended Use of Gene III's L-Ergothioneine Based on NHANES 2017-2020 Survey

Population Group	N	% Population	Consumer Only Intake (mg/day) ^a		Consumer Only Intake (mg/kg bw/day)	
			Mean ± SE	90th Percentile ± SE	Mean ± SE	90th Percentile ± SE
Children, Females and Males (2-5 years)	729	91%	55.5	105.8	3.2	6.1
Children, Females and Males (6-11 years)	1156	96%	86.3	159.3	2.6	4.8
Teenage, Female (12-18 years)	626	96%	98.8	174.3	1.7	3.0
Teenage, Male (12-18 years)	637	96%	126.1	214.4	2.0	3.4
Adults, Female (19-110 years)	3459	97%	155.2	280.8	2.1	3.8
Adults, Male (19-110 years)	3161	97%	192.4	358.6	2.2	4.1
Males and Females (2-110 years)	9768	97%	156.0	290.8	2.2	4.1

^a Calculated using the mean body weight multiplied by the cumulative intake in mg/kg bw/day. Mean body weight calculated by dividing the mean intake in mg/day by the mean intake in mg/kg bw/day, as noted in Table 8.

The cumulative intake of L-Ergothioneine on a mg/kg bw/day basis is calculated by combining the estimated intake from the intended uses for Gene III's L-Ergothioneine and an estimate of background intake from natural food sources. This approach is adopted from GRN 000734, which received a "no questions" letter from the FDA.

- The intended uses for Gene III's L-Ergothioneine are described in **Table 7** and the estimated intake is presented in **Table 8**.
- The estimate for background intake from natural sources was determined by Ramirez-Martinez et al. (2016). Ramirez-Martinez et al. (2016) estimate that in the US, the mean chronic background dietary intake of L-Ergothioneine was estimated to be 0.153 mg/kg bw/day and 0.299 mg/kg bw/day for adult and child/adolescent consumers, respectively⁴.

This results in the highest 90th percentile cumulative intake estimate on a body weight basis of 6.1 mg/kg bw/day for children aged 2 – 5 years. The highest estimated 90th percentile cumulative total daily intake on an absolute basis is 358.6 mg/day ergothioneine for adult males. The estimated 90th percentile cumulative total daily intake for the overall population is 290.8 mg/day (4.1 mg/kg bw/day).

It should be noted that all estimated intake calculations assume the worst-case consumption in that these populations would consume all fortified foods each day and that no substitution will occur. This results in an overestimation of exposure to ergothioneine from the intended uses.

3.3. Estimated Dietary Exposure to Any Other Substance That is Expected to be Formed in or On Food

This section is not applicable to Gene III's L-Ergothioneine product, as it is chemically stable under the proposed conditions of use.

3.4. Dietary Exposure to Contaminants or Byproducts

Potential contaminants of Gene III's L-Ergothioneine include microbes, heavy metals, endotoxins, residual DNA from the production organism, and histamine (a known metabolite of histidine). The specifications set for Gene III's L-Ergothioneine place limits on the maximum permissible levels of microbes and heavy metals to ensure an acceptable final product (**Table 3**). Additionally, three different batches of Gene III's L-Ergothioneine have been analyzed for histamine (results are below the limit of quantification (LOQ) of 10 mg/kg) and endotoxins (results were <0.05 EU/mg), showing below detectable residue of both histamine and endotoxins.

⁴ In GRN 000734, submitted by Blue California, the cumulative intake estimate was calculated by combining the high intended use intake estimate values with estimates of background intake of L-Ergothioneine as calculated by Ramirez-Martinez et al. (2016). Ramirez-Martinez et al. (2016) estimated the mean chronic dietary intake of L-Ergothioneine in US consumers using probabilistic estimates considering the consumption of the main dietary sources of ergothioneine (i.e., mushrooms, beans, meat organs, and oatmeal). They reported two age categories – children (noted as ages 3-11) and adults. The mean intake calculated using a probabilistic estimation procedure was 0.153 mg per kg bw per day for adult consumers and 0.299 mg per kg bw per day for adolescent consumers. No estimate of absolute intake was provided.

For the residual DNA analysis, L- Ergothioneine was dissolved in ddH₂O, and different concentrations of L-Ergothioneine (200 g/L (the limit of solubility), 100 g/L, and 10 g/L) were used as templates for PCR amplification. L-Ergothioneine samples were used for the analysis rather than DNA extracted from the L-ergothioneine because DNA could not be extracted from the samples. DNA extracted from the producing strain was diluted to various concentrations (100 ng/μL, 50 ng/μL, 10 ng/μL, 1 ng/μL, 0.1 ng/μL, 0.01 ng/μL) and was also subjected to PCR amplification to determine the sensitivity of the assay. *egtD*, *egtI*, *egtE*, and 16S rDNA were probed for in all samples. None of these genes were found in L-Ergothioneine samples at a limit of detection of 0.1 ng/ μL. The residual DNA analysis report is provided in Appendix 6.

Part 4. SELF-LIMITING LEVELS OF USE

There are no known self-limiting levels of use.

Part 5. EXPERIENCE BASED ON COMMON FOOD USE IN FOOD BEFORE 1958

L-Ergothioneine is naturally occurring in a number of foods that are part of the typical American diet, including mushrooms, grains and grain products, beans and some vegetables, eggs, animal products (including meat, organ meat, and offal), and beer (EFSA, 2016; Ey et al., 2007). However, L-Ergothioneine produced according to the method described herein does not have a history of prior use in food, therefore, experience based on common use in food before January 1, 1958, does not apply. The statutory basis for the conclusion of the GRAS status of L-Ergothioneine in this document is based on scientific procedures in accordance with 21 CFR 170.30(a)(b). The history of safe consumption of L-Ergothioneine is discussed in Part 6.

Part 6. NARRATIVE

The purpose of this section is to provide information relevant to the safety of L-Ergothioneine. A review of the historical, scientific, and regulatory literature related to the safety of L-Ergothioneine was conducted through April 14, 2025. For this purpose, PubMed was searched for information without limits on publication date and the search terms/string included (ergothioneine or L-Ergothioneine) and (safe or safety or adverse). Various government databases were also used to identify regulations across jurisdictions.

The safety of L-ergothioneine has been evaluated by the FDA in GRN 000734, which received a “no questions” letter. In addition, the EFSA assessed L-ergothioneine as a novel food and concluded that it is safe for the general population under the proposed conditions of use. As part of its assessment, EFSA identified a no-observed-adverse-effect level (NOAEL) of 800 mg/kg bw/day, based on the subchronic toxicity study by Marone et al. (2016). Given that Gene III’s L-ergothioneine is substantially equivalent to the subject evaluated by EFSA (**Table 3**), the NOAEL of 800 mg/kg bw/day is pivotal to this GRAS determination. The safety of L-Ergothioneine is further supported by a consistent body of toxicological evidence indicating that L-ergothioneine is non-genotoxic and does not exhibit reproductive or developmental toxicity. Clinical studies have also shown that L-Ergothioneine is well tolerated in humans at intake levels up to 25 mg/day.

6.1. History of Safe Consumption

L-Ergothioneine is a naturally occurring amino acid which is present primarily in mushrooms (Kushairi et al., 2020). It can only be synthesized by a limited number of microorganisms including cyanobacteria, mycobacteria, and non-yeast fungi, and is only present in human tissues via the diet (Kushairi et al., 2020). L-Ergothioneine is naturally occurring in a number of foods that are part of the typical American diet, including mushrooms, grains and grain products, beans and some vegetables, eggs, animal products (including meat, organ meat, and offal), and beer (EFSA, 2016; Ey et al., 2007). Ramirez-Martinez et al. (2016) estimated the dietary intake of L-Ergothioneine in U.S. and European consumers (France, Italy, Ireland, Finland, and Belgium) using the consumption of the main dietary sources of L-Ergothioneine (i.e., mushrooms, beans, meat organs, and oatmeal) and applying a probabilistic model. The mean chronic background intake of L-Ergothioneine was estimated at 0.153 mg/kg bw/day for adults and 0.299 mg/kg bw/day for adolescents in the U.S. In European countries, the mean intake ranged from 0.051 to 0.255 mg/kg bw/day for adults and 0.306 to 0.409 mg/kg bw/day for adolescents. Overall, U.S. consumers were estimated to have a lower overall consumption of L-Ergothioneine in their diet than European consumers.

6.2. Regulatory History

6.2.1. United States

As of April 14, 2025, a search of the FDA's GRAS Notice Inventory using the term "ergothioneine" identified two GRAS notices (**Table 10**). One received a "no questions" response from the FDA, while the other remains under review. The ergothioneine described in GRN 000734 is in the "L" conformation.

Four self-affirmed GRAS conclusions have been announced publicly, however, none of the independent GRAS assessments are publicly available for review. Oxis International reported a self-affirmed GRAS conclusion for EGT™ on April 20, 2012, in NutraIngredients (Watson, 2012). In February 2015, Tetrahedron (Paris, France) announced the self-affirmed GRAS conclusion for ERGONEINE™ (Tetrahedron, 2015). In October 2021, MitoPrime by NNB Nutrition reported their self-affirmed GRAS conclusion (Krawiec, 2021). Finally, in January 2025, Bloomage Biotech reported that their MitoEGT™ ergothioneine ingredient has been self-affirmed GRAS (Bloomage Biotech, 2025).

A search of the FDA's New Dietary Ingredient Notification (NDIN) database on April 14, 2025, identified one NDIN for ergothioneine– NDI 000823 for Ergoneine™, a synthetically produced biomimetic copy of ergothioneine found in common foods. FDA determined that Tetrahedron's Ergoneine™ did not meet the definition of a dietary ingredient because it was a synthetic copy of an amino acid that is not used in protein building.

Table 10. Summary of FDA GRAS Inventory

Substance	GRN #/ Closure Date	Intended Use	Use Level	Company	FDA Response
Ergothioneine produced by <i>Escherichia coli</i> BL-21 (DE3) expressing ergothioneine synthases from <i>Schizosaccharomyces pombe</i>	GRN 001191	Intended for use as an ingredient in cakes, cookies, pastries (including granola bars); coffee; tea; fruit drinks and ades; carbonated soft drinks; and candy containing chocolate.	5 mg/serving	Shanghai EGT Synbio Group Co., LTD	Pending
Ergothioneine	GRN 000734 / May 7, 2018	Intended for use as a nutrient in cakes, cookies, pastries (including granola bars); coffee; tea; fruit	5 mg/serving	Blue California	No questions

		drinks and ades; carbonated soft drinks; and candy containing chocolate.			
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The Dietary Supplement Label Database (DSLDD) provides examples of ingredient use in products currently (on market) or previously (off-market) available to consumers. A search of the DSLDD on April 14, 2025, using the search term “ergothioneine” identified 13 dietary supplement product labels currently listed as “on market” in the U.S. that included “L-Ergothioneine” anywhere on the label. Two products included L-Ergothioneine as the single dietary ingredient, one with 5 mg daily use recommended and one with 12.5 mg of “MitoPrime” once or twice daily recommended. One other product included L-Ergothioneine as a dietary ingredient while other products included mushroom-derived ingredients and noted that L-Ergothioneine was naturally occurring.

6.2.2. Europe

Information pertaining to a synthetic L-Ergothioneine (Ergoneine® by Tetrahedron) was reviewed by EFSA’s Panel on Dietetic Products, Nutrition and Allergies in 2016 and subsequently in 2017 to expand the safety conclusion for pregnant and lactating women and very young children. In 2016, the EFSA Panel concluded that Tetrahedron’s Ergoneine® is safe as a novel food under the proposed intended conditions of use. Data presented included specifications, representative batch data, and stability data which was determined by the panel to be sufficient and did not present any safety concerns. The product was intended for use up to 5 mg ergothioneine per serving in non-alcoholic beverages, cereal bars, milk, fresh dairy products, and chocolate, and a daily dose of 30 mg per day for adults and 20 mg per day for children in food supplements. The panel determined a NOAEL of 800 mg/kg bw/day per day, which when compared to the estimated intake levels of Tetrahedron’s Ergoneine®, provided a margin of safety of 470 for adults and 216 for children three years of age and older. For this evaluation, pregnant and breastfeeding women were not part of the target population.

In October 2017, a supplementary dietary exposure and safety assessment of synthetic L-Ergothioneine was provided by the EFSA Panel for infants and young children/toddlers as well as pregnant and breastfeeding women (EFSA, 2017). The maximum anticipated cumulative daily intakes (from fortified and dietary sources) were calculated to be 2.82 mg/kg bw/day for infants, 3.39 mg/kg bw/day for toddlers, and 1.31 mg/kg bw/day for adults including pregnant and breastfeeding women. The EFSA Panel considered that the NOAEL of 800 mg/kg bw/day used in the original assessment was also relevant for these populations and concluded that synthetic L-Ergothioneine was safe under the proposed uses and use levels for infants, young children/toddlers, and pregnant and breastfeeding women.

6.2.3. Canada

A search of the Health Canada Natural Health Products Ingredient Database (NHPID) on April 12, 2025, using the term “ergothioneine” yielded one result. Ergothioneine⁵, isolated from *S. cerevisiae*, is classified as a natural health product (NHP) under Schedule 1, item 2 (an isolate) of the Natural Health Products Regulations. It is permitted for use as a preservative in products without an antioxidant claim, with a maximum oral intake of 6 mg/day for individuals aged 14 years and older and 2.2 mg/day for those aged 1–13 years. It is also approved for topical use as a preservative and skin-conditioning agent (Health Canada, 2024).

6.3. Absorption, Distribution, Metabolism & Excretion (ADME) Studies

Orally administered L-Ergothioneine is transported into cells and tissues by the Organic Cation Transporter Novel Type 1 (OCTN1), also referred to as the Ergothioneine Transporter (ETT) due to its high level of selectivity for L-Ergothioneine over other substrates such as L-carnitine (Liu et al., 2023). ETT is expressed in several different tissues throughout the body, facilitating cellular uptake due to the hydrophilicity of L-Ergothioneine, meaning it is not passively absorbed (Zalachoras et al., 2020). ETT mRNA expression in rats indicates that L-Ergothioneine can be distributed to the liver, kidney, brain, heart, and placenta (Wu et al., 2000). However, despite the potential for placental distribution, L-Ergothioneine has not been shown to induce reproductive or developmental toxicity in rats (Forster et al., 2015; Guijarro et al., 2002). L-Ergothioneine is not rapidly metabolized or excreted — it is retained in tissues such as the liver or in circulation stored in red blood cells and is reabsorbed in the kidney (Halliwell et al., 2018; Tang et al., 2018). Details on the specific elimination pathways for L-Ergothioneine have not been identified and limited study data with purified L-Ergothioneine is available.

Pharmacokinetic (PK) studies indicate that there is limited urinary excretion of L-Ergothioneine after oral administration in healthy males. Cheah et al. (2016) investigated the pharmacokinetics of L-Ergothioneine after oral administration in healthy adult males. The participants were randomly assigned to receive placebo, 5 mg, or 25 mg of L-Ergothioneine daily for 7 days (n = 15/group). Blood and urine samples were collected on Days 1, 3, 5, 8, 14, 21, 28, and 35, and participants were asked to refrain from ergothioneine-rich foods during the study. The plasma concentration of L-Ergothioneine increased in a dose-dependent manner from Days 1 to 8, then gradually declined until Day 35. However, L-Ergothioneine concentrations remained significantly higher in the 25 mg/day group compared to controls at Day 35. Urine concentrations followed a similar pattern as plasma, increasing dose-dependently until Day 8, and declining thereafter. Urinary excretion represented < 4% of the oral dose. The whole blood concentrations of L-Ergothioneine steadily increased until Day 28 and remained at a

⁵ Health Canada does not identify if this is the L form, however, the CAS number referenced is the same as the CAS number for L-Ergothioneine as reported by PubChem.

similar concentration at Day 35, indicating significant retention in blood cells. No adverse effects were observed by the authors, and none were self-reported by any participants during the study.

Weigand-Heller et al. (2012) evaluated the bioavailability of ergothioneine from mushrooms in 10 healthy men in a randomized, placebo-controlled, crossover trial. Placebo or test meal containing 8 or 16 g of mushroom powder was administered and blood samples were taken to analyze concentrations of ergothioneine, glucose, cholesterol, triglycerides, high-density lipoprotein (HDL), low-density lipoprotein (LDL), total cholesterol, C-reactive protein (CRP) and oxygen radical absorbance capacity (ORAC). The concentration of ergothioneine in the mushroom powder was 0.4 – 2.0 mg/g, meaning that the dose of ergothioneine was up to 32 mg in the high-dose group. Ergothioneine was readily absorbed and a dose-dependent increase in red blood cell concentrations of ergothioneine was observed. A blunting of the post-prandial triglyceride response was also observed in both test groups compared to controls. This effect did not appear to be dose-dependent. All subjects completed the study, and no adverse effects were mentioned in this study. Because this study was performed with mushroom powder, rather than ergothioneine, the results of this study have limited relevance for the purpose of determining whether ergothioneine exhibits properties that would negatively affect the safety profile.

L-Ergothioneine absorption may be impacted by commensal organisms. Cheah and Halliwell (2021) reported that cell-free extracts of *E. coli*, as well as *Burkholderia* sp., *Treponema denticola*, and *Alcaligenes faecalis* express an enzyme called ergothionase, a trimethylamine lyase that converts L-Ergothioneine to thiourocanic acid and trimethylamine. It is unclear if L-Ergothioneine would be metabolized into these substances in humans. Because the gastrointestinal (GI) tracts of both rats and humans contain *E. coli*, it is possible that ergothioneine could be metabolized to thiourocanic acid and trimethylamine in the GI tracts of both species. Any effect that these metabolites would have on the toxicological profile would be reflected in the rat NOAEL.

6.4. Safety Studies

6.4.1. Acute Toxicity Studies

The oral median lethal dose (LD₅₀) of L-Ergothioneine in rats was determined to be > 2,000 mg/kg bw in a Good Laboratory Practice (GLP)-compliant study conducted according to Organization for Economic Cooperation and Development (OECD) Guideline No. 423 (Forster et al., 2015). After an overnight fast, three Sprague Dawley rats (8 weeks old) were administered a 100 mg/mL solution of L-Ergothioneine by oral gavage at 20 mL/kg bw, for a dose of 2,000 mg/kg bw. A second group of three rats were given the same dose 7 days later. No control group was used as per OECD guidelines. All animals were observed for 14 days after dosing. Clinical observations were conducted daily, body weights were measured on Days 1, 8, and 15, and postmortem examinations were conducted on all animals on Day 14. Gross examinations were conducted on the digestive tract, heart, kidneys, liver, lungs, pancreas, and spleen. None of the animals died and there were no treatment-related effects on

weight gain, morbidity, or pathology. Based on the results of this study, the LD₅₀ of L-Ergothioneine is >2,000 mg/kg bw in Sprague-Dawley rats.

6.4.2. Subacute and Subchronic Toxicity Studies

The subacute and subchronic oral toxicity of L-Ergothioneine have been evaluated in 14-, 28-, and 90-day studies in rodents (Forster et al., 2015; Marone et al., 2016). The 28- and 90-day studies were performed in accordance with GLP and OECD guidelines.

For the 14-day study conducted by Forster et al. (2015), L-Ergothioneine was given to male and female rats (numbers /group not stated) in the diet at 0, 0.3, 0.5, and 0.9%. The animals were checked twice daily for mortality and morbidity and observed once a day for clinical signs (cage-side observations). Body weight was recorded before allocation, on Day 1 and then twice a week, and food consumption was recorded twice a week. A complete macroscopic post-mortem examination was performed on all animals at termination. There were no signs of toxicity at concentrations up to and including 0.9%, which was equivalent to 662 ± 82.5 mg/kg bw/day in males and 721 ± 57.6 mg/kg bw/day for females.

Marone et al. (2016) evaluated the safety of L-Ergothioneine in a 28-day repeat dose toxicity study in Sprague-Dawley rats. The rats (n = 5/sex/group) were gavaged with 50, 200, or 500 mg/kg bw/day ergothioneine (Miranova Labs, >98% purity) or vehicle (distilled water) for 28 days. None of the animals died and no treatment-related clinical findings occurred in any of the groups. There were no effects of treatment on body weight, food intake, food conversion efficiency, or macroscopic observations at necropsy. There was a slight (10%) increase in body weight in the low-dose males compared to controls that was accompanied by an increase in food intake. This finding did not exhibit dose-dependency and was deemed unrelated to treatment by the authors. Based on these results, the authors determined that rats could tolerate L-Ergothioneine at doses ≥ 500 mg/kg bw/day. This conclusion guided their dose selection for a 90-day study.

Forster et al. (2015) conducted a combined 90-day and reproductive and developmental toxicity study in male and female Sprague Dawley rats with the same L-Ergothioneine batch that was used for their 14-day study (Tetrahedron, > 98% purity). The rats (n = 10/sex/group) were administered L-Ergothioneine (0, 0.1, 0.3, or 0.9%) in the diet for 13 weeks. Corresponding mean dose levels for these concentrations were 67, 203, and 615 mg/kg bw/day in males and 73, 238, and 721 mg/kg bw/day in females, respectively. Males were terminated at the end of the mating period and females were terminated on post-partum day 6. The results for reproductive and developmental toxicity endpoints are shown in Section 6.4.3. There were no mortalities or clinical signs or toxicity and no effects of L-Ergothioneine on body weight, organ weights or pathology, or food intake. Necropsy findings were described as commonly reported for Sprague Dawley rats of a similar age. Based on the results, NOAELs for systemic toxicity of 615 mg/kg bw/day and 721 mg/kg bw/day can be assigned for male and female rats (respectively), the highest dose levels tested.

Marone et al. (2016) conducted a 90-day repeat dose toxicity study in Sprague-Dawley rats (n = 10/sex/group). The L-Ergothioneine was sourced from the same company that was used for the 28-day study (Miranova Labs). Groups of rats were administered L-Ergothioneine at 0 (distilled water vehicle), 400, 800, or 1,600 mg/kg bw/day by oral gavage for 90 consecutive days. There were no treatment-related mortalities. One male in the middle dose group died on day 66, and this was attributed to the gavage procedure. One female in the middle dose group died on day 85 with an injury to the tail. No obvious cause of death could be determined for this animal. Males in the high-dose group exhibited soft stools or diarrhea. The authors did not consider the findings to be adverse, and no other treatment-related clinical findings were observed. There were no significant changes in ophthalmological evaluations, body weight, body weight gain, food intake, or food conversion efficiency between treatment and control groups. There also were no adverse changes in hematology, coagulation, or clinical chemistry parameters in L-Ergothioneine-treated animals. However, some statistically significant changes ($p < 0.05$) in hematology were observed: monocytes (decreased) in middle-dose males, hemoglobin (decreased) in low-dose females, hematocrit (decreased) in low- and middle-dose females, mean corpuscular volume (decreased) in all treated females, red cell distribution width and reticulocytes (increased) in high-dose females, and eosinophil count (decreased) in high-dose females compared to controls. Statistically significant changes ($p < 0.05$) in the following clinical chemistry parameters were also observed: alkaline phosphatase (increased) in high-dose males and females, glucose (decreased) in high-dose males, sodium (decreased) in middle-dose males, triglycerides (decreased) in high-dose males, cholesterol (decreased) in middle- and high-dose males, creatinine, protein, and albumin (decreased) in high-dose females, and triglycerides (increased) in high-dose females compared to controls. According to the authors, “the magnitude of the changes was considered not clinically significant, fell within historical control range of the laboratory, and were not accompanied by any other corresponding clinical or pathological change.” Therefore, the statistically significant differences in the hematology and clinical parameters listed above were not considered to be adverse. No clinically significant, treatment-related changes occurred in the urinalysis parameters and there were no treatment-related changes in absolute or relative organ weights or histopathology. Therefore, the authors determined that the NOAEL of L-Ergothioneine in the study was 1,600 mg/kg bw/day, the highest dose level tested. As discussed in Section 6.2.2, EFSA has determined that the NOAEL of ergothioneine is 800 mg/kg bw/day based on the results of this study. EFSA did not agree that some of the effects in the hematopoietic system, liver, and kidney of males/and or females administered the 1600 mg/kg bw/day dose were not adverse and chose the next lowest dose (800 mg/kg bw/day) as the NOAEL based on an absence of effects at this dose (EFSA, 2016). Gene III concurs with EFSA and derived a NOAEL of 800 mg/kg bw/day from the study.

In summary, a combined 90-day and reproductive and developmental toxicity conducted by Forster et al. (2015) determined that the NOAEL of L-Ergothioneine was 615 mg/kg bw/day and 721 mg/kg bw/day in male and female rats, respectively, the highest dose level tested. A study by Marone et al. (2016) indicated that NOAEL of L-Ergothioneine was 1,600 mg/kg bw/day in rats (the highest dose

administered). However, EFSA (2016), determined that the NOAEL of the Marone et al. (2016) study was 800 mg/kg bw/day, the next highest dose level tested. Based on the “observations from both these studies”, EFSA determined that the NOAEL for ergothioneine is 800 mg/kg bw/day. Consistent with EFSA’s conclusion, Gene III has adopted the NOAEL of 800 mg/kg bw/day as pivotal safety data for this GRAS determination.

6.4.3. Reproductive and Developmental Toxicity Studies

Reproductive and developmental toxicity studies are important for GRAS assessments because the concept of GRAS presumes safety for the general population. Effects on potentially sensitive populations such as pregnant or breastfeeding women cannot be excluded. Two studies that examined the potential for reproductive or developmental toxicity in rats were identified (Forster et al., 2015; Guijarro et al., 2002). Results of these studies are shown in **Table 11**.

Forster et al. (2015) evaluated the potential of L-Ergothioneine to induce reproductive or developmental toxicity in Sprague Dawley rats in a GLP-compliant study according to OECD Guidelines 408 and 422 (also discussed in 6.4.2). L-Ergothioneine was administered dietarily at 0, 0.1, 0.3, or 0.9% in F₀ males for 10 weeks before pairing followed by 3 weeks during pairing, and in F₀ females from 13 weeks prior to pairing until day 5 of lactation. These dietary concentrations correspond to 67, 203, and 615 mg/kg bw/day in males, 73, 238, and 721 mg/kg bw/day in females before mating, and 91, 319, and 994 mg/kg bw/day during lactation. None of the animals died. There were no effects of any dose of L-Ergothioneine on clinical observations, food intake, body weight, body weight gain during gestation, fertility, duration of gestation, pre- or post-implantation losses, litter size, mean pup body weight or weight gain, clinical signs in the pups or sex distribution. Histopathological evaluation of all F₀ animals, including reproductive organs, revealed no treatment-related findings. Based on the results, the NOAEL for reproductive or developmental toxicity was determined to be 0.9% in the diet, the highest dose tested, corresponding to 615 mg/kg bw/day in males, 721 mg/kg bw/day in females before mating, and 994 mg/kg bw/day during lactation.

Guijarro et al. (2002) assessed the effects of L-Ergothioneine on embryo development in diabetic and non-diabetic pregnant Wistar rats. Animals were assigned to four groups: control, control with ergothioneine supplementation, diabetic, and diabetic with ergothioneine supplementation. L-Ergothioneine was administered by gavage at a dose of 1.147 mg/kg bw per day from gestational Day 0 to Day 11.5. No significant effects of L-Ergothioneine were observed on plasma glucose levels or resorption rates in any group. Additionally, there were no significant differences in somite number or crown–rump length between supplemented and non-supplemented animals within either the diabetic or control groups.

In addition to the studies summarized in **Table 11**, EFSA (2017) reviewed the safety of ergothioneine for infants, young children, and pregnant or breastfeeding women and concluded that the ingredient

was safe for use in each of these populations. The EFSA report assumed that intake of ergothioneine in these groups is 2.82 mg/kg bw/day in infants, 3.39 mg/kg bw/day in young children, and 1.31 mg/kg bw/day in pregnant or breastfeeding women.

Table 11. Summary of Reproductive and Developmental Toxicity Studies

Study Design	Test Material	Dose	Results	Reference
Reproductive and developmental toxicity study in pregnant Wistar rats (normal or diabetic)	L-Ergothioneine from Oxis International (99% purity)	0 (n = 9) or 1.147 mg/kg bw/day (n = 8) by oral gavage for the first 11.5 days of pregnancy. Day 0 of gestation was the day that sperm was found in vaginal smears	Ergothioneine had no effect on the plasma glucose levels, both in diabetic and control animals. There were no significant changes in the number of somites, crown-rump length, proportion of reabsorptions, or proportion of malformed embryos between controls and normal rats given ergothioneine and there were no adverse effects on these parameters in diabetic rats administered ergothioneine.	Guijarro et al. (2002)
Reproductive and developmental toxicity study in Sprague Dawley rats following subchronic administration	L-(+)-ergothioneine from Tetrahedron (batch 2299, purity > 98%)	0, 0.1, 0.3, or 0.9% in the diet (n = 10/sex/group), corresponding to 0, 67, 203, or 615 mg/kg bw/day for males and 0, 73, 238, and 721 mg/kg bw/day for females and 91, 319, and 994 mg/kg bw/day during lactation. Dosing for males began 10 weeks before pairing and continued throughout the pairing period. For females, dosing started 13 weeks before pairing and continued through pairing, gestation, and up to day 5 of lactation.	There were no treatment-related effects on mating index, days to mating, fertility index, number of females with live pups, gestation index, number of females that delivered, duration of gestation, corpora lutea, pre- or post-implantation loss, litter size, mean pup weight, clinical signs, or repartition of dead or cannibalized pups. The pups from the high dose group tended to have a lower body weight, but this was not statistically significant and did not deviate from the historical control range. The NOAEL for reproductive and developmental toxicity was determined to be the highest dose tested, 0.9% in the diet, corresponding to 615 mg/kg bw/day in males, 721 mg/kg	Forster et al. (2015)

Study Design	Test Material	Dose	Results	Reference
			bw/day in females prior to mating, and 994 mg/kg bw/day during lactation.	

6.4.4. Genotoxicity Studies

Published genetic toxicity studies conducted with L-Ergothioneine are summarized in **Table 12**. These studies include several Ames tests, a chromosomal aberration assay, and an *in vivo* micronucleus assay. The results of all studies are negative, indicating a lack of genotoxicity associated with L-Ergothioneine exposure.

Table 12. Summary of Genotoxicity Studies

Study Type	Test Material and Dose	Result	Reference
Bacterial reverse mutation (Ames) test in <i>Salmonella typhimurium</i> , TA 98, TA 100, TA 1535, TA 1537 and <i>E. coli</i> WP2uvrA	L-(+)-ergothioneine at 1.58, 5, 15.8, 50, 158, 500, 1580, and 5000 µg/plate with and without metabolic activation	No significant increases in revertant colony numbers, no precipitation, and no toxic effects at any of the tested doses.	Marone et al. (2016)
Ames test in <i>Salmonella typhimurium</i> TA 98, TA 100, TA 1535, TA 1537, and <i>E. coli</i> WP2 pKM101 (according to OECD No. 471)	L-(+)-ergothioneine at 0.06 – 5.0 mg/plate with and without metabolic activation	No significant increases in revertant colony numbers.	EFSA (2016)
Chromosomal aberration assay in V79 Chinese hamster ovary cells (according to OECD No. 473)	L-(+)-ergothioneine at 1.25, 2.5, 3.75, and 5 mg/mL with and without metabolic activation	Ergothioneine did not induce structural chromosome aberrations and no polyploidy was observed.	Schauss et al. (2011)
Mammalian erythrocyte micronucleus assay in mice (according to OECD No. 476)	L-(+)-ergothioneine at 375, 750, and 1,500 mg/kg bw	No significant increase in micronucleated polychromatic erythrocytes at 24 or 48 hours.	Schauss et al. (2011)
Ames test in <i>Salmonella typhimurium</i> , TA 98, TA 100, TA 1535, TA 1537, and <i>E. coli</i> WP2 pKM101 (according to OECD No. 471)	L-(+)-ergothioneine at 5, 15.81, 50, 158.1, 500, 1581.1, and 5000 µg/plate with and without metabolic activation	No significant increases in revertant colony numbers.	Schauss et al. (2010)

In addition, Gene III conducted a Bacterial Reverse Mutation Test (Ames Test) with their L-Ergothioneine ingredient. The test was conducted according to GLP and the OECD Guideline 471, using plate incorporation and pre-incubation methods. This study used five histidine-requiring tester strains of *Salmonella typhimurium* (his-): TA97a, TA98, TA100, TA102, and TA1535 with and without metabolic activation (S9 mix). The test item was completely dissolved in ultra-pure water at 50 mg/mL, and test concentrations were 5000, 1500, 500, 150, 50, and 15 µg/plate. Based on a preliminary test, no cytotoxicity or precipitation was expected at these concentrations. Three replicate plates were prepared for each test concentration, negative control (ultra-pure water), and positive control for each strain. The preincubation assay involved preincubating the test strain with each test concentration, negative control (ultra-pure water), and positive control for 20 min at 35.7 – 36.2 °C prior to plating. Plate incorporation assay plates were incubated at 35.4 – 37.5 °C for 69.5 hours and preincubation assay plates were incubated at 35.4 – 37.9 °C for 48.4 hours. After incubation, the numbers of colonies in each plate were counted. The presence of precipitation or cytotoxicity of the test item was also recorded. The test material was not cytotoxic and did not produce a precipitate both with and without metabolic activation, and the positive and negative controls produced the expected results, confirming the validity of the assays. The test material did not meet any of the criteria for a positive result at any concentration in the plate incorporation or preincubation assays. Thus, L-Ergothioneine was concluded as not mutagenic under the conditions of this assay. The full report is available in Appendix 7.

6.4.5. Endocrine Disruption Studies

The Integrated Chemical Environment tool indicates that ergothioneine is not an estrogen or androgen receptor agonist or antagonist⁶

6.4.6. Clinical Studies

The safety of oral ergothioneine in humans has been evaluated in several studies. In particular, EFSA (2016) reviewed the existing human studies as part of their review of ergothioneine as a novel food. EFSA addressed potential concerns about the effects of ergothioneine on glucose homeostasis (Salt, 1931; Fraser & Jegard, 1950, as cited in EFSA, 2016; Epand, 1982; Epand et al., 1988), Crohn's disease (Peltekova et al., 2004; Taubert et al., 2005, 2009; Leung et al., 2006; Kato et al., 2010; Huff et al., 2011), and rheumatoid arthritis (Reglinski et al., 1991; Tokuhiko et al., 2003; Taubert et al., 2006, as cited in EFSA, 2016). The EFSA panel concluded that no relationship could be determined between dietary or supplemental ergothioneine and the development of these conditions. This conclusion was based on the observational nature of the studies, which assessed ergothioneine levels in the study populations rather than through controlled intervention trials. Regarding direct human exposure to ergothioneine, only one short-term oral intake study was reviewed by EFSA. Based on the study by Cheah et al. (2016), the panel determined that no safety concerns were associated with short-term oral

⁶ <https://ice.ntp.niehs.nih.gov/Search>. Identifier is 497-30-3 (The CAS number for L-Ergothioneine). Query performed on March 18, 2025.

exposure to ergothioneine in humans. More recent clinical trials further support that doses up to 25 mg/day of L-Ergothioneine are well tolerated in humans and are discussed below. Because the effect of substances on organ pathology cannot be determined from clinical studies and clinical studies are conducted on a limited number of subjects, the results of clinical studies typically cannot be used as the pivotal basis for the safety of food ingredients but they can provide supportive safety information.

Calvo et al. (2016) conducted a retrospective analysis of a clinical trial in which 37 adults (24 females and 13 males) consumed 100 g of white button mushrooms per day for 16 weeks, with a 4-week follow-up period. Serum samples were analyzed for L-Ergothioneine and inflammatory and oxidative stress biomarkers that have been associated with diabetes. Serum ergothioneine increased from baseline (317 ng/mL) to 16 weeks (677 ng/mL) and remained elevated at 20 weeks (496 ng/mL). L- Ergothioneine administration did not adversely affect any of the parameters measured in the study.

Cheah et al. (2016) conducted a randomized, double-blind, placebo-controlled trial with L-Ergothioneine in 45 healthy young males. Subjects received placebo or L-Ergothioneine (5 or 25 mg/day) for 7 days. No adverse effects (AE) were observed, and none of the subjects reported side effects. Additionally, there were no significant differences between groups in liver function (albumin, bilirubin, alanine transaminase, aspartate transaminase, alkaline phosphatase, gamma-glutamyl-transpeptidase, and lactate dehydrogenase), glucose, or lipid profile (low-density lipoprotein, high-density lipoprotein, total cholesterol, and triglycerides). The results indicate that L-Ergothioneine was well tolerated at 25 mg/day in healthy males.

Katsube et al. (2022) conducted a randomized, double-blind, placebo-controlled study with L-Ergothioneine in 92 healthy adults (26 females and 66 males). The subjects received placebo (food-grade dextrin) or 20 mg of L-Ergothioneine (Tetrahedron, purity >99.5%) daily for 4 weeks. No treatment-related adverse events were observed during the study and all biochemical parameters remained within normal limits. L-Ergothioneine did not adversely affect any of the endpoints in the study. The results show that L-Ergothioneine was well tolerated at 20 mg/day over 4 weeks.

Yau et al. (2024) conducted a randomized, double-blind, placebo-controlled trial to evaluate the effects of L-ergothioneine in 19 elderly individuals with mild cognitive impairment. Participants received either a placebo (99% microcrystalline cellulose, 1% magnesium stearate) or 25 mg of L-Ergothioneine (ERGOLD, formerly Tetrahedron, purity >99%) three times per week for one year. No significant changes were observed in whole blood parameters, liver function markers, renal function markers, or fasting glucose levels throughout the study. Additionally, no adverse reactions or clinically relevant changes were reported over the year-long administration period, suggesting that L-ergothioneine was well tolerated in elderly individuals.

In general, EFSA's review of available human studies found no evidence linking the consumption of L-Ergothioneine to concerns such as glucose homeostasis, Crohn's disease, or rheumatoid arthritis. As these studies were observational in nature and lacked safety endpoints or reports of adverse effects, they are considered of limited value for the safety evaluation of L-Ergothioneine. Additionally, the current body of human data, including more recent clinical studies, supports that L-Ergothioneine is well tolerated at doses up to 25 mg/day in humans, with no adverse effects observed.

6.5. Safety of Production Organism

The production strain used for L-ergothioneine manufacturing is a genetically modified *E. coli* K-12 MG1655, engineered to express three biosynthetic enzymes that catalyze the conversion of histidine to L-ergothioneine. The safety of the production strain is assessed based on the established safety profile of the host strain, the safety of the inserted genes, and the genetic modification process. The decision tree by Pariza and Johnson (2001) was also used to determine the safety of the production organism.

Additionally, the manufacturing process includes a pasteurization step designed to eliminate production strain. To confirm the absence of the production organism, *E. coli* testing is included in the product specifications. Batch analyses of five non-consecutive batches (**Table 4**) demonstrated that *E. coli* was not detected per 1 g of the final product. Gene III also tested three non-consecutive batches for residual DNA, and the results demonstrated that residual DNA was not detected in the final product (Appendix 6). Thus, there is reasonable certainty that the production organism does not pose safety concerns to the consumer.

6.5.1. Safety of Host Strain

The host strain, *E. coli* K-12 MG1655, is classified as a Biosafety Level 1 microorganism by the ATCC. *E. coli* K-12 has been widely used as a laboratory organism for over 50 years and is among the most extensively characterized microorganisms (Bachmann, 1972; Jensen, 1993). The complete genome of *E. coli* K-12 has been sequenced, confirming the absence of toxigenic potential (Blattner et al., 1997; Hayashi et al., 2006). It is classified as a Risk Group 1 organism under the National Institutes of Health (NIH) Guidelines, indicating that it is not associated with disease in healthy adult humans (NIH, 2024). Beyond laboratory research, *E. coli* K-12 has a well-documented history of safe use in the food and pharmaceutical industries. Both *E. coli* K-12 and its derivatives have been utilized in the production of specialty chemicals and pharmaceuticals and were exempted from review under the Toxic Substances Control Act (TSCA) by the U.S. EPA (U.S. EPA, 1997). The FDA affirmed the GRAS status of chymosin, a food enzyme preparation derived from a genetically modified *E. coli* K-12 strain, in 1990 (21 CFR §184.1685) (Flamm, 1991; Olempska-Beer et al., 2006). In addition, *E. coli* K-12 and its derivatives have been used as production organisms in multiple GRAS notices (e.g., GRN 000624, GRN 000659, GRN 000735, GRN 000745, GRN 001010, and GRN 001106), all of which

received a “no questions” letter. Specifically, *E. coli* K-12 MG1655 has been employed in GRAS submissions for various food ingredients, including ergothioneine (GRN 000734), 2'-O-fucosyllactose (GRN 000749 and GRN 000897), 3-fucosyllactose (GRN 000951), and 6'-sialyllactose (GRN 001075), each receiving a "no questions" letter from the FDA. The EFSA has also evaluated several food ingredients and enzymes produced using *E. coli* K-12 and its derivatives. These include 2'-fucosyllactose/difucosyllactose mixtures (EFSA NDA, 2019), lacto-N-fucopentaose I (LNFP-I)/2'-fucosyllactose (2'-FL) mixtures (EFSA NDA, 2023a), 3-fucosyllactose (3-FL) (EFSA NDA, 2023b; EFSA NDA, 2021), and d-psicose 3-epimerase (EFSA CEP, 2021). EFSA concluded that these substances are safe or do not raise safety concerns under their intended conditions of use.

E. coli K-12 is widely recognized as a safe, non-pathogenic, and non-toxigenic microorganism with extensive applications in food biotechnology (Gorbach, 1978; Mühldorfer & Hacker, 1994; U.S. EPA, 1997). It is frequently used as a reference strain for studying the virulence factors of pathogenic *E. coli*, as it is non-pathogenic (Blanc-Potard et al., 2002; Kaper et al., 2004). *E. coli* K-12 and its derivatives do not colonize the mammalian gastrointestinal tract, produce toxins such as Shiga toxin, or persist in soil and water (U.S. EPA, 1997).

In summary, *E. coli* K-12 is a non-pathogenic and non-toxigenic microorganism that has been used as a production organism for food ingredients, which have either received a "no questions" letter from the FDA or been deemed safe by EFSA.

6.5.2. Safety of Inserted Genes

The three inserted genes—*egtI*, *egtE*, and *egtD*—are sourced from *N. crassa* and *M. smegmatis* MC² 155. The *egtI* gene is derived from *N. crassa*, a non-pathogenic filamentous fungus (Santos et al., 2018). *N. crassa* is non-toxic and has a well-documented history of human consumption (Havlik et al., 2017; Bartholomai et al., 2022). It has traditionally been used in East Asia for fermenting soybean byproducts in China, producing oncom in Indonesia, and in traditional foods in Brazil and Borneo (Bartholomai et al., 2022). No evidence of pathogenicity or toxigenicity has been reported in either its long history of food use or laboratory studies (Perkins & Davis, 2000).

The *egtD* and *egtE* genes originate from *M. smegmatis* MC² 155, a bacterium classified as a Biosafety Level 1 organism by ATCC. *M. smegmatis* is a saprophytic bacterium commonly found in soil, water, and sebaceous secretions (Czubat et al., 2020). It is widely recognized as non-pathogenic (Warner et al., 2005; Czubat et al., 2020; Sparks et al., 2023). A PubMed search found no evidence suggesting that *M. smegmatis* poses a risk of toxigenicity.

6.5.2.1. Potential Allergenicity

The potential allergenicity of the inserted genes was assessed using the AllergenOnline and Allergome databases. A multi-tiered approach was applied in AllergenOnline, incorporating (1) direct sequence comparison using FASTA alignment to known allergens with >35% sequence identity and an E-value <1, (2) an 80-amino-acid sliding window search, and (3) identification of contiguous 8-mer sequence matches with known allergens. The analysis was further supplemented by cross-reactivity evaluations using the Allergome database.

For *egtI*, AllergenOnline identified a FASTA alignment match with Proj1 (Group 2468), showing 26.3% identity over a 99-amino-acid alignment, with an E-value of 0.9. No matches exceeding the 35% identity threshold were found in the sliding window analysis, and no 8-mer matches were detected. The Allergome analysis identified potential cross-reactivity with Cav p 4.0101, Fel d 2, and Alta 3.0101, exhibiting sequence identities of 55%, 51%, and 58%, respectively. These matches involved short sequences (<80 amino acids) associated with structural motifs (e.g., β -sheets and α -helices) rather than immune recognition sites. These findings suggest a low direct allergenicity risk and a low-to-moderate potential for cross-reactivity based on sequence similarities within conserved structural motifs.

For *egtD*, AllergenOnline did not identify any significant matches using FASTA alignment. The sliding window search confirmed the absence of relevant alignments, and no 8-mer matches were detected. The Allergome analysis indicated weak sequence similarities with the apple allergen Mal d 1 (42% identity over 76 amino acids) and the dust mite allergen Der p 1 (39% identity over 68 amino acids). These alignments were localized to β -sheet-rich conserved domains and signal peptide regions, which are not associated with IgE-binding epitopes. Collectively, the data indicate a low direct allergenicity risk and a low likelihood of cross-reactivity.

For *egtE*, no significant matches were identified in AllergenOnline using FASTA alignment. The sliding window analysis found no alignments exceeding the identity threshold, and no 8-mer sequence matches were detected. The Allergome analysis yielded results consistent with those for *egtD*, showing sequence similarities with Mal d 1 (42% identity over 76 amino acids) and Der p 1 (39% identity over 68 amino acids). These matches were confined to short sequence segments and structural regions with minimal potential for immune recognition. The overall assessment indicates negligible direct allergenicity risk and a low potential for cross-reactivity.

In conclusion, the inserted genes *egtD*, *egtI*, and *egtE* present a low risk of direct allergenicity. AllergenOnline did not identify any sequence alignments exceeding established thresholds for allergenic potential. While sequence similarities with known allergens were observed in Allergome, these involved short structural motifs with limited relevance to immune recognition. Overall, the findings support the conclusion that the inserted genes pose no significant allergenic risk.

6.5.2.2. Potential Toxicogenicity

The toxigenicity of the inserted genes was evaluated using ToxinPred2 and the T3DB database. ToxinPred2 predictions were conducted using a hybrid algorithm (RF+BLAST+MERCI) with a classification cutoff of 0.6 and an E-value threshold of $\leq 10^{-6}$. Additionally, sequences were aligned against known toxin sequences in T3DB using FASTA alignment, with a homology threshold set at $E \leq 1 \times 10^{-5}$.

ToxinPred2 classified all tested sequences (*egtI*, *egtD*, and *egtE*) as non-toxic, with toxicity scores ranging from 0.08 to 0.15 and confidence levels of 93–96%. No N-terminal signal peptides or toxin domains were detected. Similarly, T3DB analysis did not identify significant homology between the inserted genes and known toxins.

In conclusion, the absence of toxin domains, low toxicity scores, and lack of homology with known toxins collectively indicate that these genes do not pose a significant toxicity risk.

Collectively, the non-toxicogenic and non-pathogenic nature of the source organisms, along with *in silico* evaluation results indicating no allergenic or toxigenic risk for the inserted genes, supports the conclusion that the insertion of these genes encoding enzymes involved in L-Ergothioneine biosynthesis does not pose a safety concern in food production.

6.5.3. Safety of Genetic Modification

The production strain for Gene III's L-Ergothioneine is developed from *E. coli* K-12 MG1655. A conventional molecular recombinant method was employed to integrate the target genes *egtI*, *egtE*, and *egtD* for the production of L-Ergothioneine. As detailed in Section 2.2.2, the expression plasmid and the genes encoding the enzymes are well characterized. The genes were carefully designed before being integrated into the plasmid. The molecular manipulations were conducted precisely in such a way as to ensure the exclusion of any unintended genes. The modification process also does not introduce any new sequences that would lead to the generation of toxic secondary metabolites.

Based on the comprehensive characterization of the plasmid and inserted genes, and because all molecular manipulation in the production strain via integration of recombinant plasmid is done accurately, there is reasonable certainty that the genetic modification process does not pose safety risks to the consumer.

6.5.4. Decision Tree

The safety of Gene III's genetically modified *E. coli* K-12 MG 1655 production strain is assessed using the Pariza and Johnson (2001) decision tree. The assessment is as follows:

1. Is the production strain genetically modified? (If yes, go to 2. If no, go to 6) **YES**
2. Is the production strain modified using rDNA techniques? (If yes, go to 3. If no, go to 5.) **YES**
3. Issues relating to the introduced DNA are addressed in 3a–3e.
 - 3a. Do the expressed enzyme product(s) which are encoded by the introduced DNA have a history of safe use in food? (If yes, go to 3c. If no, go to 3b) **YES**
 - 3c. Is the test article free of transferable antibiotic resistance gene DNA? (If yes, go to 3e. If no, go to 3d) **YES**
 - 3e. Is all other introduced DNA well characterized and free of attributes that would render it unsafe for constructing microorganisms to be used to produce food-grade products? (If yes, go to 4. If no, go to 12) **YES**
4. Is the introduced DNA randomly integrated into the chromosome? (If yes, go to 5. If no, go to 6) **NO**
6. Is the production strain derived from a safe lineage, as previously demonstrated by repeated assessment via this evaluation procedure? (If yes, the test article is ACCEPTED. If no, go to 7) **YES, the test article is ACCEPTED.**

In conclusion, Gene III's production strain for L-Ergothioneine is safe for production use.

6.6. GRAS Criteria

FDA defines “safe” or “safety” as it applies to food ingredients as:

“...reasonably certainty in the minds of competent scientists that the substance is not harmful under the intended conditions of use.”⁷

Amplification is provided in that the conclusion of safety is to include probable consumption of the substance in question, the cumulative effect of the substance and appropriate safety factors. It is the FDA's operational definition of safety that serves as the framework against which this evaluation is provided.

Furthermore, in discussing GRAS criteria, FDA notes that:

“...General recognition of safety requires common knowledge, throughout the expert scientific community knowledgeable about the safety of substances directly or indirectly added to food, that

⁷ See 21 CFR 170.3 (e)(i) and 81 FR 54959 Available at: <https://www.federalregister.gov/documents/2016/08/17/2016-19164/substances-generally-recognized-as-safe>.

there is reasonable certainty that the substance is harmful under the conditions of its intended use.”
“Common knowledge can be based on either “scientific procedures” or on experience based on common use of a substance in food prior to January 1, 1958.”

FDA discusses in more detail what is meant by the requirement of general knowledge and acceptance of pertinent information within the scientific community, i.e., the so-called “common knowledge element”, in terms of the two following component elements:

- Data and information relied upon to establish safety must be generally available, and this is most commonly established by utilizing published, peer-reviewed scientific journals; and
- There must be a basis to conclude that there is consensus (but not unanimity) among qualified scientists about the safety of the substance for its intended use, and this is established by relying upon secondary scientific literature such as published review articles, textbooks, or compendia, or by obtaining opinions of experts or opinions from authoritative bodies, such as JECFA and the National Academy of Sciences.

General recognition of safety based upon scientific procedures requires the same quantity and quality of scientific evidence as is required to obtain approval of a food additive. General recognition of safety through scientific procedures shall be based upon the application of generally available and accepted scientific data, information, or methods, which ordinarily are published, as well as the application of scientific principles, and may be corroborated by the application of unpublished scientific data, information, or methods.

The apparent imprecision of the terms “appreciable”, “at the time” and “reasonably certainty” demonstrates that the FDA recognized the impossibility of providing absolute safety in this or any other area (Lu, 1988; Renwick, 1990; Rulis and Levitt, 2009).

As noted below, this safety assessment to ascertain GRAS status for L-Ergothioneine for the specified food uses meets FDA criteria for reasonable certainty of no harm by considering both the technical and common knowledge elements.

6.6.1. Common Knowledge Elements for GRAS Conclusions

The first common knowledge element for a GRAS conclusion requires that data and information relied upon to establish safety must be generally available. This is most commonly established by utilizing studies published in peer-reviewed scientific journals. The second common knowledge element for a GRAS conclusion requires that consensus exists within the broader scientific community.

6.6.1.1. Public Availability of Scientific Information

The regulatory framework for determining whether a substance is GRAS is outlined in 21 CFR 170.30, which states that GRAS status through scientific procedures shall ordinarily be based upon published studies, which may be corroborated by unpublished studies and other data and information. These criteria have been applied to the existing data for L-Ergothioneine. This GRAS evaluation satisfies the first common knowledge element, as the scientific information that is the basis for the GRAS determination for L-Ergothioneine is publicly available.

Pivotal information for the safety of L-Ergothioneine in rats has been published (Marone et al., 2016) and supporting information in rats (Forster et al., 2015) and from clinical studies also has been published (Calvo et al., 2016; Cheah et al., 2016; Katsube et al., 2022; Yau et al., 2024).

Furthermore, L-Ergothioneine has previously been determined GRAS (GRN 000734) and received a “no questions” letter from the FDA. EFSA (2016) has also determined that L-Ergothioneine is safe in food at up to 30 mg/day in adults and 20 mg/day in children. Gene III’s L-Ergothioneine is substantially equivalent to the L-Ergothioneine subject of GRN 000734 and EFSA’s opinion as shown in **Table 3**.

6.6.2. Scientific Consensus

The second common knowledge element for a GRAS conclusion requires that there be a basis to conclude that consensus exists among qualified scientists about the safety of the substance for its intended use.

Gene III affirms that L-Ergothioneine is manufactured under cGMP conditions with raw materials and processing aids that meet the appropriate food grade regulations. Gene III’s specifications for L-Ergothioneine meet or exceed specifications for L-Ergothioneine ingredients that have been notified as GRAS (GRN 000734) or have been determined safe by EFSA, and batch-to-batch consistency has been demonstrated against these specifications.

Gene III’s L-Ergothioneine is intended to be used as an ingredient in beverages (coffee, tea, fruit drinks and ades, carbonated soft drinks, sport and energy drinks, nutritional beverages), protein and nutritional powders, bakery and other baked goods (nutrition bars, cereal bars, sweet bakery products) and candy at levels ranging from 8.3 – 150 mg/100 g of food, with the intention of providing 30 – 150 mg of L-Ergothioneine per serving. It will not be used in infant formulas or USDA-regulated foods.

The estimated mean and 90th percentile intake of Gene III’s L-Ergothioneine for the overall population is 145.4 mg/day (2.05 mg/kg bw/day) and 285.0 mg/day (3.92 mg/kg bw/day), respectively. Based on the consumption analysis, the population with the highest absolute consumption (adult males) has mean and 90th percentile EDIs of L-Ergothioneine at 182.8 mg/day and 338.1 mg/day, respectively.

The population with the highest consumption on a body weight basis is children aged 2 – 5 years; the mean and 90th percentile EDI in this age group are 2.87 mg/kg bw/day and 5.77 mg/kg bw/day, respectively.

Estimated 90th percentile cumulative total daily intake of L-Ergothioneine for the overall population is 290.8 mg/day (4.1 mg/kg bw/day). On a cumulative basis, the population with the highest absolute consumption (adult males) has a 90th percentile EDI of 358.6 mg/day (4.1 mg/kg bw/day). The cumulative 90th percentile EDI for the population with the highest consumption on a body weight basis (children aged 2 – 5 years) is 6.1 mg/kg bw/day.

In vivo and *in vitro* genotoxicity studies, including an unpublished Ames study using Gene III's L-Ergothioneine, consistently show that L-Ergothioneine is not genotoxic. Furthermore, in an EFSA evaluation of a petition for the use of synthetic L-Ergothioneine as a novel food ingredient, two subchronic toxicity studies in rats (Forster et al., 2015; Marone et al., 2016) were evaluated to establish a NOAEL for L- ergothioneine. The Forster et al. (2015) study was an OECD guideline combined 90-day and reproductive and developmental toxicity study that established L-Ergothioneine NOAELs of 615 mg/kg bw/day and 721 mg/kg bw/day in male and female rats, respectively (the highest dose levels tested), and the study by Marone et al. (2016) was a 90-day OECD guideline study in male and female rats with a NOAEL of 800 mg/kg bw/day, according to EFSA.

The NOAEL of 800 mg/kg bw/day from the Marone et al. (2016) study was used by EFSA to determine a safe level of intake of L-Ergothioneine for children as well as adults and is used for the GRAS Determination of Gene III's L-Ergothioneine. Based on similar specifications and undetectable levels of contaminants that could affect toxicity, Gene III's L-Ergothioneine is substantially equivalent to the L-Ergothioneine that was the subject of EFSA's evaluation. An acceptable daily intake (ADI) of 8 mg/kg bw/day for humans is calculated after applying a 100-fold uncertainty factor⁸ to the rat NOAEL to account for interspecies and intraspecies variation. The 90th percentile cumulative intake estimates for L-Ergothioneine are below the ADI when Gene III's L-Ergothioneine is added to food as indicated.

EFSA has determined that L-Ergothioneine is safe in food at up to 30 mg/day in adults and 20 mg/day in children, but these intakes were proposed by an applicant and safety at higher levels was not evaluated. Similar intake levels have been shown to be safe in clinical studies with adult subjects.

These intakes are less than the estimated mean and 90th percentile intake levels for Gene III's L-Ergothioneine in all age groups evaluated. However, the EDIs for Gene III's L-Ergothioneine are likely

⁸ To determine an ADI for food additives, a 100-fold safety factor is historically applied to the NOAEL derived from animal experiments. The safety factor is comprised of a 10-fold factor to allow for interspecies differences and a 10-fold factor to account for possible differences between individual humans (Renwick, 1991).

to be overestimated because they assume the worst-case consumption in that these populations would consume all fortified foods each day and that no substitution will occur.

In summary, a compelling case can be made that scientific consensus exists regarding the safety of L-Ergothioneine under intended use. Gene III Biotechnology Co., Ltd. maintains that other appropriately qualified scientists would also conclude that L-Ergothioneine is GRAS under the intended use given the available evidence and regulatory framework using accepted toxicological principles.

Part 7. LIST OF SUPPORTING DATA AND INFORMATION IN THE GRAS NOTICE

7.1. List of Appendices

Appendix 1: Genomic Sequencing Report

Appendix 2: Statement of Food Grade Material Status

Appendix 3: Statement of cGMP Compliance

Appendix 4: Certificate of Analysis (CoA) of L-Ergothioneine

Appendix 5: Report on Estimated Daily Intake of L-Ergothioneine

Appendix 6: Analysis of Residual DNA in L-Ergothioneine

Appendix 7: Bacteria Reverse Mutation Test Report of L-Ergothioneine

7.2. List of Acronyms

°C	Celsius
µg	Micrograms
µL	Microliters
ADME	Absorption, Distribution, Metabolism and Excretion
AE	Adverse effects
Al	Aluminum
ATCC	American Type Culture Collection
bw	Body weight
CCTCC	China Center for Type Culture Collection
CDC	Center for Disease Control and Prevention
CEP	Panel on Food Contact Materials, Enzymes and Processing Aids
cGMP	Current Good Manufacturing Practices
CoAs	Certificates of analysis
DSLDD	Dietary supplement label database
EFSA	European Food Safety Authority
Egt1	Ergothioneine biosynthesis protein
EgtD	Histidine methyltransferase
EgtE	Hercynylcysteine sulfoxide lyase
ETT	Ergothioneine transporter
FD&C Act	Federal Food Drug and Cosmetics Act
FDA	U.S. Food and Drug Administration
g	Grams
GC	Gas Chromatography
GI	Gastrointestinal
GLP	Good Laboratory Practice
GRAS	Generally Recognized as Safe

GRN	GRAS Notification
kg	Kilograms
L	Liters
LD ₅₀	Median lethal dose
LDL	Low-density lipoprotein
LOQ	Limit of quantification
mg	Milligrams
MPa	Megapascals
NCBI	National Center for Biotechnology Information
NDA	Panel on Nutrition, Novel Foods and Food Allergens
NDIN	New Dietary Ingredient Notification
ng	Nanograms
NHANES	National Health and Nutrition Examination Survey
NHP	Natural Health Product
NHPID	Natural Health Products Ingredient Database
NIH	National Institutes of Health
nm	Nanometers
NOAEL	No observed adverse effect level
OCTN1	Organic cation transporter novel type 1
OECD	Organisation for Economic Co-operation and Development
ORAC	Oxygen radical absorbance capacity
PCR	Polymerase Chain Reaction
PE	Polyethylene
PET	Polyethylene terephthalate
PFAS	per- and poly-fluoroalkyl substances
PK	Pharmacokinetics
RO	Reverse Osmosis
rpm	Revolutions per minute
SAM	S-adenosyl-L-methionine
TSCA	Toxic Substances Control Act
U.S. EPA	United States Environmental Protection Agency
U.S.	United States
UV	Ultraviolet

7.3. List of References

Bachmann, B. J. (1972). Pedigrees of some mutant strains of *Escherichia coli* K-12. *Bacteriological Reviews*, 36(4), 525–557. <https://doi.org/10.1128/br.36.4.525-557.1972>

Bartholomai, B. M., Ruwe, K. M., Thurston, J., Jha, P., Scaife, K., Simon, R., Abdelmoteleb, M., Goodman, R. E., & Farhi, M. (2022). Safety evaluation of *Neurospora crassa* mycoprotein for use as a novel meat alternative and enhancer. *Food and Chemical Toxicology*, 168, 113342. <https://doi.org/10.1016/j.fct.2022.113342>

Biotech, B. (2025, January 24). Bloomage MitoEGTTM achieves Self-Affirmed U.S. GRAS Certification. *GlobeNewswire News Room*. <https://www.globenewswire.com/news-release/2025/01/24/3014870/0/en/Bloomage-MitoEGTTM-Achieves-Self-Affirmed-U-S-GRAS-Certification.html>

Blanc-Potard, A., Tinsley, C., Scaletsky, I., Bouguenec, C. L., Guignot, J., Servin, A. L., Nassif, X., & Bernet-Camard, M. (2002). Representational Difference Analysis between Afa/Dr Diffusely Adhering *Escherichia coli* and Nonpathogenic *E. coli* K-12. *Infection and Immunity*, 70(10), 5503–5511. <https://doi.org/10.1128/iai.70.10.5503-5511.2002>

Blattner, F. R., Plunkett, G., Bloch, C. A., Perna, N. T., Burland, V., Riley, M., Collado-Vides, J., Glasner, J. D., Rode, C. K., Mayhew, G. F., Gregor, J., Davis, N. W., Kirkpatrick, H. A., Goeden, M. A., Rose, D. J., Mau, B., & Shao, Y. (1997). The Complete Genome Sequence of *Escherichia coli* K-12. *Science*, 277(5331), 1453–1462. <https://doi.org/10.1126/science.277.5331.1453>

Browning, D. F., Hobman, J. L., & Busby, S. J. W. (2023). Laboratory strains of *Escherichia coli* K-12: things are seldom what they seem. *Microbial Genomics*, 9(2). <https://doi.org/10.1099/mgen.0.000922>

Calvo, M. S., Mehrotra, A., Beelman, R. B., Nadkarni, G., Wang, L., Cai, W., Goh, B. C., Kalaras, M. D., & Uribarri, J. (2016). A Retrospective Study in Adults with Metabolic Syndrome: Diabetic Risk Factor Response to Daily Consumption of *Agaricus bisporus* (White Button Mushrooms). *Plant Foods for Human Nutrition*, 71(3), 245–251. <https://doi.org/10.1007/s11130-016-0552-7>

CDC (2022) *National Health and Nutrition Examination Survey (NHANES) 2017-2018 Dietary Data: Centers for Disease Control and Prevention (CDC) National Center for Health Statistics*. Available at: <https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Dietary&Cycle=2017-2020>.

Cheah, I. K., & Halliwell, B. (2011). Ergothioneine; antioxidant potential, physiological function and role in disease. *Biochimica Et Biophysica Acta (BBA) - Molecular Basis of Disease*, 1822(5), 784–793. <https://doi.org/10.1016/j.bbadis.2011.09.017>

Cheah, I. K., & Halliwell, B. (2021). Ergothioneine, recent developments. *Redox Biology*, 42, 101868. <https://doi.org/10.1016/j.redox.2021.101868>

Cheah, I. K., Tang, R. M., Yew, T. S., Lim, K. H., & Halliwell, B. (2016). Administration of pure ergothioneine to healthy human subjects: uptake, metabolism, and effects on biomarkers of oxidative damage and inflammation. *Antioxidants and Redox Signaling*, 26(5), 193–206. <https://doi.org/10.1089/ars.2016.6778>

Czubat, B., Minias, A., Brzostek, A., Żaczek, A., Struś, K., Zakrzewska-Czerwińska, J., & Dziadek, J. (2020). Functional Disassociation Between the Protein Domains of MSMEG_4305 of *Mycobacterium smegmatis* (*Mycobacterium smegmatis*) in vivo. *Frontiers in Microbiology*, 11. <https://doi.org/10.3389/fmicb.2020.02008>

EFSA Panel on Food Contact Materials, Enzymes and Processing Aids (CEP), Lambré, C., Baviera, J. M. B., Bolognesi, C., Cocconcelli, P. S., Crebelli, R., Gott, D. M., Grob, K., Lampi, E., Mengelers, M., Mortensen, A., Riviére, G., Steffensen, I., Tlustos, C., Van Loveren, H., Vernis, L., Zorn, H., Glandorf, B., Herman, L., Andryszkiewicz, M., . . . Chesson, A. (2021). Safety evaluation of the food enzyme d-psicose 3-epimerase from the genetically modified *Escherichia coli* strain K-12 W3110 (pWKLP). *EFSA Journal*, 19(4). <https://doi.org/10.2903/j.efsa.2021.6565>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K. I., Kearney, J., Maciuk, A., Mangelsdorf, I., McArdle, H. J., Naska, A., Pelaez, C., Pentieva, K., Siani, A., Thies, F., Tsabouri, S., Vinceti, M., Cubadda, F., Engel, K. H., Frenzel, T., Heinonen, M., . . . Knutsen, H. K. (2019). Safety of 2'-fucosyllactose/difucosyllactose mixture as a novel food pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, 17(6). <https://doi.org/10.2903/j.efsa.2019.5717>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bohn, T., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K. I., Maciuk, A., Mangelsdorf, I., McArdle, H. J., Naska, A., Pentieva, K., Siani, A., Thies, F., Tsabouri, S., Vinceti, M., Aguilera-Gómez, M., Cubadda, F., Frenzel, T., Heinonen, M., Maradona, M. P., . . . Knutsen, H. K. (2023a). Safety of lacto-N-fucopentaose I/2'-fucosyllactose (LNFP-I/2'-FL) mixture as a novel food pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, 21(12). <https://doi.org/10.2903/j.efsa.2023.8412>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bohn, T., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K. I., Maciuk, A., Mangelsdorf, I., McArdle, H. J., Naska, A., Pelaez, C., Pentieva, K., Siani, A., Thies, F., Tsabouri, S., Vinceti, M., Aguilera-Gómez, M., Cubadda, F., Frenzel, T., Heinonen, M., . . . Knutsen, H. K. (2023b). Safety of 3-fucosyllactose (3-FL) produced by a derivative strain of *Escherichia coli* K-12 DH1 as a novel food pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, 21(6). <https://doi.org/10.2903/j.efsa.2023.8026>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K. I., Kearney, J., Maciuk, A., Mangelsdorf, I., McArdle, H. J., Naska, A., Pelaez, C., Pentieva, K., Siani, A., Thies, F., Tsabouri, S., Vinceti, M., Cubadda, F., Frenzel, T., Heinonen, M., Marchelli, R., . . . Knutsen, H. K. (2021). Safety of 3-FL (3-Fucosyllactose) as a novel food pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, 19(6). <https://doi.org/10.2903/j.efsa.2021.6662>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bresson, J., Burlingame, B., Dean, T., Fairweather-Tait, S., Heinonen, M., Hirsch-Ernst, K. I., Mangelsdorf, I., McArdle, H. J., Naska, A., Neuhaus-Berthold, M., Nowicka, G., Pentieva, K., Sanz, Y., Siani, A., Sjödin, A., Stern, Anchor Center for Certification (ACC)

M., Tomé, D., Vinceti, M., . . . Van Loveren, H. (2017). Statement on the safety of synthetic l-ergothioneine as a novel food – supplementary dietary exposure and safety assessment for infants and young children, pregnant and breastfeeding women. *EFSA Journal*, 15(11). <https://doi.org/10.2903/j.efsa.2017.5060>

EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bresson, J., Burlingame, B., Dean, T., Fairweather-Tait, S., Heinonen, M., Hirsch-Ernst, K. I., Mangelsdorf, I., McArdle, H. J., Naska, A., Neuhäuser-Berthold, M., Nowicka, G., Pentieva, K., Sanz, Y., Siani, A., Sjödin, A., Stern, M., Tomé, D., Vinceti, M., . . . Van Loveren, H. (2016). Safety of synthetic l-ergothioneine (Ergoneine®) as a novel food pursuant to Regulation (EC) No 258/97. *EFSA Journal*, 14(11). <https://doi.org/10.2903/j.efsa.2016.4629>

Epand, R. (1982). The role of dietary ergothioneine in the development of diabetes mellitus. *Medical Hypotheses*, 9(2), 207–213. [https://doi.org/10.1016/0306-9877\(82\)90137-2](https://doi.org/10.1016/0306-9877(82)90137-2)

Epand, R. M., Epand, R. F., & Wong, S. C. (1988). Study of the Ergothioneine Concentration in the Blood of Individuals with Diabetes Mellitus. *Clinical Chemistry and Laboratory Medicine (CCLM)*, 26(10). <https://doi.org/10.1515/cclm.1988.26.10.623>

Ey, J., Schömig, E., & Taubert, D. (2007). Dietary sources and antioxidant effects of ergothioneine. *Journal of Agricultural and Food Chemistry*, 55(16), 6466–6474. <https://doi.org/10.1021/jf071328f>

FDA (2016a) GRN No. 624 D-allulose 3-epimerase from *Arthrobacter globiformis* M30 produced in *Escherichia coli*. Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=624>

FDA (2016b) GRN No. 659 Lacto-N-neotetraose. Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=659>

FDA (2018) GRN No. 745 Steviol glycosides consisting primarily of rebaudioside M. Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=745>

FDA (2018a) GRN No. 735 2'-Fucosyllactose. Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=735>

FDA (2018b) GRN No. 734 Ergothionine. Available at: <https://www.cfsanappsexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=734>.

FDA (2018c) GRN No. 749 2'-O-fucosyllactose. Available at: <https://www.cfsanappsexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=749>.

FDA (2020) GRN No. 897 2-O-fucosyllactose. Available at: <https://www.cfsanappsexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=897>.

FDA (2021) GRN No. 951 3-fucosyllactose. Available at: <https://www.cfsanappsexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=951>.

FDA (2022) GRN No. 1010 Rebaudioside M obtained by enzymatic treatment of steviol glycosides purified from the leaves of *Stevia rebaudiana* (Bertoni). Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=1010>

FDA (2023a) GRN No. 1075 6'-sialyllactose sodium salt. Available at: <https://www.cfsanappsexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=1075>.

FDA (2023b) GRN No. 1106 Rebaudioside I obtained by enzymatic treatment of steviol glycosides purified from the leaves of *Stevia rebaudiana* (Bertoni) Bertoni (rebaudioside I). Available at: <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=GRASNotices&id=1106>

Flamm, E. L. (1991). How FDA approved Chymosin: A case history. *Nature Biotechnology*, 9(4), 349–351. <https://doi.org/10.1038/nbt0491-349>

Forster, R., Spézia, F., Papineau, D., Sabadie, C., Erdelmeier, I., Moutet, M., & Yadan, J. (2015). Reproductive safety evaluation of L-Ergothioneine. *Food and Chemical Toxicology*, 80, 85–91. <https://doi.org/10.1016/j.fct.2015.02.019>

Gorbach, S. L. (1978). Recombinant DNA: an Infectious Disease perspective. *The Journal of Infectious Diseases*, 137(5), 615–623. <https://doi.org/10.1093/infdis/137.5.615>

Guijarro, M., Indart, A., Aruoma, O., Viana, M., & Bonet, B. (2002). Effects of ergothioneine on diabetic embryopathy in pregnant rats. *Food and Chemical Toxicology*, 40(12), 1751–1755. [https://doi.org/10.1016/s0278-6915\(02\)00177-1](https://doi.org/10.1016/s0278-6915(02)00177-1)

Halliwell, B., Cheah, I. K., & Tang, R. M. Y. (2018). Ergothioneine – a diet-derived antioxidant with therapeutic potential. *FEBS Letters*, 592(20), 3357–3366. <https://doi.org/10.1002/1873-3468.13123>

Havlik, D., Brandt, U., Bohle, K., & Fleißner, A. (2017). Establishment of *Neurospora crassa* as a host for heterologous protein production using a human antibody fragment as a model product. *Microbial Cell Factories*, 16(1). <https://doi.org/10.1186/s12934-017-0734-5>

Hayashi, K., Morooka, N., Yamamoto, Y., Fujita, K., Isono, K., Choi, S., Ohtsubo, E., Baba, T., Wanner, B. L., Mori, H., & Horiuchi, T. (2006). Highly accurate genome sequences of *Escherichia coli* K-12 strains MG1655 and W3110. *Molecular Systems Biology*, 2(1). <https://doi.org/10.1038/msb4100049>

Health Canada. (2024). *Chemical substance - Ergothioneine*. Retrieved August 23, 2024, from <http://webprod.hc-sc.gc.ca/nhp/nd-bdipsn/ingredReq.do?id=13222&lang=eng>

Huff, C. D., Witherspoon, D. J., Zhang, Y., Gatenbee, C., Denson, L. A., Kugathasan, S., Hakonarson, H., Whiting, A., Davis, C. T., Wu, W., Xing, J., Watkins, W. S., Bamshad, M. J., Bradfield, J. P., Bulayeva, K., Simonson, T. S., Jorde, L. B., & Guthery, S. L. (2011). Crohn's Disease and Genetic Hitchhiking at IBD5. *Molecular Biology and Evolution*, 29(1), 101–111. <https://doi.org/10.1093/molbev/msr151>

- Jensen, K. F. (1993). The *Escherichia coli* K-12 “wild types” W3110 and MG1655 have an rph frameshift mutation that leads to pyrimidine starvation due to low pyrE expression levels. *Journal of Bacteriology*, 175(11), 3401–3407. <https://doi.org/10.1128/jb.175.11.3401-3407.1993>
- Kan, A., Gelfat, I., Emani, S., Praveschotinunt, P., & Joshi, N. S. (2020). Plasmid Vectors for *in Vivo* Selection-Free Use with the Probiotic *E. coli* Nissle 1917. *ACS Synthetic Biology*, 10(1), 94–106. <https://doi.org/10.1021/acssynbio.0c00466>
- Kaper, J. B., Nataro, J. P., & Mobley, H. L. T. (2004). Pathogenic *Escherichia coli*. *Nature Reviews Microbiology*, 2(2), 123–140. <https://doi.org/10.1038/nrmicro818>
- Kato, Y., Kubo, Y., Iwata, D., Kato, S., Sudo, T., Sugiura, T., Kagaya, T., Wakayama, T., Hirayama, A., Sugimoto, M., Sugihara, K., Kaneko, S., Soga, T., Asano, M., Tomita, M., Matsui, T., Wada, M., & Tsuji, A. (2010). Gene knockout and metabolome Analysis of Carnitine/Organic Cation Transporter OCTN1. *Pharmaceutical Research*, 27(5), 832–840. <https://doi.org/10.1007/s11095-010-0076-z>
- Katsube, M., Watanabe, H., Suzuki, K., Ishimoto, T., Tatebayashi, Y., Kato, Y., & Murayama, N. (2022). Food-derived antioxidant ergothioneine improves sleep difficulties in humans. *Journal of Functional Foods*, 95, 105165. <https://doi.org/10.1016/j.jff.2022.105165>
- Krawiec, S. (2021, October 18). NNB announced independent conclusion of GRAS status for its ergothioneine ingredient. *Nutritional Outlook*. <https://www.nutritionaloutlook.com/view/nnb-announced-independent-conclusion-of-gras-status-for-its-ergothioneine-ingredient>
- Kushairi, N., Phan, C. W., Sabaratnam, V., Naidu, M., & David, P. (2020). Dietary amino acid ergothioneine protects HT22 hippocampal neurons against H2O2-induced neurotoxicity via antioxidative mechanism. *PharmaNutrition*, 13, 100214. <https://doi.org/10.1016/j.phanu.2020.100214>
- Leung, E., Hong, J., Fraser, A. G., Merriman, T. R., Vishnu, P., & Krissansen, G. W. (2006). Polymorphisms in the organic cation transporter genes SLC22A4 and SLC22A5 and Crohn’s disease in a New Zealand Caucasian cohort. *Immunology and Cell Biology*, 84(2), 233–236. <https://doi.org/10.1111/j.1440-1711.2006.01423.x>
- Liu, H., Tang, W., Wang, X., Jiang, J., Zhang, W., & Wang, W. (2023). Safe and effective antioxidant: The biological mechanism and potential pathways of ergothioneine in the skin. *Molecules*, 28(4), 1648. <https://doi.org/10.3390/molecules28041648>
- Lu, F. C. (1988). Acceptable daily intake: Inception, evolution, and application. *Regulatory Toxicology and Pharmacology*, 8(1), 45–60. [https://doi.org/10.1016/0273-2300\(88\)90006-2](https://doi.org/10.1016/0273-2300(88)90006-2)
- Marone, P. A., Trampota, J., & Weisman, S. (2016). A safety evaluation of a Nature-Identical L-Ergothioneine in Sprague Dawley rats. *International Journal of Toxicology*, 35(5), 568–583. <https://doi.org/10.1177/1091581816653375>

- Mühldorfer, I., & Hacker, J. (1994). Genetic aspects of *Escherichia coli* virulence. *Microbial Pathogenesis*, 16(3), 171–181. <https://doi.org/10.1006/mpat.1994.1018>
- National Institutes of Health. (2024). *NIH GUIDELINES FOR RESEARCH INVOLVING RECOMBINANT OR SYNTHETIC NUCLEIC ACID MOLECULES (NIH GUIDELINES)*. Office of Science Policy. Retrieved March 19, 2025, from https://osp.od.nih.gov/wp-content/uploads/NIH_Guidelines.pdf
- Olempska-Beer, Z. S., Merker, R. I., Ditto, M. D., & DiNovi, M. J. (2006). Food-processing enzymes from recombinant microorganisms—a review. *Regulatory Toxicology and Pharmacology*, 45(2), 144–158. <https://doi.org/10.1016/j.yrtph.2006.05.001>
- Pariza, M. W., & Johnson, E. A. (2001). Evaluating the safety of microbial enzyme preparations used in food processing: Update for a new century. *Regulatory Toxicology and Pharmacology*, 33(2), 173–186. <https://doi.org/10.1006/rtp.2001.1466>
- Peltekova, V. D., Wintle, R. F., Rubin, L. A., Amos, C. I., Huang, Q., Gu, X., Newman, B., Van Oene, M., Cescon, D., Greenberg, G., Griffiths, A. M., St George-Hyslop, P. H., & Siminovitch, K. A. (2004). Functional variants of OCTN cation transporter genes are associated with Crohn disease. *Nature Genetics*, 36(5), 471–475. <https://doi.org/10.1038/ng1339>
- Perkins, D. D., & Davis, R. H. (2000). Evidence for safety of *Neurospora* species for academic and commercial uses. *Applied and Environmental Microbiology*, 66(12), 5107–5109. <https://doi.org/10.1128/aem.66.12.5107-5109.2000>
- Ramirez-Martinez, A., Wesolek, N., Yadan, J., Moutet, M., & Roudot, A. (2015). Intake assessment of L-ergothioneine in some European countries and in the United States. *Human and Ecological Risk Assessment an International Journal*, 22(3), 667–677. <https://doi.org/10.1080/10807039.2015.1104241>
- Reglinski, J., Smith, W., Wilson, R., Buchanan, L., McKillop, J., Thomson, J., Brzeski, M., Marabani, M., & Sturrock, R. (1991). Clinical analysis in intact erythrocytes using 1H spin echo NMR. *Clinica Chimica Acta*, 201(1–2), 45–57. [https://doi.org/10.1016/0009-8981\(91\)90023-6](https://doi.org/10.1016/0009-8981(91)90023-6)
- Renwick, A. G. (1990). Acceptable daily intake and the regulation of intense sweeteners. *Food Additives & Contaminants*, 7(4), 463–475. <https://doi.org/10.1080/02652039009373909>
- Renwick, A. G. (1993). Data-derived safety factors for the evaluation of food additives and environmental contaminants. *Food Addit Contam*, 10(3), 275–305. <https://doi.org/10.1080/02652039309374152>
- Rulis, A. M., & Levitt, J. A. (2008). FDA'S food ingredient approval process. *Regulatory Toxicology and Pharmacology*, 53(1), 20–31. <https://doi.org/10.1016/j.yrtph.2008.10.003>
- Salt, H. B. (1931). The ergothioneine content of the blood in health and disease. *Biochemical Journal*, 25(5), 1712–1719. <https://doi.org/10.1042/bj0251712>

- Santos, F. C., Lobo, G. M., Fernandes, A. S., Videira, A., & De Almeida, R. F. M. (2018). Changes in the Biophysical Properties of the Cell Membrane Are Involved in the Response of *Neurospora crassa* to Staurosporine. *Frontiers in Physiology*, 9. <https://doi.org/10.3389/fphys.2018.01375>
- Schauss, A. G., Béres, E., Vértési, A., Frank, Z., Pasics, I., Endres, J., Aruoma, O. I., & Hirka, G. (2011). The effect of ergothioneine on clastogenic potential and mutagenic activity. *International Journal of Toxicology*, 30(4), 405–409. <https://doi.org/10.1177/1091581811405856>
- Schauss, A. G., Vértési, A., Endres, J. R., Hirka, G., Clewell, A., Qureshi, I., & Pasics, I. (2010). Evaluation of the safety of the dietary antioxidant ergothioneine using the bacterial reverse mutation assay. *Toxicology*, 278(1), 39–45. <https://doi.org/10.1016/j.tox.2010.07.015>
- Soma, Y., Takahashi, M., Fujiwara, Y., Shinohara, T., Izumi, Y., Hanai, T., & Bamba, T. (2021). Design of Synthetic Quorum Sensing Achieving Induction Timing-Independent Signal Stabilization for Dynamic Metabolic Engineering of *E. coli*. *ACS Synthetic Biology*, 10(6), 1384–1393. <https://doi.org/10.1021/acssynbio.1c00008>
- Sparks, I. L., Derbyshire, K. M., Jacobs, W. R., & Morita, Y. S. (2023). *Mycobacterium smegmatis*: The Vanguard of Mycobacterial Research. *Journal of Bacteriology*, 205(1). <https://doi.org/10.1128/jb.00337-22>
- Tang, R. M. Y., Cheah, I. K., Yew, T. S. K., & Halliwell, B. (2018). Distribution and accumulation of dietary ergothioneine and its metabolites in mouse tissues. *Scientific Reports*, 8(1). <https://doi.org/10.1038/s41598-018-20021-z>
- Taubert, D., Grimberg, G., Jung, N., Rubbert, A. & Schömig, E. (2005). Functional role of the 503F variant of the organic cation transporter OCTN1 in Crohn's disease. *Gut*, 54(10), 1505–1506. <https://doi.org/10.1136/gut.2005.076083>
- Taubert, D., Jung, N., Goeser, T., & Schomig, E. (2009). Increased ergothioneine tissue concentrations in carriers of the Crohn's disease risk-associated 503F variant of the organic cation transporter OCTN1. *Gut*, 58(2), 312–314. <https://doi.org/10.1136/gut.2008.164418>
- Tetrahedron. (2015). *Tetrahedron's Ergoneine self-affirmed GRAS*. NewHope. Retrieved August 23, 2024, from <http://www.newhope.com/specialty/tetrahedrons-ergoneine-self-affirmed-gras>
- Tokuhiro, S., Yamada, R., Chang, X., Suzuki, A., Kochi, Y., Sawada, T., Suzuki, M., Nagasaki, M., Ohtsuki, M., Ono, M., Furukawa, H., Nagashima, M., Yoshino, S., Mabuchi, A., Sekine, A., Saito, S., Takahashi, A., Tsunoda, T., Nakamura, Y., & Yamamoto, K. (2003). An intronic SNP in a RUNX1 binding site of SLC22A4, encoding an organic cation transporter, is associated with rheumatoid arthritis. *Nature Genetics*, 35(4), 341–348. <https://doi.org/10.1038/ng1267>
- U.S. Environmental Protection Agency. (2018, August 23). *Reducing PFAS in drinking water with treatment technologies*. <https://www.epa.gov/sciencematters/reducing-pfas-drinking-water-treatment-technologies>

U.S. EPA. (1997). *ATTACHMENT I--FINAL RISK ASSESSMENT OF ESCHERICHIA COLI K-12 DERIVATIVES*. Retrieved March 19, 2025, from <https://www.epa.gov/sites/default/files/2015-09/documents/fra004.pdf>

Warner, D. F., Etienne, G., Wang, X., Matsoso, L. G., Dawes, S. S., Soetaert, K., Stoker, N. G., Content, J., & Mizrahi, V. (2005). A derivative of *Mycobacterium smegmatis* mc2155 that lacks the duplicated chromosomal region. *Tuberculosis*, 86(6), 438–444. <https://doi.org/10.1016/j.tube.2005.10.001>

Watson, E. (2012). *Biotech firm launches bid to turn L-Ergothioneine into household name as 'EGT'*. NutraIngredients-USA. Retrieved August 23, 2024, from <https://www.nutraingredients-usa.com/Industry/Biotech-firm-launches-bid-to-turn-L-Ergothioneine-into-household-name-as-EGT>

Weigand-Heller, A. J., Kris-Etherton, P. M., & Beelman, R. B. (2011). The bioavailability of ergothioneine from mushrooms (*Agaricus bisporus*) and the acute effects on antioxidant capacity and biomarkers of inflammation. *Preventive Medicine*, 54, S75–S78. <https://doi.org/10.1016/j.ypmed.2011.12.028>

Wu, X., George, R. L., Huang, W., Wang, H., Conway, S. J., Leibach, F. H., & Ganapathy, V. (2000). Structural and functional characteristics and tissue distribution pattern of rat OCTN1, an organic cation transporter, cloned from placenta. *Biochimica Et Biophysica Acta (BBA) - Biomembranes*, 1466(1–2), 315–327. [https://doi.org/10.1016/s0005-2736\(00\)00189-9](https://doi.org/10.1016/s0005-2736(00)00189-9)

Yau, Y. F., Cheah, I. K., Mahendran, R., Tang, R. M., Chua, R. Y., Goh, R. E. S., Feng, L., Li, J., Kua, E. H., Chen, C., & Halliwell, B. (2024). Investigating the efficacy of ergothioneine to delay cognitive decline in mild cognitively impaired subjects: A pilot study. *Journal of Alzheimer S Disease*. <https://doi.org/10.1177/13872877241291253>

Zalachoras, I., Hollis, F., Ramos-Fernández, E., Trovo, L., Sonnay, S., Geiser, E., Preitner, N., Steiner, P., Sandi, C., & Morató, L. (2020). Therapeutic potential of glutathione-enhancers in stress-related psychopathologies. *Neuroscience & Biobehavioral Reviews*, 114, 134–155. <https://doi.org/10.1016/j.neubiorev.2020.03.015>

Appendix 1 – Genomic Sequencing Report

Gene III Biotechnology Co., Ltd

East side of 7th Floor, Building 4, No.6 Xuzhuang Road,
Xuanwu District, Nanjing, Jiangsu, China
Phone: 400-809-2959

GeneIII Ⅲ

Report on bacterial resequencing

Project number: T40266

Project Name: Bacterial Resequencing

Conclusion date: August 2, 2024



Gene III Biotechnology Co., Ltd

East side of 7th Floor, Building #, No.6 Xuzhuang Road,
Xiazhenwu District, Nanjing, Jiangsu, China
Phone: 400-800-2959

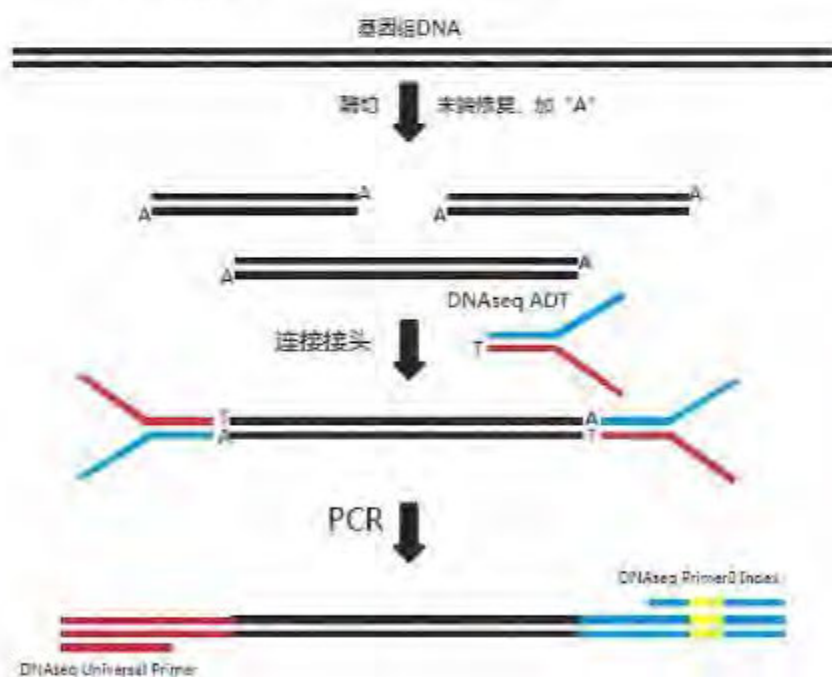
GeneIII

Introduction

Gene III Biotechnology Co., Ltd. contracted Synbio Technologies on July 1, 2024 to perform an analysis of their genetically modified *E. coli* K12 MG1655. Analyses included sequencing and genomic analysis for virulence factors, drug resistance genes, secondary metabolite genes and potential to produce allergenic proteins. A summary of the results is presented below.

1. Database construction and sequencing

Library construction: After the sample detection is qualified, use the Enzymatic Universal DNAseq Library Prep Kit (KAITAI-BIO, AT4107) to build the library. Add a break repair reagent to break the DNA into short fragments and perform end repair with "A". After purifying the recovered product, connect the sequencing adapter (DNA Universal ADT). PCR amplify the DNA, and use AMPure XP beads to screen and recover the target product from 300bp to 600bp for sequencing. The schematic diagram for construction is as follows:



Library quality inspection: After the library construction is completed, Qubit is used for preliminary quantification, and the library is diluted to 1ng/ul. Then, Qseq100 DNA Analyzer is used to detect the insert size of the library. After the insert size meets expectations, Q-PCR method is used to

Gene III Biotechnology Co., Ltd

East side of 7th Floor, Building 3 (No. 6 Xuzhuang Road),
Xuanwu District, Nanjing, Jiangsu, China
Phone: 400-630-2859



accurately determine the effective concentration of the library (effective concentration of the library $>2\text{nM}$) to ensure the quality of the library.

On machine sequencing: After passing the library inspection, different libraries are pooled according to the requirements of effective concentration and target off machine data volume, and then sequenced using the NovaSeq 6000 platform.

2 Bioinformatics analysis process

Information analysis involves the following steps:

- (1) Raw offline data processing: This step filters reads with low sequencing quality values, retains high-quality reads, and the filtered data is called Clean Data;
- (2) Reference genome alignment: Align Clean reads to the reference genome and compare the results for statistical analysis;
- (3) Genomic comparative analysis: This step compares the differences between the sample and the reference genome at the genomic level

3 Project and sample information

Project type: Bacterial resequencing

Project number: T40266

Species information: *Escherichia coli* K-12 substr. MG1655

Sample Name: G3_egg (SeqName: S24B0719)

Reference genome link for *E. coli* K-12 substr. MG1655: https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000005845.2/

4 Results and Analysis

4.1 Data Quality Control

4.1.1 Raw sequencing data

The raw image data obtained from second-generation high-throughput sequencing instruments (such as Illumina HiSeq TM 2500) is converted into sequence data through base calling, which we

Gene III Biotechnology Co., Ltd

East side of 7th Floor (Building #), No. 6 Xuzhuang Road,
Xueshi District, Nanjing, Jiangsu, China
Phone: 400-600-2939



refer to as raw data or raw reads. The results are stored in FASTQ file format, including the sequence of reads and the sequencing quality of bases.

The information statistics of the raw data from the sequencing machine are as follows:

Original offline data statistics table

Sample	Raw length(bp)	Insert size(bp)	Raw reads(num)	Raw bases(bp)	Q20(%)	Q30(%)	GC(%)
G5_cgt	150	269	7,464,506	1,119,675,900	97.70	97.59	49.65

Explanation: Sample represents the name of the sample; Read length represents the length of the original data; Insert size represents the length of the inserted segment; Raw reads (bases) represent the amount of raw data; Q20 and Q30 represent the percentage of base numbers in valid data with sequencing error rates less than 0.01 (mass value greater than 20) and 0.001 (mass value greater than 30); GC (%) represents the GC content of the sample.

4.1.2 Clean data

The raw data obtained from sequencing is likely to contain splices or reads with lower base quality, which will affect the subsequent analysis results. These reads need to be removed before information analysis. The higher the quality of the base, the higher the sequencing accuracy. The main condition for the quality control sequencing data is Q20. If Q20>85%, it means that the sequencing accuracy of more than 85% of the bases is 99%. Connectors are removed from offline data and low-quality data are filtered. The filtering steps are as follows:

- (1) Remove the reads contaminated with the connector (the alkaline base of the connector contaminated in the reads is greater than 5bp. For dual end sequencing, if one end is contaminated with the connector, remove the reads at both ends);
- (2) Remove low-quality reads (bases with a quality value Q<15 in reads account for more than 30% of the length of reads. For dual end sequencing, if one end is a low-quality read, both end reads will be removed);
- (3) Remove Reads containing more than 5% N (for dual end sequencing, if one end contains more than 5% N, both end Reads will be removed).

The statistical information of the filtered data is shown in the table below:

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Cleandata Data statistics table

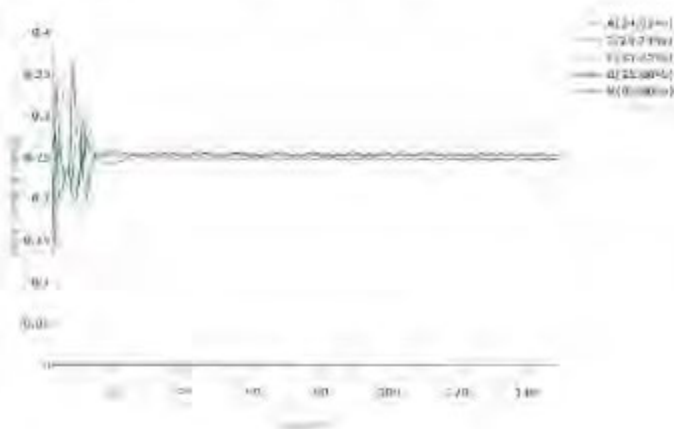
Sample	Clean	Clean	Low quality	Duplication	Data use	Q20(%)	Q30(%)
Name	reads(num)	bases(bp)	reads(num)	reads rate (%)	rate(%)		
G3_egt	7,309,528	1,087,515,696	53,744	23.47	97.92	97.63	92.99

Explanation: Sample represents the name of the sample; Clean reads (bases) represents the amount of effective data filtered out; Low quality reads represent low-quality data; Duplication reads rate represents the proportion of duplicate data; Data use rate represents the ratio of effective data volume; Q20 and Q30 represent the percentage of base numbers in valid data with sequencing error rates less than 0.01 (mass value greater than 20) and 0.001 (mass value greater than 30);

Clean data Base distribution map:

The distribution of GC content is used to detect the presence of AT and GC separation phenomena, which may be caused by sequencing or library construction and can to some extent affect the accuracy of analysis. In theory, the content of G and C bases, as well as A and T bases, should be equal for each sequencing cycle, and the entire sequencing process should remain stable and horizontal. Due to random primer amplification bias and other reasons, there is often a significant fluctuation in the first 6-7 bases of each read obtained during sequencing, which is a normal situation. In the genome project, when the construction of the database is relatively uniform, the boundaries of the four colors representing different bases should fluctuate very little, almost in a straight line.

The GC distribution of Clean Reads for each sample is as follows:



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Explanation: Base content distribution map: The horizontal axis represents the position of the reads where the base is located, and the vertical axis represents the content of each base at a certain position.

4.2 Reference genome alignment

Use BWA software to align Clean reads to the reference genome, SAMTOOLS software to sort and label PCR repeat sequences on the files, and finally use BamDeal software to statistically analyze the alignment results. The alignment rate and coverage of sequencing data are the foundation of project analysis. The sample alignment rate reflects the similarity between the sequencing data and the reference genome for *E. coli* K-12 substr. MG 1655, and the coverage can directly reflect the homology between the sequencing data and the reference sequence. The comparison results of each sample are shown in the table:

Comparison rate, sequencing depth, and coverage depth statistics of samples

Sample	Mapping Rate	Properly paired	Depth(X)	Coverage(%)
G3_cgl	99.56	92.85	172.37	99.24

Explanation: Sample represents the name of the sample; Mapping Rate represents the ratio of reads aligned to the reference genome to the total clean reads; Properly paid represents the ratio of the number of paired reads aligned to the reference genome to the total clean reads; Depth (X) represents the sequencing depth, which is the ratio of the base number on the alignment to the effective genome size (excluding N); Coverage (%) represents the proportion of sequences obtained through sequencing to the entire genome.

4.3 Genomic analysis for virulence factors, drug resistance genes, secondary metabolite genes and potential to produce allergenic proteins

4.3.1 Virulence factor (VFDB) annotation of bacterial pathogens

The database analyzed by Virulence factors is VFDB (version 2021), and the VFDB database link is www.mgc.ac.cn/VFs.

The full name of the VFDB database is Virulence Factors of Pathogenic Bacteria, which is a database dedicated to studying pathogenic factors of bacteria, chlamydia, and mycoplasma. It contains 26 genera, a total of 459 pathogenic factors, 24 pathogenic islands, and 2505 genes related to virulence factors. Using BLAST software, compare the amino acid sequences of the target species with the VFDB database, combine the gene and corresponding virulence factor functional annotation information of the target species, and obtain annotation results.

Toxicity factors include bacterial toxins, cell surface proteins that regulate bacterial attachment, carbohydrates, proteins, and hydrolytic enzymes that protect bacterial cell surfaces. In some species,

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East side of 7th Floor, Building 4, No. 6 Xuezhong Road,
Xuanwu District, Nanjing, Jiangsu, China
Phone: 400-800-2539



different combinations of independent factors can cause different diseases. Further comparative genomic analysis can reveal more information about the evolution of organisms and pathogens. Analysis was conducted on July 3, 2024, and three genes were annotated in G3_egt ($e_value < 0.0001$, identity 100%) as shown below. The identified genes ($yagZ/ccpA^1$, $fepB^2$ and $fimE^3$) are intrinsic to *E. coli* K12 subst. MG 1655.

Parameter Name	Matched Genes		
Gene_id	NP_414827.1	NP_415124.1	NP_418733.1
Subject_id	VFG002414	VFG000924	VFG000872
Locus_id	NP_286010	NP_752610	NP_757240
Description	(yagZ/ccpA) E. coli common pilus structural subunit EcpA [ECP (VF0404)] [Escherichia coli O157:H7 str. EDL933]	(fepB) ferrienterobactin ABC transporter periplasmic binding protein [Enterobactin (VF0228)] [Escherichia coli CFT073]	(fimE) Type 1 fimbriae Regulatory protein fimE [Type 1 fimbriae (VF0221)] [Escherichia coli CFT073]
identity	100	100	100
alignment_length	195	318	198
mismatches	0	0	0
gap_openings	0	0	0
q.start	1	1	1
q.end	195	318	198
s.start	1	1	1
s.end	195	318	198
e_value	3.09E-130	4.45E-223	9.32E-147
score	361	606	403

4.3.2 Drug resistance gene (CARD) annotation

The database for antimicrobial resistance gene analysis is CARD (version 2020), and the CARD database link is: <https://card.mcmaster.ca/download/>

CARD (Comprehensive Antimicrobial Resistance Database) searches for target proteins that include antibiotic related data and antibiotic resistance genes. CARD integrates different molecular and sequence data, providing a unique organizational principle in the form of ARO (Antimicrobial Resistance Ontology), and can quickly identify new unannotated genome sequences and infer antibiotic resistance genes.

¹ NCBI reference sequence: NP_414827.1, Available at https://www.ncbi.nlm.nih.gov/protein/NP_414827.1

² NCBI reference sequence: NP_415124.1, Available at https://www.ncbi.nlm.nih.gov/protein/NP_415124.1

³ NCBI reference sequence: NP_418733.1 Available at https://www.ncbi.nlm.nih.gov/protein/NP_418733.1

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Analysis was conducted on July 3, 2024, and a total of 42 genes were annotated in G3_egt (e_value<0.0001, identity 100%). Gene III Biotechnology Co., Ltd. has determined that these genes are intrinsic to *E. coli* K12.

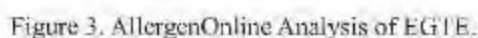
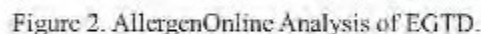
4.3.3 Allergen Online Annotations

Allergen Online, also known as FARRP Allergen Protein Database, it is a free peer-reviewed searchable database of allergen lists and sequences designed to help identify proteins that may have a potential risk of allergic cross reactivity. Database link: www.allergenonline.com, maintained by the Food Allergy Research and Resource Program (FARRP) and constantly updated to reflect the latest scientific research and discoveries. The latest version (V22 released as of May 25, 2023) contains 2290 taxonomic proteomes and a total of 952 peer-reviewed sequences.

Analysis was conducted on August 2, 2024, and no matches of greater than 35% identity were found for EGT1, EGTD, and EGTE over 80mer sliding windows (Figures 1-3).

No Matches of Greater than 35% Identity Found

Figure 1. AllergenOnline Analysis for EGT1.



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4.3.4 Analysis of secondary metabolism gene clusters

Secondary metabolites are substances synthesized by microorganisms during a certain growth period using primary metabolites as precursors. *E. coli* has been shown to produce secondary metabolites with antibacterial activity.⁴

The antiSMASH program (version 7.1.0) was used to query the genome for regions associated with secondary metabolite production, with a testing date of July 1, 2024.

The analysis results showed a 100% similarity with enterobactin in region 19.1, while no other highly similar secondary metabolite regions were found (see below). Enterobactin is an iron-binding protein known to be produced by *E. coli* K12 subst. MG 1655⁵ and is therefore considered intrinsic to the strain.



⁴ Altaee, N. et al. (2016). International Journal of Current Pharmaceutical Review and Research; 7(6), 393-399.

⁵ Washington-Hughes, C. (2019). Microbiology Open 8(1): e00691.

Available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6460284/>

Appendix 2 – Statement of Food Grade Material Status

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Xuanwu District, Nanjing, Jiangsu, China
Phone: 400-800-2959

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Statement

We hereby certify that all raw materials, processing aids and food contact substances used in the manufacture of L-Ergothioneine are used in accordance with existing U.S. regulations, are the subjects of effective food contact notifications, or were concluded to be generally recognized as safe for their respective uses.

Gene III Biotechnology Co., Ltd



Appendix 3 - Statement of cGMP Compliance

北大医药重庆大新药业股份有限公司
CHONGQING DAXIN PHARMACEUTICAL CO., LTD.

地址: 中国重庆北碚

电话 TEL: (86-23) 61301056

邮箱 Email: drug_registration@daxinpharm.cn

传真 FAX: (86-23) 61301713

ADD: No.21, Founder Road, Shuitu Town, BeiBei District, Chongqing of China, 400714

April 1, 2024

Statement

Name of product: L-Ergothioneine

Manufacturer name and address: No.21 Founder Road, Shuitu Town, Beibei District, Chongqing 400714, China.

We Chongqing Daxin Pharmaceutical Co., Ltd. hereby declare that steviol glycoside has evaluated and meet the relevant requirements of food safety system FSSC22000 and ISO22000 certification. L-ergothioneine is manufactured in the same location and under the same conditions as the steviol glycoside. Therefore, L-Ergothioneine is manufactured in compliance with current Good Manufacturing Practices(cGMP).

Sincerely



Mr. Zhang Xinxu
Deputy Quality Director
Chongqing Daxin Pharmaceutical Co., Ltd.



Ju Hongying
Quality Manager
Gene III Biotechnology Co., Ltd.

Appendix 4 – Certificate of Analysis (CoA) of L-Ergothioneine

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Xuanwu District, Nanjing, Jiangsu, China
Phone: +86-250-29359

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Certificate of Analysis

NO.: GS20230720

Product Name: L-Ergothioneine		Batch Number: EGT20230720
Production Date: 2023.07.20		Expiry date: 2025.07.19
Item	Standard	Result
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform
Optical rotation	$[\alpha]_D^{20} \geq (+)122^{\circ}$ (C=1, H ₂ O)	+127.0°
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%
Loss on Drying	$\leq 0.5\%$	0.04%
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	32.5ppm
Lead	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Arsenic	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Mercury	$\leq 0.1\text{mg/kg}$	$< 0.01\text{mg/kg}$
Cadmium	$\leq 0.1\text{mg/kg}$	$< 0.05\text{mg/kg}$
Total aerobic bacteria count	$\leq 1000 \text{ CFU/g}$	$< 10\text{CFU/g}$
Molds and Yeasts	$\leq 50 \text{ CFU/g}$	$< 10\text{CFU/g}$
<i>Escherichia coli</i>	Negative/g	Negative
<i>Salmonella</i>	Negative/25g	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative
Conclusion: up to standard		

Report by: [redacted] Reviewed by: [redacted] Approved by: [redacted]



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Certificate of Analysis

NO.: GS20240207

Product Name: L-Ergothioneine		Batch Number: EGT20240207
Production Date: 2024.02.27		Expiry date: 2026.02.26
Item	Standard	Result
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform
Optical rotation	$[\alpha]_D^{25} \geq (+)122^{\circ}$ (C=1, H ₂ O)	-126.0°
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%
Loss on Drying	$\leq 0.5\%$	0.05%
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	197ppm
Lead	$\leq 0.5\text{mg/kg}$	$< 0.05\text{ mg/kg}$
Arsenic	$< 0.5\text{mg/kg}$	$< 0.05\text{ mg/kg}$
Mercury	$\leq 0.1\text{mg/kg}$	$< 0.01\text{mg/kg}$
Cadmium	$\leq 0.1\text{mg/kg}$	$< 0.05\text{mg/kg}$
Total aerobic bacteria count	$\leq 1000\text{ CFU/g}$	$< 10\text{CFU/g}$
Molds and Yeasts	$\leq 50\text{ CFU/g}$	$< 10\text{CFU/g}$
<i>Escherichia coli</i>	Negative/g	Negative
<i>Salmonella</i>	Negative/25g	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative
Conclusion: up to standard		

Report by:

Reviewed by

Approved by:



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Certificate of Analysis

NO.: GS20250107

Product Name: L-Ergothioneine		Batch Number: EGT20250107
Production Date: 2025.01.11		Expiry date: 2027.01.10
Item	Standard	Result
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform
Optical rotation	$[\alpha]_D^{25} \geq (+)122^{\circ}$ (C=1, H ₂ O)	+127.0°
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%
Loss on Drying	$\leq 0.5\%$	0.04%
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	410 ppm
Lead	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Arsenic	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Mercury	$\leq 0.1\text{mg/kg}$	$< 0.01\text{mg/kg}$
Cadmium	$\leq 0.1\text{mg/kg}$	$< 0.05\text{mg/kg}$
Total aerobic bacteria count	$\leq 1000 \text{ CFU/g}$	$< 10\text{CFU/g}$
Molds and Yeasts	$\leq 50 \text{ CFU/g}$	$< 10\text{CFU/g}$
<i>Escherichia coli</i>	Negative/g	Negative
<i>Salmonella</i>	Negative/25g	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative
Conclusion: up to standard		

Report by:

Reviewed by:

Approved by:

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Xuanwu District, Nanjing, Jiangsu, China
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Certificate of Analysis

NO.: GS20250115

Product Name: L-Ergothioneine		Batch Number: EGT20250115
Production Date: 2025.01.23		Expiry date: 2027.01.22
Item	Standard	Result
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform
Optical rotation	$[\alpha]_D^{25} \geq (+)122^{\circ}$ (C=1, H ₂ O)	+128.0°
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%
Loss on Drying	$\leq 0.5\%$	0.14%
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	422 ppm
Lead	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Arsenic	$\leq 0.5\text{mg/kg}$	$< 0.05 \text{ mg/kg}$
Mercury	$\leq 0.1\text{mg/kg}$	$< 0.01\text{mg/kg}$
Cadmium	$\leq 0.1\text{mg/kg}$	$< 0.05\text{mg/kg}$
Total aerobic bacteria count	$\leq 1000 \text{ CFU/g}$	$< 10\text{CFU/g}$
Molds and Yeasts	$\leq 50 \text{ CFU/g}$	$< 10\text{CFU/g}$
<i>Escherichia coli</i>	Negative/g	Negative
<i>Salmonella</i>	Negative/25g	Negative
<i>Staphylococcus aureus</i>	Negative/25g	Negative
Conclusion: up to standard		

Report by:

Reviewed by:

Approved by:

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Certificate of Analysis

NO.: GS20250117

Product Name: L-Ergothioneine		Batch Number: EGT20250117	
Production Date: 2025.01.25		Expiry date: 2027.01.24	
Item	Standard	Result	
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities	
Identification	HPLC chromatogram matches the standard sample	Conform	
Optical rotation	$[\alpha]_D^{25} \geq (+)122^{\circ}$ (C=1, H ₂ O)	+128.0°	
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%	
Loss on Drying	$\leq 0.5\%$	0.14%	
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	440 ppm	
Lead	$\leq 0.5\text{mg/kg}$	$< 0.05\text{ mg/kg}$	
Arsenic	$\leq 0.5\text{mg/kg}$	$< 0.05\text{ mg/kg}$	
Mercury	$\leq 0.1\text{mg/kg}$	$< 0.01\text{mg/kg}$	
Cadmium	$\leq 0.1\text{mg/kg}$	$< 0.05\text{mg/kg}$	
Total aerobic bacteria count	$\leq 1000\text{ CFU/g}$	$< 10\text{CFU/g}$	
Molds and Yeasts	$\leq 50\text{ CFU/g}$	$< 10\text{CFU/g}$	
<i>Escherichia coli</i>	Negative/g	Negative	
<i>Salmonella</i>	Negative/25g	Negative	
<i>Staphylococcus aureus</i>	Negative/25g	Negative	
Conclusion: up to standard			
Report by:		Reviewed by:	
		Approved by:	

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质控部
检验专用章

Appendix 5 – Report on Estimated Daily Intake of L-Ergothioneine

Proposed Use and Food Codes Utilized for Intake Calculation

The dietary exposure distributions were calculated using the Creme Food Safety® model¹, a scientific cloud-based software service designed and developed to calculate dietary intakes of foods, chemicals, and nutrients in populations of consumers. This is achieved by linking food consumption data from the What We Eat In America (WWEIA) portion of the National Health and Nutrition Examination Survey (NHANES) to the appropriate food composition and chemical occurrence data using a number of validated and published models, available upon request from Crème Global (<https://www.cremeglobal.com/>). Calculations for this intake analysis were completed using deterministic (single points) input data. Output calculation types include daily average intakes, acute exposures, as well as population statistics such as mean, percentiles, standard errors, and confidence intervals. Results are output for "Consumers Only" (i.e., consumers of the food / substance of interest), and Total Population (consumers and non-consumers). Results of the exposure assessment are given in absolute terms (mg/day) as well as relative to the consumer's body weight (mg/kg bw/day). The per unit of bodyweight exposure is calculated on a subject level using the bodyweight recorded by the NHANES data. The National Center for Health Statistics, Centers for Disease Control and Prevention note that less than 30 individuals reporting may be statistically unreliable for estimating the mean intake and less than 80 individuals reporting may be statistically unreliable for estimating the 90th percentile intake.¹

The proposed uses of L-ergothioneine are provided below. Gene III Biotech's L-Ergothioneine is intended for use in cakes, cookies, pastries, granola bars, nutrition bars, coffee, tea, fruit drinks and "ades", soft drinks, sport and energy drinks, candy, nutritional beverages, and protein and nutritional powders. The intended use level of ergothioneine ranges from 30 - 150 mg/serving in each of the food categories. The intended use levels and categories incorporate all of those listed in Table 4 of [GRN 734](#) in section 3, page 15. The intended use of Gene III Biotech's L-Ergothioneine, along with inclusion levels, is summarized in Table 1. The food categories and decision criteria for food code selection to match the intended use in GRN 734 are provided in Table 2.

The food codes utilized to calculate the estimated daily intake were selected from the NHANES 2017-2020 pre-pandemic survey. Food codes were selected that most appropriately match the intended use. The final food code selections for intended use are found in Table 7.

¹ CDC (1996) *ANALYTIC AND REPORTING GUIDELINES: The Third National Health and Nutrition Examination Survey, NHANES III (1988-1994)*. Hyattsville, Maryland: National Center for Health Statistics, Centers for Disease Control and Prevention (CDC). Available at: <https://wwwn.cdc.gov/Nchs/data/nhanes/analyticguidelines/88-94-analytic-reporting-guidelines.pdf>

Table 1. Intended Food Categories for Gene III Biotech’s L-Ergothioneine

Food Categories	NHANES Food Code Description	Use Level (mg/serving) ¹	Use Level (mg/100 g)
Cakes, Cookies, Pastries	Included all relevant codes to match GRN 734 (described in Table 2)	30	34.6-85.7
Nutrition Bars, Cereal Bars	included all "cereal bars" which includes breakfast bars; included all "nutrition bars"	30	75
Coffee	Included all relevant codes to match GRN 734 (described in Table 2)	60	16.7
Tea	Included all relevant codes to match GRN 734 (described in Table 2)"	60	16.7
Fruit drinks and ades	Included all relevant codes to match GRN 734 (described in Table 2)	30	12.5
Carbonated soft drinks	Include all (+diet) soft drinks; carbonated fruit drink and punches made with soda; carbonated waters	60	16.7
Sport and Energy Drinks	Included all (+diet) sport and energy drinks	30	8.3
Candy	Included all of "candy containing chocolate" and "candy not containing chocolate"	30	100
Gummy Candy Surrogate	Selection gelatin food codes to serve as surrogate for gummy candys	150	120
Nutritional Beverages	All "nutritional beverages"	60	25
Protein and Nutritional Powders	All "protein and nutritional powders"	60	150

¹ Serving size determined from Reference Amount Customarily Consumed (RACC) - 21 CFR § 101.12.

Assumption that 1 g = 1 ml for RACC.

g - Grams; mg – Milligrams; mL- Milliliters

Table 2. WWEIA/NHANES Food Categories Chosen to Match GRN 734

Food Categories	GRN 734 Description	NHANES Food Code Description
Cakes, Cookies, Pastries (including granola bars)	Includes yeast-type sweet rolls, yeast- and crumb- or quick-bread-type coffee cakes, croissants, cakes, cookies, pies, cobblers, turnovers, danish pastries, doughnuts, breakfast bars and tarts, granola bars, and sweet crackers.	Included all of "cakes and pies" except raw cake batter, strudels, crisps, baklava and basbousa, empanada and coconut cream cake "Puerto Rican style", dessert pizza, ice cream cake, snack cakes, cheesecakes, tiramisu; Included all of "Doughnuts, sweet rolls, pastries"; included all cookies; included all "cereal bars" which includes breakfast bars; included two codes from "nutrition bars" for cereal/granola bars;
Coffee	Includes decaffeinated and regular coffee made from ground or instant coffee, coffee mixes, and coffee substitutes.	Included all other codes from "Coffee"; dry powders from "not included in a food category"; included frozen and iced coffee
Tea	Includes decaffeinated and regular tea obtained ready to drink or made from leaves or from instant tea mixes with or without lemon, sugar, or artificial sweetener; and herb and other teas.	Included all "tea"; included two dry powder codes from "not included in a food category"

Food Categories	GRN 734 Description	NHANES Food Code Description
Fruit drinks and ades	Includes regular and low-calorie fruit drinks, punches, and ades, including those made from powdered mix and frozen concentrate. Excludes fruit juices, which are tabulated under Fruits, and carbonated fruit drinks, which are tabulated under "carbonated soft drinks."	Included two codes of flavored water; included "light" fruit juice - 40-50% juice; included dry powders (4 codes) from "not included in a food category"; included fruit drinks from "other diet drinks";
Carbonated soft drinks	Includes regular and low-calorie carbonated soft drinks, such as colas, fruit-flavored and cream sodas, ginger ale, root beer, and carbonated soft drinks containing fruit juice; carbonated fruit juice drinks; and sweetened and unsweetened carbonated water.	Include all (+diet) soft drinks; carbonated fruit drink and punches made with soda; carbonated waters
Candy containing chocolate	Includes plain milk chocolate, chocolate candy bars, and all chocolate-flavored or chocolate-covered candy reported separately. One ounce of candy weighs 28 g.	Included all of "candy containing chocolate", codes from "candy not containing chocolate" that reference chocolate flavor

¹ Serving size determined from Reference Amount Customarily Consumed (RACC) - 21 CFR § 101.12.

Assumption that 1 g = 1 ml for RACC.

g - Grams; mg – Milligrams; mL- Milliliters

Estimated Daily Intake of Gene III Biotech's L-Ergothioneine from Intended Uses

The estimated daily intake (EDI) provided in Table 3 (absolute intake) and Table 4 (intake per kg of bodyweight) are calculated based on the proposed use of Gene III Biotech's L-Ergothioneine use in conventional foods for ages 2+ years.

Table 3. Estimated Daily Intake of Gene III Biotech's L-Ergothioneine (mg/day)

Population Group	Number of users	% Pop.	Ingredient Intake Per Capita (mg/day)		Ingredient Intake – All Consumers (mg/day)	
			Mean ±SE	90th Percentile ±SE	Mean ±SE	90th Percentile ±SE
2-5 y	729	91%	46.2 ± 1.6	94.5 ± 3.6	49.8 ± 1.5	95.5 ± 3.8
6-11 y	1156	96%	72.0 ± 1.6	140.5 ± 3.8	75.0 ± 1.4	141.5 ± 3.5
12-18 F	626	96%	79.5 ± 2.5	148.4 ± 4.9	83.1 ± 2.2	149.4 ± 5.2
12-18 M	637	96%	99.6 ± 3.8	207.2 ± 8.8	105.3 ± 3.5	207.8 ± 8.5
19-110 F	3459	97%	145.4 ± 1.8	269.4 ± 4.3	147.8 ± 1.8	272.8 ± 4.1
19-110 M	3161	97%	178.7 ± 2.3	337.2 ± 7.1	182.8 ± 2.3	338.1 ± 8.5
2-110 M+F	9768	97%	141.5 ± 1.3	281.0 ± 3.2	145.4 ± 1.2	285.0 ± 3.3

* The % Pop. (population) represents the number of individuals in the specific demographic category that reported eating at least one of the intended use foods on at least one day of the survey.

Abbreviations: kg – kilograms; mg – milligrams; SE – standard error; y – year

Table 4. Estimated Daily Intake of Gene III Biotech’s L-Ergothioneine (mg/kg bw/day)

Population Group	Number of users	% Pop.	Ingredient Intake Per Capita (mg/kg bw/day)		Ingredient Intake – All Consumers (mg/kg bw/day)	
			Mean $\pm$ SE	90th Percentile $\pm$ SE	Mean $\pm$ SE	90th Percentile $\pm$ SE
2-5 y	729	91%	2.66 $\pm$ 0.09	5.53 $\pm$ 0.33	2.87 $\pm$ 0.08	5.77 $\pm$ 0.35
6-11 y	1156	96%	2.17 $\pm$ 0.05	4.50 $\pm$ 0.13	2.26 $\pm$ 0.05	4.54 $\pm$ 0.12
12-18 F	626	96%	1.36 $\pm$ 0.04	2.65 $\pm$ 0.12	1.43 $\pm$ 0.05	2.72 $\pm$ 0.13
12-18 M	637	96%	1.58 $\pm$ 0.06	3.10 $\pm$ 0.10	1.67 $\pm$ 0.06	3.14 $\pm$ 0.11
19-110 F	3459	97%	1.96 $\pm$ 0.03	3.64 $\pm$ 0.06	2.00 $\pm$ 0.03	3.66 $\pm$ 0.08
19-110 M	3161	97%	2.04 $\pm$ 0.03	3.87 $\pm$ 0.08	2.09 $\pm$ 0.03	3.90 $\pm$ 0.10
2-110 M+F	9768	97%	2.00 $\pm$ 0.02	3.87 $\pm$ 0.06	2.05 $\pm$ 0.02	3.92 $\pm$ 0.06

*The % Pop. (population) represents the number of individuals in the specific demographic category that reported eating at least one of the intended use foods on at least one day of the survey.

Abbreviations: bw – bodyweight; kg – kilograms; mg – milligrams; SE – standard error; y – year

Estimated Daily Intake of L-Ergothioneine from Cumulative Uses

The EDIs provided below in mg/kg bw/day (Table 5) are calculated by adding the estimated intake amount of L-Ergothioneine from background diet as estimated by Ramirez-Martinez et al. (2016), and the estimated intake amount of L-Ergothioneine from Gene III Biotech’s intended use. The EDIs provided below in mg/day (Table 6) are calculated using the mean body weight multiplied by the cumulative intake in mg/kg bw/day (Table 5). The mean body weight is calculated by dividing the mean intake in mg/day (Table 3) by the mean intake in mg/kg bw/day (Table 4) for consumers only. The ages included in this analysis are ages 2+ years.

Table 5. Estimated Cumulative Daily Intake of Ingredient (mg/kg bw/day)

Population Group	Number of users	% Pop.	Ingredient Intake Per Capita (mg/kg bw/day)		Ingredient Intake – All Consumers (mg/kg bw/day)	
			Mean	90th Percentile	Mean	90th Percentile
2-5 y	729	91%	3.0	5.8	3.2	6.1
6-11 y	1156	96%	2.5	4.8	2.6	4.8
12-18 F	626	96%	1.7	3.0	1.7	3.0
12-18 M	637	96%	1.9	3.4	2.0	3.4
19-110 F	3459	97%	2.1	3.8	2.1	3.8
19-110 M	3161	97%	2.2	4.0	2.2	4.1
2-110 M+F	9768	97%	2.1	4.0	2.2	4.1

*The % Pop. (population) represents the number of individuals in the specific demographic category that reported eating at least one of the intended use foods on at least one day of the survey.

Abbreviations: bw – bodyweight; kg – kilograms; mg – milligrams; y – year

Table 6. Estimated Cumulative Daily Intake of L-Ergothioneine (mg/day)

Population Group	Number of users	% Pop.*	Mean Body Weight**	Ingredient Intake Per Capita (mg/day)		Ingredient Intake – All Consumers (mg/day)	
				Mean***	90th Percentile***	Mean***	90th Percentile***
2-5 y	729	91%	17.3	52.1	100.7	55.5	105.8
6-11 y	1156	96%	32.9	82.9	159.3	86.3	159.3
12-18 F	626	96%	57.8	99.4	175.4	98.8	174.3
12-18 M	637	96%	62.4	119.8	214.3	126.1	214.4
19-110 F	3459	97%	73.8	155.8	281.9	155.2	280.8
19-110 M	3161	97%	87.3	192.7	350.4	192.4	358.6
2-110 M+F	9768	97%	69.4	148.6	283.0	156.0	290.8

* The % Pop. (population) represents the number of individuals in the specific demographic category that reported eating at least one of the intended use foods on at least one day of the survey.

**Mean body weight calculated by dividing the mean intake in mg/day (Table 3) by the mean intake in mg/kg bw/day (Table 4) for consumers only.

***Calculated using the mean body weight multiplied by the cumulative intake in mg/kg bw/day (Table 5).

Abbreviations: mg – milligrams; y – year

Table 7. 2017-2020 Pre-pandemic NHANES Food Codes Used for Intended Use Analysis

Food Code	Main Food Description
53100100	Cake or cupcake, NFS
53101100	Cake, angel food
53101200	Cake, angel food, with icing or filling
53101250	Cake, angel food, with fruit and icing or filling
53102100	Cake or cupcake, applesauce, without icing or filling
53102200	Cake or cupcake, apple
53102600	Cake or cupcake, banana, without icing or filling
53102700	Cake or cupcake, banana
53102800	Cake or cupcake, Black Forest
53103000	Cake, Boston cream pie
53104100	Cake or cupcake, carrot, without icing or filling
53104260	Cake or cupcake, carrot
53104300	Cake, carrot, diet
53104400	Cake or cupcake, coconut
53105270	Cake or cupcake, chocolate with chocolate icing, bakery
53105275	Cake or cupcake, chocolate, no icing
53105300	Cake or cupcake, German chocolate
53105500	Cake, chocolate, with icing, diet
53106500	Cake, cream
53109300	Cake, Dobos Torte

53110000	Cake, fruit cake
53111000	Cake or cupcake, gingerbread
53113000	Cake, jelly roll
53114000	Cake or cupcake, lemon, without icing or filling
53114100	Cake or cupcake, lemon
53115100	Cake or cupcake, marble, without icing or filling
53115200	Cake or cupcake, marble
53115310	Cake or cupcake, nut, without icing or filling
53115320	Cake or cupcake, nut, with icing or filling
53115410	Cake or cupcake, oatmeal
53115450	Cake or cupcake, peanut butter
53116000	Cake, pound
53116020	Cake, pound, with icing or filling
53116270	Cake, pound, chocolate
53116390	Cake, pound, reduced fat, cholesterol free
53116500	Cake or cupcake, pumpkin, without icing or filling
53116510	Cake or cupcake, pumpkin
53116550	Cake or cupcake, raisin-nut
53116570	Cake, Ravani
53116600	Cake, rice flour, without icing or filling
53116650	Cake, Quezadilla, El Salvadorian style
53117100	Cake or cupcake, spice, without icing or filling
53117200	Cake or cupcake, spice
53118100	Cake, sponge
53118200	Cake, sponge, with icing or filling
53118300	Cake, sponge, chocolate
53118410	Rum cake, without icing
53118500	Cake, torte
53118550	Cake, tres leche
53119000	Cake, pineapple, upside down
53120270	Cake or cupcake, white with white icing, bakery
53120275	Cake or cupcake, white, without icing or filling
53121270	Cake or cupcake, white with chocolate icing, bakery
53121275	Cake or cupcake, white, no icing
53122070	Cake, shortcake, biscuit type, with whipped cream and fruit
53122080	Cake, shortcake, biscuit type, with fruit
53123070	Cake, strawberry shortcake
53123080	Cake, shortcake, sponge type, with fruit
53123500	Cake, shortcake, with whipped topping and fruit, diet
53124110	Cake or cupcake, zucchini
53300100	Pie, NFS
53300170	Pie, individual size or tart, NFS
53300180	Pie, fried, NFS

53301000	Pie, apple
53301070	Pie, apple, fast food
53301080	Pie, apple, fried pie
53301500	Pie, apple, one crust
53302000	Pie, apricot, two crust
53302070	Pie, apricot, individual size or tart
53302080	Pie, apricot, fried pie
53303000	Pie, blackberry, two crust
53303070	Pie, blackberry, individual size or tart
53303500	Pie, berry
53303510	Pie, berry, not blackberry, blueberry, boysenberry, huckleberry, raspberry, or strawberry; one crust
53303570	Pie, berry, not blackberry, blueberry, boysenberry, huckleberry, raspberry, or strawberry, individual size or tart
53304000	Pie, blueberry
53304070	Pie, blueberry, individual size or tart
53305000	Pie, cherry
53305010	Pie, cherry, one crust
53305070	Pie, cherry, individual size or tart
53305080	Pie, cherry, fried pie
53305700	Pie, lemon
53305720	Pie, lemon, not cream or meringue, individual size or tart
53305750	Pie, lemon, fried pie
53306000	Pie, mince, two crust
53307000	Pie, peach
53307050	Pie, peach, one crust
53307070	Pie, peach, individual size or tart
53307080	Pie, peach, fried pie
53307500	Pie, pear, two crust
53307570	Pie, pear, individual size or tart
53308000	Pie, pineapple, two crust
53308070	Pie, pineapple, individual size or tart
53309000	Pie, raisin, two crust
53309070	Pie, raisin, individual size or tart
53310000	Pie, raspberry, one crust
53310050	Pie, raspberry, two crust
53311000	Pie, rhubarb, two crust
53312000	Pie, strawberry
53313000	Pie, strawberry-rhubarb, two crust
53314000	Pie, strawberry, individual size or tart
53340000	Pie, apple-sour cream
53340500	Pie, cherry, made with cream cheese and sour cream
53341000	Pie, banana cream

53341070	Pie, banana cream, individual size or tart
53341500	Pie, buttermilk
53341750	Pie, chess
53342000	Pie, chocolate cream
53342070	Pie, chocolate cream, individual size or tart
53343000	Pie, coconut cream
53343070	Pie, coconut cream, individual size or tart
53344000	Pie, custard
53344070	Pie, custard, individual size or tart
53344200	Mixed fruit tart filled with custard or cream cheese
53345000	Pie, lemon cream
53345070	Pie, lemon cream, individual size or tart
53346000	Pie, peanut butter cream
53346500	Pie, pineapple cream
53347000	Pie, pumpkin
53347070	Pie, pumpkin, individual size or tart
53347500	Pie, sour cream, raisin
53347600	Pie, squash
53348000	Pie, strawberry cream
53348070	Pie, strawberry cream, individual size or tart
53360000	Pie, sweet potato
53365000	Pie, vanilla cream
53370000	Pie, chiffon, not chocolate
53371000	Pie, chiffon, chocolate
53373000	Pie, black bottom
53381000	Pie, lemon meringue
53381070	Pie, lemon meringue, individual size or tart
53382000	Pie, chocolate-marshmallow
53385000	Pie, pecan
53385070	Pie, pecan, individual size or tart
53385500	Pie, oatmeal
53386000	Pie, pudding, flavors other than chocolate
53387000	Pie, Toll house chocolate chip
53390000	Pie, shoo-fly
53390100	Pie, tofu with fruit
53391000	Pie shell
53391100	Pie shell, graham cracker
53391150	Pie shell, chocolate
53391200	Vanilla wafer dessert base
53410100	Cobbler, apple
53410200	Cobbler, apricot
53410300	Cobbler, berry
53410500	Cobbler, cherry

53410800	Cobbler, peach
53410850	Cobbler, pear
53410880	Cobbler, plum
53410900	Cobbler, rhubarb
53450000	Turnover, fruit
53450300	Turnover or dumpling, berry
53450500	Turnover or dumpling, cherry
53450800	Turnover or dumpling, lemon
53451000	Turnover or dumpling, peach
53451500	Turnover, guava
53451750	Turnover, pumpkin
53105262	Cake or cupcake, chocolate with white icing, bakery
53105264	Cake or cupcake, chocolate with white icing, from mix
53105272	Cake or cupcake, chocolate with chocolate icing, from mix
53105310	Cake or cupcake, gluten free
53105396	Cake, chocolate, flourless
53116520	Cake or cupcake, red velvet
53118110	Cake or cupcake, strawberry
53120272	Cake or cupcake, white with white icing, from mix
53121272	Cake or cupcake, white with chocolate icing, from mix
53300200	Tart, all types
53305800	Pie, key lime
53710400	Cereal or granola bar (General Mills Fiber One Chewy Bar)
53710500	Cereal or granola bar (Kellogg's Nutri-Grain Cereal Bar)
53710502	Cereal or granola bar (Kellogg's Nutri-Grain Yogurt Bar)
53710504	Cereal or granola bar (Kellogg's Nutri-Grain Fruit and Nut Bar)
53710600	Milk 'n Cereal bar
53710700	Cereal or granola bar (Kellogg's Special K bar)
53710810	Cereal or granola bar (KIND Fruit and Nut Bar)
53710900	Cereal or granola bar (General Mills Nature Valley Chewy Trail Mix)
53710902	Cereal or granola bar, with yogurt coating (General Mills Nature Valley Chewy Granola Bar)
53710904	Cereal or granola bar (General Mills Nature Valley Sweet and Salty Granola Bar)
53710906	Cereal or granola bar (General Mills Nature Valley Crunchy Granola Bar)
53711000	Cereal or granola bar (Quaker Chewy Granola Bar)
53711002	Cereal or granola bar (Quaker Chewy 90 Calorie Granola Bar)
53711004	Cereal or granola bar (Quaker Chewy 25% Less Sugar Granola Bar)
53711006	Cereal or granola bar (Quaker Chewy Dipps Granola Bar)
53711100	Cereal or granola bar (Quaker Granola Bites)
53712000	Snack bar, oatmeal
53712100	Cereal or Granola bar, NFS
53712200	Cereal or granola bar, lowfat, NFS
53712210	Cereal or granola bar, nonfat
53713000	Cereal or granola bar, reduced sugar, NFS

53713010	Cereal or granola bar, fruit and nut
53713100	Cereal or granola bar, peanuts , oats, sugar, wheat germ
53714200	Cereal or granola bar, chocolate coated, NFS
53714210	Cereal or granola bar, with coconut, chocolate coated
53714220	Cereal or granola bar with nuts, chocolate coated
53714230	Cereal or granola bar, oats, nuts, coated with non-chocolate coating
53714250	Cereal or granola bar, coated with non-chocolate coating
53714300	Cereal or granola bar, high fiber, coated with non-chocolate yogurt coating
53714400	Cereal or granola bar, with rice cereal
53714500	Breakfast bar, NFS
53714510	Breakfast bar, date, with yogurt coating
53714520	Breakfast bar, cereal crust with fruit filling, lowfat
51187020	Anisette toast
53201000	Cookie, NFS
53202000	Cookie, almond
53203000	Cookie, applesauce
53203500	Cookie, biscotti
53205260	Cookie, bar, with chocolate
53206000	Cookie, chocolate chip
53206020	Cookie, chocolate chip, made from home recipe or purchased at a bakery
53206030	Cookie, chocolate chip, reduced fat
53206100	Cookie, chocolate chip sandwich
53206500	Cookie, chocolate, made with rice cereal
53206550	Cookie, chocolate, made with oatmeal and coconut, no bake
53207000	Cookie, chocolate or fudge
53207020	Cookie, chocolate or fudge, reduced fat
53207050	Cookie, chocolate, with chocolate filling or coating, fat free
53208000	Cookie, marshmallow, chocolate-covered
53208200	Cookie, marshmallow pie, chocolate covered
53209005	Cookie, chocolate, with icing or coating
53209010	Cookie, sugar wafer, chocolate-covered
53209015	Cookie, chocolate sandwich
53209020	Cookie, chocolate sandwich, reduced fat
53209100	Cookie, chocolate, sandwich, with extra filling
53209500	Cookie, chocolate and vanilla sandwich
53210000	Cookie, chocolate wafer
53210900	Cookie, graham cracker with chocolate and marshmallow
53211000	Cookie bar, with chocolate, nuts, and graham crackers
53215500	Cookie, coconut
53220000	Cookie, fruit-filled bar
53220010	Cookie, fruit-filled bar, fat free
53220030	Cookie, fig bar
53220040	Cookie, fig bar, fat free

53222010	Cookie, fortune
53222020	Cookie, cone shell, ice cream type, wafer or cake
53223000	Cookie, gingersnaps
53223100	Cookie, granola
53224000	Cookie, ladyfinger
53224250	Cookie, lemon bar
53225000	Cookie, macaroon
53226000	Cookie, marshmallow, with coconut
53226500	Cookie, marshmallow, with rice cereal, no bake
53226550	Cookie, marshmallow, with rice cereal and chocolate chips
53226600	Cookie, marshmallow and peanut butter, with oat cereal, no bake
53228000	Cookie, meringue
53230000	Cookie, molasses
53231000	Cookie, Lebkuchen
53231400	Cookie, multigrain, high fiber
53233000	Cookie, oatmeal
53233010	Cookie, oatmeal, with raisins
53233040	Cookie, oatmeal, reduced fat, NS as to raisins
53233050	Cookie, oatmeal sandwich, with creme filling
53233060	Cookie, oatmeal, with chocolate chips
53233080	Cookie, oatmeal sandwich, with peanut butter and jelly filling
53233100	Cookie, oatmeal, with chocolate and peanut butter, no bake
53234000	Cookie, peanut butter
53234100	Cookie, peanut butter, with chocolate
53234250	Cookie, peanut butter with rice cereal, no bake
53235000	Cookie, peanut butter sandwich
53235500	Cookie, with peanut butter filling, chocolate-coated
53235600	Cookie, Pfeffernusse
53236000	Cookie, Pizzelle
53236100	Cookie, pumpkin
53237000	Cookie, raisin
53237010	Cookie, raisin sandwich, cream-filled
53237500	Cookie, rum ball, no bake
53238000	Cookie, sandwich-type, not chocolate or vanilla
53239000	Cookie, shortbread
53239010	Cookie, shortbread, reduced fat
53239050	Cookie, shortbread, with icing or filling
53239100	Pocky
53240000	Cookie, animal
53240010	Cookie, animal, with frosting or icing
53241500	Cookie, butter or sugar
53241510	Marie biscuit
53241600	Cookie, butter or sugar, with fruit and/or nuts

53242000	Cookie, sugar wafer
53242500	Cookie, toffee bar
53243000	Cookie, vanilla sandwich
53243010	Cookie, vanilla sandwich, extra filling
53243050	Cookie, vanilla sandwich, reduced fat
53244010	Cookie, butter or sugar, with chocolate icing or filling
53244020	Cookie, butter or sugar, with icing or filling other than chocolate
53246000	Cookie, tea, Japanese
53247000	Cookie, vanilla wafer
53247050	Cookie, vanilla wafer, reduced fat
53247500	Cookie, vanilla with caramel, coconut, and chocolate coating
53251100	Cookie, rugelach
53260030	Cookie, chocolate chip, sugar free
53260200	Cookie, oatmeal, sugar free
53260300	Cookie, sandwich, sugar free
53260400	Cookie, sugar or plain, sugar free
53260500	Cookie, sugar wafer, sugar free
53260600	Cookie, peanut butter, sugar free
53261000	Cookie, gluten free
54102010	Graham crackers
54102015	Graham crackers (Teddy Grahams)
54102020	Graham crackers, chocolate covered
54102060	Crackers, Cuban
54102100	Graham crackers, reduced fat
54102200	Graham crackers, sandwich, with filling
51160000	Roll, sweet, no frosting
51160100	Roll, sweet, cinnamon bun, no frosting
51160110	Roll, sweet, cinnamon bun, frosted
51161000	Pan Dulce, with fruit, no frosting
51161020	Roll, sweet, with fruit, frosted
51161030	Roll, sweet, with fruit, frosted, diet
51161050	Roll, sweet, frosted
51161250	Pan Dulce, no topping
51161270	Pan Dulce, with sugar topping
51161280	Pan Dulce, with raisins and icing
51165000	Coffee cake, yeast type
51166000	Croissant
51166100	Croissant, cheese
51166200	Croissant, chocolate
51166500	Croissant, fruit
51167000	Brioche
51168000	Bread, Spanish coffee
51188100	Pannetone

53415120	Fritter, fruit
53415200	Fritter, banana
53415220	Fritter, berry
53420000	Cream puff, eclair, custard or cream filled, NS as to icing
53420100	Cream puff, eclair, custard or cream filled, not iced
53420200	Cream puff, eclair, custard or cream filled, iced
53420210	Cream puff, eclair, custard or cream filled, iced, reduced fat
53420250	Cream puff, no filling or icing
53420310	Wheat flour fritter, without syrup
53420400	Sopaipilla, without syrup or honey
53420410	Sopaipilla with syrup or honey
53430700	Tamale, sweet
53430750	Tamale, sweet, with fruit
53452100	Pastry, fruit-filled
53452120	Pastry, made with bean or lotus seed paste filling, baked
53452130	Pastry, made with bean paste and salted egg yolk filling, baked
53452150	Pastry, Chinese, made with rice flour
53452170	Pastry, cookie type, fried
53452200	Pastry, Italian, with cheese
53452400	Pastry, puff
53452420	Pastry, puff, custard or cream filled, iced or not iced
53452450	Cheese pastry puffs
53452500	Pastry, mainly flour and water, fried
53500100	Breakfast pastry, NFS
53510000	Danish pastry, plain or spice
53510100	Danish pastry, with fruit
53511000	Danish pastry, with cheese
53520000	Doughnut, NFS
53520100	Doughnut, cake type, plain
53520120	Doughnut, chocolate
53520130	Doughnut, cake type, powdered sugar
53520135	Doughnut, cake type, with icing
53520140	Doughnut, cake type, chocolate icing
53520160	Doughnut, chocolate, with chocolate icing
53520170	Doughnut holes
53520200	Churros
53520510	Beignet
53521110	Doughnut, yeast type
53521130	Doughnut, yeast type, with chocolate icing
53521140	Doughnut, jelly
53521210	Doughnut, custard-filled
53521230	Doughnut, custard-filled, with icing
53530000	Breakfast tart

53530010	Breakfast tart, lowfat
53610100	Coffee cake, crumb or quick-bread type
53610170	Coffee cake, crumb or quick-bread type, with fruit
53610200	Coffee cake, crumb or quick-bread type, cheese-filled
55801000	Funnel cake with sugar
55801010	Funnel cake with sugar and fruit
58124210	Pastry, cheese-filled
51160200	Pan dulce, NFS
53415110	Fritter, plain
53720100	Nutrition bar (Balance Original Bar)
53720200	Nutrition bar (Clif Bar)
53720210	Nutrition bar (Clif Kids Organic Zbar)
53720300	Nutrition bar (PowerBar)
53720400	Nutrition bar (Slim Fast Original Meal Bar)
53720500	Nutrition bar (Snickers Marathon Protein Bar)
53720600	Nutrition bar (South Beach Living Meal Bar)
53720610	Nutrition bar (South Beach Living High Protein Bar)
53720700	Nutrition bar (Tiger's Milk)
53720800	Nutrition bar (Zone Perfect Classic Crunch)
53729000	Nutrition bar or meal replacement bar, NFS
53710800	Cereal or granola bar (Kashi Chewy)
53710802	Cereal or granola bar (Kashi Crunchy)
92100000	Coffee, NS as to type
92100500	Coffee, NS as to brewed or instant
92101000	Coffee, brewed
92101500	Coffee, brewed, blend of regular and decaffeinated
92101600	Coffee, Turkish
92101610	Coffee, espresso
92101630	Coffee, espresso, decaffeinated
92101700	Coffee, brewed, flavored
92101800	Coffee, Cuban
92101810	Coffee, macchiato
92101820	Coffee, macchiato, sweetened
92101850	Coffee, cafe con leche
92101851	Coffee, cafe con leche, decaffeinated
92101900	Coffee, Latte
92101901	Coffee, Latte, nonfat
92101903	Coffee, Latte, with non-dairy milk
92101904	Coffee, Latte, flavored
92101905	Coffee, Latte, nonfat, flavored
92101906	Coffee, Latte, with non-dairy milk, flavored
92101910	Coffee, Latte, decaffeinated
92101911	Coffee, Latte, decaffeinated, nonfat

92101913	Coffee, Latte, decaffeinated, with non-dairy milk
92101917	Coffee, Latte, decaffeinated, flavored
92101918	Coffee, Latte, decaffeinated, nonfat, flavored
92101919	Coffee, Latte, decaffeinated, with non-dairy milk, flavored
92101920	Frozen coffee drink
92101921	Frozen coffee drink, nonfat
92101923	Frozen coffee drink, with non-dairy milk
92101925	Frozen coffee drink, with whipped cream
92101926	Frozen coffee drink, nonfat, with whipped cream
92101928	Frozen coffee drink, with non-dairy milk and whipped cream
92101930	Frozen coffee drink, decaffeinated
92101931	Frozen coffee drink, decaffeinated, nonfat
92101933	Frozen coffee drink, decaffeinated, with non-dairy milk
92101935	Frozen coffee drink, decaffeinated, with whipped cream
92101936	Frozen coffee drink, decaffeinated, nonfat, with whipped cream
92101938	Frozen coffee drink, decaffeinated, with non-dairy milk and whipped cream
92101950	Coffee, Cafe Mocha
92101955	Coffee, Cafe Mocha, nonfat
92101960	Coffee, Cafe Mocha, with non-dairy milk
92101965	Coffee, Cafe Mocha, decaffeinated
92101970	Coffee, Cafe Mocha, decaffeinated, nonfat
92101975	Coffee, Cafe Mocha, decaffeinated, with non-dairy milk
92102000	Frozen mocha coffee drink
92102010	Frozen mocha coffee drink, nonfat
92102020	Frozen mocha coffee drink, with non-dairy milk
92102030	Frozen mocha coffee drink, with whipped cream
92102040	Frozen mocha coffee drink, nonfat, with whipped cream
92102050	Frozen mocha coffee drink, with non-dairy milk and whipped cream
92102060	Frozen mocha coffee drink, decaffeinated
92102070	Frozen mocha coffee drink, decaffeinated, nonfat
92102080	Frozen mocha coffee drink, decaffeinated, with non-dairy milk
92102090	Frozen mocha coffee drink, decaffeinated, with whipped cream
92102100	Frozen mocha coffee drink, decaffeinated, nonfat, with whipped cream
92102110	Frozen mocha coffee drink, decaffeinated, with non-dairy milk and whipped cream
92102400	Iced Coffee, brewed
92102401	Iced Coffee, brewed, decaffeinated
92102450	Iced Coffee, pre-lightened and pre-sweetened
92102500	Coffee, Iced Latte
92102501	Coffee, Iced Latte, nonfat
92102502	Coffee, Iced Latte, with non-dairy milk
92102503	Coffee, Iced Latte, flavored
92102504	Coffee, Iced Latte, nonfat, flavored
92102505	Coffee, Iced Latte, with non-dairy milk, flavored

92102510	Coffee, Iced Latte, decaffeinated
92102511	Coffee, Iced Latte, decaffeinated, nonfat
92102512	Coffee, Iced Latte, decaffeinated, with non-dairy milk
92102513	Coffee, Iced Latte, decaffeinated, flavored
92102514	Coffee, Iced Latte, decaffeinated, nonfat, flavored
92102515	Coffee, Iced Latte, decaffeinated, with non-dairy milk, flavored
92102600	Coffee, Iced Cafe Mocha
92102601	Coffee, Iced Cafe Mocha, nonfat
92102602	Coffee, Iced Cafe Mocha, with non-dairy milk
92102610	Coffee, Iced Cafe Mocha, decaffeinated
92102611	Coffee, Iced Cafe Mocha, decaffeinated, nonfat
92102612	Coffee, Iced Cafe Mocha, decaffeinated, with non-dairy milk
92103000	Coffee, instant, reconstituted
92104000	Coffee, instant, 50% less caffeine, reconstituted
92111000	Coffee, NS as to brewed or instant, decaffeinated
92111010	Coffee, brewed, decaffeinated
92114000	Coffee, instant, decaffeinated, reconstituted
92121000	Coffee, instant, pre-lightened and pre-sweetened with sugar, reconstituted
92121001	Coffee, instant, decaffeinated, pre-lightened and pre-sweetened with sugar, reconstituted
92121010	Coffee, instant, pre-sweetened with sugar, reconstituted
92121020	Coffee, mocha, instant, pre-lightened and pre-sweetened with sugar, reconstituted
92121030	Coffee, mocha, instant, pre-lightened and pre-sweetened with low calorie sweetener, reconstituted
92121040	Coffee, instant, pre-lightened and pre-sweetened with low calorie sweetener, reconstituted
92121041	Coffee, instant, decaffeinated, pre-lightened and pre-sweetened with low calorie sweetener, reconstituted
92121050	Coffee, mocha, instant, decaffeinated, pre-lightened and pre-sweetened with low calorie sweetener, reconstituted
92130000	Coffee, pre-lightened and pre-sweetened with sugar
92130001	Coffee, decaffeinated, pre-lightened and pre-sweetened with sugar
92130005	Coffee, pre-lightened and pre-sweetened with low calorie sweetener
92130006	Coffee, decaffeinated, pre-lightened and pre-sweetened with low calorie sweetener
92130010	Coffee, pre-lightened
92130011	Coffee, decaffeinated, pre-lightened
92130020	Coffee, pre-sweetened with sugar
92130021	Coffee, decaffeinated, pre-sweetened with sugar
92130030	Coffee, pre-sweetened with low calorie sweetener
92130031	Coffee, decaffeinated, pre-sweetened with low calorie sweetener
92152000	Coffee and chicory, brewed
92152010	Coffee and chicory, brewed, decaffeinated
92161000	Coffee, Cappuccino
92161001	Coffee, Cappuccino, nonfat
92161002	Coffee, Cappuccino, with non-dairy milk
92162000	Coffee, Cappuccino, decaffeinated

92162001	Coffee, Cappuccino, decaffeinated, nonfat
92162002	Coffee, Cappuccino, decaffeinated, with non-dairy milk
92171000	Coffee, bottled/canned
92171010	Coffee, bottled/canned, light
92201010	Coffee substitute
92202010	Chicory beverage
92203000	Cereal beverage
92203110	Cereal beverage with beet roots, from powdered instant
92191100	Coffee, instant, not reconstituted
92191105	Coffee, instant, 50% less caffeine, not reconstituted
92191200	Coffee, instant, decaffeinated, not reconstituted
92191400	Coffee, instant, pre-sweetened with sugar, not reconstituted
92192000	Coffee, mocha, instant, pre-lightened and pre-sweetened with sugar, not reconstituted
92192030	Coffee, mocha, instant, pre-lightened and pre-sweetened with low calorie sweetener, not reconstituted
92192040	Coffee, mocha, instant, decaffeinated, pre-lightened and pre-sweetened with low calorie sweetener, not reconstituted
92193000	Coffee, instant, pre-lightened and pre-sweetened with sugar, not reconstituted
92193005	Coffee, instant, decaffeinated, pre-lightened and pre-sweetened with sugar, not reconstituted
92193020	Coffee, instant, pre-lightened and pre-sweetened with low calorie sweetener, not reconstituted
92193025	Coffee, instant, decaffeinated, pre-lightened and pre-sweetened with low calorie sweetener, not reconstituted
92291300	Coffee substitute, dry powder
92302000	Tea, hot, leaf, black
92302500	Tea, hot, leaf, black, decaffeinated
92303010	Tea, hot, leaf, green
92303100	Tea, hot, leaf, green, decaffeinated
92304100	Tea, hot, leaf, oolong
92305010	Tea, iced, instant, black, unsweetened
92305040	Tea, iced, instant, black, pre-sweetened with sugar
92305050	Tea, iced, instant, black, decaffeinated, pre-sweetened with sugar
92305090	Tea, iced, instant, black, pre-sweetened with low calorie sweetener
92305110	Tea, iced, instant, black, decaffeinated, pre-sweetened with low calorie sweetener
92305180	Tea, iced, instant, black, decaffeinated, unsweetened
92305900	Tea, iced, instant, green, unsweetened
92305910	Tea, iced, instant, green, pre-sweetened with sugar
92305920	Tea, iced, instant, green, pre-sweetened with low calorie sweetener
92306000	Tea, hot, herbal
92306090	Tea, hot, hibiscus
92306700	Tea, hot, chamomile
92306800	Tea, hot, with milk
92306850	Tea, ginger

92306910	Tea, bubble
92306920	Tea, kombucha
92307500	Iced Tea / Lemonade juice drink
92307510	Iced Tea / Lemonade juice drink, light
92307520	Iced Tea / Lemonade juice drink, diet
92308000	Tea, iced, brewed, black, pre-sweetened with sugar
92308010	Tea, iced, brewed, black, pre-sweetened with low calorie sweetener
92308020	Tea, iced, brewed, black, unsweetened
92308030	Tea, iced, brewed, black, decaffeinated, pre-sweetened with sugar
92308040	Tea, iced, brewed, black, decaffeinated, pre-sweetened with low calorie sweetener
92308050	Tea, iced, brewed, black, decaffeinated, unsweetened
92308500	Tea, iced, brewed, green, pre-sweetened with sugar
92308510	Tea, iced, brewed, green, pre-sweetened with low calorie sweetener
92308520	Tea, iced, brewed, green, unsweetened
92308530	Tea, iced, brewed, green, decaffeinated, pre-sweetened with sugar
92308540	Tea, iced, brewed, green, decaffeinated, pre-sweetened with low calorie sweetener
92308550	Tea, iced, brewed, green, decaffeinated, unsweetened
92309000	Tea, iced, bottled, black
92309010	Tea, iced, bottled, black, decaffeinated
92309020	Tea, iced, bottled, black, diet
92309030	Tea, iced, bottled, black, decaffeinated, diet
92309040	Tea, iced, bottled, black, unsweetened
92309050	Tea, iced, bottled, black, decaffeinated, unsweetened
92309500	Tea, iced, bottled, green
92309510	Tea, iced, bottled, green, diet
92309520	Tea, iced, bottled, green, unsweetened
92307000	Tea, iced, instant, black, unsweetened, dry
92307400	Tea, iced, instant, black, pre-sweetened, dry
94100200	Water, non-carbonated, flavored
94100300	Water beverage, fruit flavored
42403010	Coconut water, unsweetened
42404010	Coconut water, sweetened
92510610	Fruit juice drink
92510650	Tamarind drink
92510955	Lemonade, fruit juice drink
92510960	Lemonade, fruit flavored drink
92511000	Lemonade, frozen concentrate, not reconstituted
92511015	Fruit flavored drink
92511250	Fruit juice beverage, 40-50% juice, citrus
92513000	Slush frozen drink
92530410	Fruit flavored drink, with high vitamin C
92530510	Cranberry juice drink, with high vitamin C
92530610	Fruit juice drink, with high vitamin C

92530950	Vegetable and fruit juice drink, with high vitamin C
92531030	Fruit juice drink (Sunny D)
92541010	Fruit flavored drink, powdered, reconstituted
92542000	Fruit flavored drink, with high vitamin C, powdered, reconstituted
92550030	Fruit juice drink, with high vitamin C, light
92550035	Fruit juice drink, light
92550040	Fruit juice drink, diet
92550110	Cranberry juice drink, with high vitamin C, light
92550200	Grape juice drink, light
92550350	Orange juice beverage, 40-50% juice, light
92550360	Apple juice beverage, 40-50% juice, light
92550370	Lemonade, fruit juice drink, light
92550380	Pomegranate juice beverage, 40-50% juice, light
92550405	Vegetable and fruit juice drink, with high vitamin C, light
92552020	Fruit juice drink, reduced sugar (Sunny D)
92552030	Fruit juice drink (Capri Sun)
92582100	Fruit juice drink, with high vitamin C, plus added calcium
92582110	Fruit juice drink, added calcium (Sunny D)
92612010	Sugar cane beverage
92900100	Fruit flavored drink, with high vitamin C, powdered, not reconstituted
92900110	Fruit flavored drink, powdered, not reconstituted
92900200	Fruit flavored drink, powdered, not reconstituted, diet
92900300	Sports drink, dry concentrate, not reconstituted
92550400	Vegetable and fruit juice drink, with high vitamin C, diet
92550610	Fruit flavored drink, with high vitamin C, diet
92550620	Fruit flavored drink, diet
92552000	Fruit flavored drink, with high vitamin C, powdered, reconstituted, diet
92552010	Fruit flavored drink, powdered, reconstituted, diet
92400100	Soft drink, NFS, diet
92410320	Soft drink, cola, diet
92410350	Soft drink, cola, decaffeinated, diet
92410370	Soft drink, pepper type, diet
92410400	Soft drink, pepper type, decaffeinated, diet
92410420	Soft drink, cream soda, diet
92410520	Soft drink, fruit flavored, diet, caffeine free
92410560	Soft drink, fruit flavored, caffeine containing, diet
92410620	Soft drink, ginger ale, diet
92410720	Soft drink, root beer, diet
92410820	Soft drink, chocolate flavored, diet
92411610	Soft drink, cola, fruit or vanilla flavored, diet
92411620	Soft drink, cola, chocolate flavored, diet
92400000	Soft drink, NFS
92410310	Soft drink, cola

92410315	Soft drink, cola, reduced sugar
92410340	Soft drink, cola, decaffeinated
92410360	Soft drink, pepper type
92410390	Soft drink, pepper type, decaffeinated
92410410	Soft drink, cream soda
92410510	Soft drink, fruit flavored, caffeine free
92410550	Soft drink, fruit flavored, caffeine containing
92410610	Soft drink, ginger ale
92410710	Soft drink, root beer
92410810	Soft drink, chocolate flavored
92411510	Soft drink, cola, fruit or vanilla flavored
92411520	Soft drink, cola, chocolate flavored
95312400	Energy drink, low calorie (Monster)
95312410	Energy drink, sugar free (Monster)
95312500	Energy drink, sugar free (Mountain Dew AMP)
95312550	Energy drink, sugar free (No Fear)
95312555	Energy drink, sugar-free (NOS)
95312600	Energy drink, sugar-free (Red Bull)
95312700	Energy drink, sugar free (Rockstar)
95312800	Energy drink, sugar free (Vault)
95313200	Energy drink, sugar free
95322200	Sports drink, low calorie (Gatorade G2)
95322500	Sports drink, low calorie (Powerade Zero)
95323000	Sports drink, low calorie
95310200	Energy drink (Full Throttle)
95310400	Energy drink (Monster)
95310500	Energy drink (Mountain Dew AMP)
95310550	Energy drink (No Fear)
95310555	Energy drink (No Fear Motherload)
95310560	Energy drink (NOS)
95310600	Energy drink (Red Bull)
95310700	Energy drink (Rockstar)
95310750	Energy drink (SoBe Energize Energy Juice Drink)
95310800	Energy drink (Vault)
95311000	Energy Drink
95312560	Energy drink (Ocean Spray Cran-Energy Juice Drink)
95312900	Energy drink (XS)
95312905	Energy drink (XS Gold Plus)
95320200	Sports drink (Gatorade G)
95320500	Sports drink (Powerade)
95321000	Sports drink, NFS
95330100	Fluid replacement, electrolyte solution
95330500	Fluid replacement, 5% glucose in water

92410110	Water, tonic
92410210	Water, carbonated, plain
92410250	Water, carbonated, flavored
92510720	Fruit punch, made with fruit juice and soda
92510730	Fruit punch, made with soda, fruit juice, and sherbet or ice cream
91700500	M&M's Almond Chocolate Candies
91701010	Almonds, chocolate covered candy
91703040	Caramel candy, chocolate covered
91703050	Caramel with nuts and cereal, chocolate covered
91703060	Caramel with nuts, chocolate covered
91703070	Rolo
91703150	Toblerone, milk chocolate with honey and almond nougat
91703200	TWIX Caramel Cookie Bars
91703250	TWIX Chocolate Fudge Cookie Bars
91703300	TWIX Peanut Butter Cookie Bars
91703400	Whatchamacallit
91703600	Espresso coffee beans, chocolate-covered
91705010	Chocolate candy
91705020	Chocolate candy with cereal
91705030	Kit Kat
91705040	Chocolate, milk, with nuts, not almond or peanuts
91705050	Milk chocolate candy, with fruit and nuts
91705060	Milk chocolate candy, with almonds
91705070	Chocolate, milk, with peanuts
91705090	Chocolate candy with fondant and caramel
91705200	Chocolate chips
91705300	Dark chocolate candy
91705310	Chocolate, sweet or dark, with almonds
91705500	Mexican chocolate, tablet
91706000	Coconut candy, chocolate covered
91707010	Fondant, chocolate covered
91709000	Gumdrops, chocolate covered
91713010	Fudge, chocolate, chocolate-coated
91713020	Fudge, chocolate, chocolate-coated, with nuts
91713030	Fudge, chocolate
91713040	Fudge, chocolate, with nuts
91715000	Fudge, caramel and nut, chocolate-coated candy
91715100	SNICKERS Bar
91715200	Baby Ruth
91715300	100 GRAND Bar
91716110	Halvah, chocolate covered
91718050	Honey-combed hard candy with peanut butter, chocolate covered
91718100	Butterfinger

91718110	Butterfinger Crisp
91718200	Chocolate-flavored sprinkles
91723010	Marshmallow, chocolate covered
91726110	Nougat, with caramel, chocolate covered
91726130	MILKY WAY Bar
91726140	MILKY WAY MIDNIGHT Bar
91726150	MARS Almond Bar
91726410	Nougat, chocolate covered
91726420	3 MUSKETEERS Bar
91726425	3 Musketeers Truffle Crisp Bar
91727010	Nuts, chocolate covered, not almonds or peanuts
91731000	Peanuts, chocolate covered candy
91731010	M&M's Peanut Chocolate Candies
91731060	M&M's Peanut Butter Chocolate Candies
91733200	Peanut Bar, chocolate covered candy
91734000	Peanut butter, chocolate covered
91734100	Reese's Peanut Butter Cup
91734300	Reese's Sticks
91734400	Reese's Fast Break
91734450	Reese's Crispy Crunchy Bar
91739010	Raisins, chocolate covered
91746010	Sugar-coated chocolate discs
91746100	Chocolate candy, candy shell
91746120	Sixlets
91746150	Easter egg, candy coated chocolate
91746200	M&M's Pretzel Chocolate Candies
91760100	Toffee, chocolate covered
91760200	Toffee, chocolate-coated, with nuts
91760500	Truffles
91770030	Dietetic or low calorie candy, chocolate covered
91705005	Chocolate candy, other, NFS
91705012	Chocolate candy with nuts, other, NFS
91705015	Chocolate candy with nuts
91705080	Chocolate candy, cookie filled
91705290	Dark chocolate candy, other, NFS
91705312	Dark chocolate candy with nuts, other, NFS
91705315	Dark chocolate candy with nuts
91705440	Chocolate candy, fudge
91705450	Chocolate candy, caramel filled
91705460	Chocolate candy, caramel filled with nuts
91705470	Chocolate candy, coconut filled
91705480	Chocolate candy, cream filled
91705510	Chocolate candy, nougat filled

91705520	Chocolate candy, nougat filled with nuts
91705530	Chocolate candy, peanut butter filled
91705550	Chocolate candy with dried fruit
91706010	Chocolate candy, sugar free
91746110	Chocolate candy, candy shell with nuts
44201000	Carob chips
91700010	Candy, NFS
91701020	Almonds, sugar-coated
91701030	Almonds, yogurt-covered
91702010	Butterscotch morsels
91703010	Caramel, chocolate-flavored roll
91703020	Caramel, flavor other than chocolate
91703030	Caramel, with nuts
91703080	Caramel, all flavors, sugar free
91703500	Nuts, carob-coated
91705400	White chocolate candy
91705410	Chocolate, white, with almonds
91705420	Chocolate, white, with cereal
91705430	Kit Kat White
91706100	Coconut candy, no chocolate covering
91706400	Coconut candy, Puerto Rican style
91707000	Fondant
91708000	Fruit peel, candied
91708010	Date candy
91708020	Soft fruit confections
91708030	Fruit leather and fruit snacks candy
91708040	Fun Fruits Creme Supremes
91708070	Tamarind candy
91708100	Fruit snacks candy, with high vitamin C
91708150	Yogurt covered fruit snacks candy, with added vitamin C
91708160	Yogurt covered fruit snacks candy rolls, with high vitamin C
91713050	Fudge, peanut butter
91713060	Fudge, peanut butter, with nuts
91713070	Fudge, vanilla
91713080	Fudge, vanilla, with nuts
91713090	Fudge, divinity
91713100	Fudge, brown sugar, penuche
91716010	Halvah, plain
91718000	Honey-combed hard candy with peanut butter
91718300	Ladoo, round ball
91721000	Candy, licorice
91723000	Candy, marshmallow
91723020	Marshmallow, candy-coated

91726000	Nougat, plain
91728000	Candy, nougat with nuts
91728500	Sugared pecans, sugar and egg white coating
91731100	Peanuts, sugar-coated
91731150	Peanuts, yogurt covered
91732000	Peanut bar
91732100	Planters Peanut Bar
91733000	Candy, peanut brittle
91734200	Reese's Pieces
91734500	Peanut butter morsels
91735000	Pralines
91736000	Pineapple candy, Puerto Rican style
91739600	Raisins, yogurt covered
91742010	Sesame Crunch, Sahadi
91745010	Candy, gummy
91745020	Hard candy
91745040	Butterscotch hard candy
91745100	Skittles
91750000	Candy, taffy
91760000	Toffee, plain
91760700	Wax candy, liquid filled
91770000	Dietetic or low calorie candy, NFS
91770010	Dietetic or low calorie gumdrops
91770020	Dietetic or low calorie hard candy
91770050	Dietetic or low calorie mints
91800100	Chewing gum, NFS
91801000	Chewing gum
91802000	Chewing gum, sugar free
91705250	Candy, sprinkles
91706020	Candy, non chocolate, other, NFS
91723030	Candy, caramel
91745000	Candy, mint
91745025	Candy, hard
91745030	Candy, lollipop
91745035	Cough drops
91745050	Candy, cotton
91745110	Candy, fruit flavored pieces
91746300	Candy, fruit snacks
91746350	Candy, fruit leather
91770060	Candy, non chocolate, sugar free
91501010	Gelatin dessert
91501020	Gelatin dessert with fruit
91501100	Gelatin salad with vegetables

91511010	Gelatin dessert, sugar free
91511020	Gelatin dessert, sugar free, with fruit
95101000	Nutritional drink or shake, ready-to-drink (Boost)
95101010	Nutritional drink or shake, ready-to-drink (Boost Plus)
95102000	Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)
95103000	Nutritional drink or shake, ready-to-drink (Ensure)
95103010	Nutritional drink or shake, ready-to-drink (Ensure Plus)
95104000	Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)
95105000	Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)
95106000	Nutritional drink or shake, ready-to-drink (Muscle Milk)
95106010	Nutritional drink or shake, ready-to-drink, light (Muscle Milk)
95110000	Nutritional drink or shake, ready-to-drink (Slim Fast)
95110010	Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)
95110020	Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)
95120000	Nutritional drink or shake, ready-to-drink, NFS
95120010	Nutritional drink or shake, high protein, ready-to-drink, NFS
95120020	Nutritional drink or shake, high protein, light, ready-to-drink, NFS
95120050	Nutritional drink or shake, liquid, soy-based
95201000	Nutritional powder mix (Carnation Instant Breakfast)
95201010	Nutritional powder mix, sugar free (Carnation Instant Breakfast)
95201200	Nutritional powder mix (EAS Whey Protein Powder)
95201300	Nutritional powder mix (EAS Soy Protein Powder)
95201500	Nutritional powder mix, high protein (Herbalife)
95201600	Nutritional powder mix (Isopure)
95201700	Nutritional powder mix (Kellogg's Special K20 Protein Water)
95202000	Nutritional powder mix (Muscle Milk)
95202010	Nutritional powder mix, light (Muscle Milk)
95210000	Nutritional powder mix (Slim Fast)
95210010	Nutritional powder mix, sugar free (Slim Fast)
95210020	Nutritional powder mix, high protein (Slim Fast)
95220000	Nutritional powder mix, NFS
95220010	Nutritional powder mix, high protein, NFS
95230000	Nutritional powder mix, whey based, NFS
95230010	Nutritional powder mix, protein, soy based, NFS
95230020	Nutritional powder mix, protein, light, NFS
95230030	Nutritional powder mix, protein, NFS

ⁱ A note on Creme Food Safety® model: The system supports both deterministic (single points) input data and probabilistic input data. Probabilistic data can be represented by parametric or empirical distributions, integrated in the analysis using Monte Carlo simulations. If the type of modelling used is probabilistic, it means the input data may be represented as probability distributions when required (rather than using point estimates only) and the

simulation process uses conditional distribution sampling. When using probabilistic modelling and a population-based approach, the output from the model – the systemic exposure results – are themselves a distribution. In the Creme Food Data Science[®] model, standard errors of statistics are calculated using a resampling technique called bootstrapping. For example, a mean value can be estimated from the collected sample data which is assumed to be representative of the total population. Using the bootstrap method allows a distribution of the mean values to be generated and used to assess the accuracy of the estimated statistic (in this case, the mean value). This is performed by sampling with replacement from the data set in question several times, generating a number of different estimates of each statistic. The standard error of the mean is then the standard deviation of the mean values obtained from the large number of bootstrap samples.

Appendix 6 – Analysis of Residual DNA in L-Ergothioneine

Primers	Sequence (5' to 3')
Egt1-F	CTTTAAGAAGGAGATATACCATGCCGTCAGCGGAAAGC
Egt1-R	GATCTCAGTGGTGGTGGTGGTGGTCTCGAGCAGATCTCTAACGAC
EgtD-F	GAAGGAGATATACCATGGGCACATTGCTACTAGCTAATTATTAGC
EgtD-R	GATCTCAGTGGTGGTGGTGGTGGTGGTCTCGAGACGCACAGCCAGGCTAA

EgIE-F	CTTTAAGAAGGAGATATACCATGGGCCTAGCTCAGCAATGGAG
EgIE-R	GATCTCAGTGGTGGTGGTGGTGGTCTCGAGCGGAGCCTCACGTAAC
27F	AGAGTTTGATCCTGGCTCAG
1492R	GGTTACCTTGTACGACTT

Reaction system

ddH ₂ O	20 µL
2 x Taq Plus Master Mix	25 µL
Primer 1 (10 µM)	2 µL
Primer 2 (10 µM)	2 µL
Template DNA	1 µL

PCR amplification procedure

Step	Temp	Time
Initial denaturation	95 °C	3 min
	95 °C	15 sec
32 Cycles	60 °C	15 sec
	72 °C	60 sec/kb
Final extension	72 °C	5 min
Hold	4 °C	∞

(3) Preparation of agarose gel

TAE buffer is used as solvent to prepare 1% agarose gels with YeaGreen nucleic acid gel stain.

4. Results

Four gel electrophoresis pictures were generated on behalf of the PCR amplification results of Egt1, EgtD, and EgtE, with 16S rDNA as internal positive control.



Figure 1. Gene Egt1 (2686 bp) amplification including 6 concentration gradients of production strain and 3 concentration gradients of L-Ergothioneine product. M: DNA marker; 1-6: PCR products of positive control, the DNA template concentrations are 100 ng/uL, 50 ng/uL, 10 ng/uL, 1 ng/uL, 0.1 ng/uL, 0.01 ng/uL; 7-9: PCR products of L-Ergothioneine product (Batch No.: EGT20230903), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 10-12: PCR products of L-Ergothioneine product (Batch No.: EGT20231201), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 13-15: PCR products of L-Ergothioneine product (Batch No.: EGT20240207), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively.



Figure 2. Gene EgtD (1021 bp) amplification including 6 concentration gradients of production strain and 3 concentration gradients of L-Ergothioneine product. M: DNA marker; 1-6: PCR products of positive control, the DNA template concentrations are 100 ng/uL, 50 ng/uL, 10 ng/uL, 1 ng/uL, 0.1 ng/uL, 0.01 ng/uL; 7-9: PCR products of L-Ergothioneine product (Batch No.: EGT20230903), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 10-12: PCR products of L-Ergothioneine product (Batch No.: EGT20231201), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 13-15: PCR products of L-Ergothioneine product (Batch No.: EGT20240207), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively.



Figure 3. Gene EgtE (1168 bp) amplification including 6 concentration gradients of production strain and 3 concentration gradients of L-Ergothioneine product. M: DNA marker; 1-6: PCR products of positive control, the DNA template concentrations are 100 ng/μL, 50 ng/μL, 10 ng/μL, 1 ng/μL, 0.1 ng/μL, 0.01 ng/μL; 7-9: PCR products of L-Ergothioneine product (Batch No.: EGT20230903), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 10-12: PCR products of L-Ergothioneine product (Batch No.: EGT20231201), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 13-15: PCR products of L-Ergothioneine product (Batch No.: EGT20240207), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively.



Figure 4. 16S rDNA (1465 bp) amplification by using universal primers of 27F and 1492R, including 6 concentration gradients of production strain and 3 concentration gradients of L-Ergothioneine product. M: DNA marker; 1-6: PCR products of positive control, the DNA template concentrations are 100 ng/μL, 50 ng/μL, 10 ng/μL, 1 ng/μL, 0.1 ng/μL, 0.01 ng/μL; 7-9: PCR products of L-Ergothioneine product (Batch No.: EGT20230903), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 10-12: PCR products of L-Ergothioneine product (Batch No.: EGT20231201), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively; 13-15: PCR products of L-Ergothioneine product (Batch No.: EGT20240207), the corresponding L-Ergothioneine concentrations are 200 g/L, 100 g/L, and 10 g/L respectively.

5. Conclusion

The detection limit of PCR amplification for these four genes is 0.1 ng/μL. Residual DNA has been shown to be absent in the L-Ergothioneine products.

Appendix 7 – Bacterial Reverse Mutation Test Report of L- Ergothioneine



Study No.: S240800050

FINAL REPORT

STUDY TITLE

Bacterial Reverse Mutation Test with L-Ergothioneine

STUDY PLAN No.: SP240025

STUDY No.: S240800050

STUDY DIRECTOR

Wei Min

FINAL REPORT DATE

July 05, 2024

SPONSOR

Gene III Biotechnology Co., Ltd.

TEST FACILITY

Safety Evaluation Center of Shenyang SYRICAL Testing Co., Ltd.
No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province,
the People's Republic of China (Post code: 110141)
Telephone: 86-24-62353324 Fax: 86-24-62353328



Study No.: S240800050

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Study No.: S240800050

GLP COMPLIANCE STATEMENT

STUDY TITLE: Bacterial Reverse Mutation Test with L-Ergothioneine

STUDY NO.: S240800050

SPONSOR: Gene III Biotechnology Co., Ltd.

The study described in this report was conducted in compliance with OECD Principles of Good Laboratory Practice [ENV/MC/CHEM (98)17] as revised in 1997.

The data supplied by sponsor including the stability data, the compound analysis data and source of the test item are beyond assurance of this report.

In this study, no analysis was carried out to determine the homogeneity, concentration and stability of the test item formulations. As being prepared, the test item was weighed accurately and mixed with the solvent thoroughly. In addition, the test item was applied to the test system within four hours after being formulated. So it was assumed that the formulations were prepared exactly, homogeneously and stable for this duration. This exception is considered not to affect the purpose or integrity of the study.

The study was conducted according to the study plan and related SOPs. There was no circumstance that affected the quality or integrity of the study data, and this final report accurately reflects the data obtained during the course of the study.

STUDY DIRECTOR

Wei Min

Safety Evaluation Center of Shenyang SYRICI Testing Co., Ltd.

In Vitro Toxicology Department

2024-07-05

DATE

TEST FACILITY MANAGER

Zhang Xuan

Safety Evaluation Center of Shenyang SYRICI Testing Co., Ltd.

2024-07-05

DATE



Study No.: S240800050

QUALITY ASSURANCE STATEMENT

STUDY TITLE: Bacterial Reverse Mutation Test with L-Ergothioneine

STUDY NO.: S240800050

In compliance with OECD Principles of Good Laboratory Practice [ENV/MC/CHEM (98)17] revised in 1997, study plan and the test facility's SOPs, the study plan, critical phase(s), raw data and final report of the study have been inspected by Quality Assurance Unit (QAU). All observations were reported promptly in writing to the Study Director (SD) and Test Facility Management (TFM). The specifications of the inspection performed are presented below:

Items of audit	Dates of audit	Dates reported to SD	Dates reported to TFM
Study plan	MAY.27.2024 _ MAY.27.2024	MAY.27.2024	MAY.27.2024
Preparation of test item*	APR.10.2024 _ RPR.10.2024	APR.10.2024	APR.10.2024
Plate-incorporation*	APR.10.2024 _ APR.10.2024	APR.10.2024	APR.10.2024
Colonies count*	APR.12.2024 _ APR.12.2024	APR.12.2024	APR.12.2024
Raw data/draft report I	JUL.03.2024 _ JUL.03.2024	JUL.03.2024	JUL.03.2024
Raw data/final report	JUL.05.2024 _ JUL.05.2024	JUL.05.2024	JUL.05.2024

*This phase was process-based inspection

According to the inspection, the above phases are considered to be an accurate description of the performance of the study. The final report is also considered to be a true and accurate reflection of raw data.

In addition, QAU also conducts regular facility-based inspections in accordance with the Quality Assurance Programme, to comprehensively evaluate the GLP compliance of the overall facilities and activities of the test facility.

Quality Assurance Unit

Ma Juman

Safety Evaluation Center of Shenyang SYRICI Testing Co., Ltd.

2024-07-05

DATE



Study No.: S240800050

SUMMARY

Introduction. The study was performed to evaluate the capability of L-Ergothioneine to induce reverse mutations in the genome of the histidine requiring tester strains of *Salmonella typhimurium* in comparison to the concurrent negative control (solvent control) with and without the metabolic activation system, and the method was designed in compliance with OECD Guideline for the Testing of Chemicals No. 471 “Bacterial Reverse Mutation Test.” (Corrected; June 26, 2020).

Method. In this study, five histidine requiring tester strains of *Salmonella typhimurium* (*his-*) including TA97a, TA98, TA100, TA102 and TA1535 were treated with L-Ergothioneine solutions of different concentrations, using the plate incorporation method and the pre-incubation method, with the concurrent negative control (solvent control) and positive controls, both in the presence and absence of the metabolic activation system (S9 mix). Based on the results of the preliminary test (Study No.:S24080005A) that had been performed in this lab before, in the first test, 6 doses (the concentration in final treatment mixture) were selected for each tester strain, including 5000, 1500, 500, 150, 50 and 15 $\mu\text{g}/\text{plate}$ with and without metabolic activation. Ultra pure water (H_2O) was used as the solvent and concurrent solvent control, and each group was plated in triplicate. After that, the validation test (pre-incubation method) was conducted using the same doses and controls as those used in the first test, and each group was plated in triplicate. All plates were inverted and incubated for approximately 48~72 hours at $37 \pm 3^\circ\text{C}$ after each treatment. After incubation, the number of the revertant colonies was observed and counted in each plate, and the precipitate and cytotoxicity of the test item were observed at the same time.

Results. In two tests, the results of the viable count showed that the density of living bacteria in the fresh culture of each tester strain was more than 1×10^9 cells/mL. At the same time, all results of the concurrent solvent control and positive controls met the requirements of this test, so the sensitivity of the test system and the efficacy of the metabolic activation system were validated.



Study No.: S240800050

The results of test item precipitate showed that no precipitate was observed on the surface of the minimal glucose agar (MGA) plates at any dose of each tester strains before and after incubation, with and without metabolic activation in the first test. In the validation test, the same precipitate results were obtained as those in the first test.

The results of cytotoxicity showed that no cytotoxicity was observed at any dose level in 5 tester strains with and without metabolic activation in the first test. In the validation test, the similar cytotoxicity results were observed as those in the first test.

The results of colony counting showed that the mean number of revertant colonies at each dose level was less than two times of that of the concurrent solvent control in TA97a, TA98, TA100, TA102 and less than three times of that of the concurrent solvent control in TA1535 with and without metabolic activation in the first test and the validation test.

Conclusion. The results of the first test and the validation test were negative in this study. Thus, the test item, L-Ergothioneine, was considered to be non-mutagenic in the bacterial reverse mutation test using the histidine requiring tester strains of *Salmonella typhimurium* under the conditions of this study.



Study No.: S210800050

1. STUDY TITLE

Bacterial Reverse Mutation Test with L-Ergothioneine

2. STUDY OBJECTIVE

This study was conducted to evaluate L-Ergothioneine for its capability to induce reverse mutations, either with or without the metabolic activation system (S9 mix), in the genome of the histidine requiring strains of *Salmonella typhimurium*, in order to predict the possibility of genotoxicity and mutagenic effect.

3. GUIDELINE

OECD Guidelines for the Testing of Chemicals, No. 471 "Bacterial Reverse Mutation Test" (Corrected: June 26, 2020).

4. GLP COMPLIANCE

OECD Principles on Good Laboratory Practice [ENV/MC/CHEM (98)17] as revised in 1997.

5. TEST FACILITY

Test facility: Safety Evaluation Center of Shenyang SYRICI Testing Co., Ltd.
Address: No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province, P.
R. China (Post code: 110141)
Telephone: 86-24-62353324
Fax: 86-24-62353328

6. SPONSOR AND REPRESENTATIVE

Sponsor: Gene III Biotechnology Co., Ltd.
Address: East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu
District Nanjing, Jiangsu 210023 China
Contact person: Ju Hongying
Telephone: 18052038505



Study No.: S240800050

E-mail: juhongying@geneiii.com.cn; 275080046@qq.com

7. TEST FACILITY MANGAGER, STUDY DIRECTOR AND PARTICIPANTS

Test facility manager (FM): Zhang Xuan

Address: No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province, P. R. China (Post code: 110141)

Telephone: 86-24-62353450

Email: zhangxuan1@sinochem.com

Study Director (SD): Wei Min

Address: No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province, P. R. China (Post code: 110141)

Telephone: 86-24-62353472

Email: weimin@sinochem.com

Responsible for test operation: Huo Jinling

Address: No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province, P. R. China (Post code: 110141)

Telephone: 86-24-62353472

E-mail: huojinling@sinochem.com

Responsible for Test item administration: Pang Yanan

Address: No. 600 Shenliao Road, Tiexi District, Shenyang, Liaoning Province, P. R. China (Post code: 110141)

Telephone: 86-24-62353452

E-mail: pangyanan@sinochem.com

Study report authors: Wei Min, Huo Jinling

Main study participants: Huo Jinling, Ma Xiaoguang, Kong Shengyu



Study No.: S240800050

8. QUALITY ASSURANCE

The study plan, raw data and final report have been audited by QAU. According to Quality assurance Programmes, QAU have carried out process-based inspections to the study practices. QAU will issue a QA statement to be included in the final report.

9. TEST ITEM, SOLVENT AND CONTROL ITEMS

9.1 TEST ITEM

Name: L-Ergothioneine

Test item code in the facility: 2024E005

Active ingredient: L-Ergothioneine, EGT

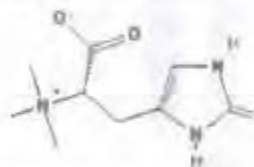
CAS No.: 497-30-3

Chemical Name: L-Ergothioneine

Molecular formula: $C_9H_{15}N_3O_2S$

Molecular weight: 229.298 g/mol

Structural formula:



Batch No.: EGT20240207

Appearance/Color/State: White powder, no peculiar smell, no visible impurities

Content or purity (Measured result): 99.9% (supplied by Sponsor)

Solubility: Soluble in water

Stability: The test item is stable when stored and used at normal ambient temperature.

Manufacturer: Gene III Biotechnology Co., Ltd.

Address: East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu District Nanjing, Jiangsu 210023 China

Manufacture date: Feb. 27, 2024

Arrival date: Apr. 11, 2024

Expiry Date: Feb. 26, 2026



Study No.: S240800050

Receiving weight of test item (Gross weight): 10.0g

Storage conditions: Room temperature (15~25 °C), dry, avoid light

Storage site: Stored in test item storage room of this facility

Handling Precautions: The strict precautions, including work clothes, gloves, face mask, goggle and cap have been worn when handling the test item. Hands were washed with soap and water after using the test item.

The data relating to the stability, the compound analysis and source of the test item were provided by the Sponsor, and are the responsibility of the Sponsor. A copy of the Certification of Analysis (COA), as provided by the Sponsor, has been included in the report(See annex 5).

Disposal of remaining test item: After the test, the remaining test item will be collective disposal by test facility.

9.2 SOLVENT (SOLVENT CONTROL)

Name: Ultra pure water

CAS No.: 7732-18-5

Common name/Abbreviated name: H₂O

Purpose: Used as the solvent for test item preparation and the concurrent solvent control for all tester strains.

Justification for choice: According to the test results of test item solubility (Study No.: S24080005A), ultra pure water was selected as the solvent and H₂O is also accredited by OECD Guidelines for the Testing of Chemicals, No. 471 (June 26, 2020).

Supplier: Sartorius arium[®] mini ultra-pure water machine in this facility (Z-9)

Batch No.: 20240531-01 and 20240604-01

Purity/Content: 100%

Appearance/Color/State: Colorless transparent liquid

Grade: Sterile (0.22µm aqueous membrane filtration)

Storage condition: Room temperature (RT, 15~25 °C)

Expiry date: Preparation just before use and not saved after using.

9.3 POSITIVE CONTROL ITEMS

1) Name: Fenaminosulf



Study No.: S240800050

CAS No.: 140-56-7

Common name / Abbreviated name: Dexon

Code in this facility: C12021002

Purpose: Used in the tester strains TA97a, TA98 and TA102 without metabolic activation system (S9-).

Supplier: AccuStandard, Inc., USA

Batch No.: 4030165-01

Content or purity(HPLC): 91.0%

Appearance: Yellow-brown crystalline solid

Storage condition: RT

Storage site: Stored in the positive cabinet in room A220 of this facility.

Expiry date: Jan. 18, 2026

2) Name: 2-Aminofluorene

CAS No.: 153-78-6

Common name/ Abbreviated name: 2-AF

Code in this facility: C12023034

Purpose: Used in the tester strains TA97a, TA98 and TA100 with metabolic activation system (S9+).

Supplier: Shanghai Aladdin Biochemical Technology Co., Ltd.

Batch No.: J2116394

Content or purity(Nonaqueous titration): 98.22160%

Appearance: White to light brown powder or crystals

Storage condition: RT

Storage site: Stored in the positive cabinet in room A220 of this facility.

Expiry date: Oct. 22, 2025 (Date of next testing)

3) Name: Sodium azide

CAS No.: 26628-22-8

Common name/ Abbreviated name: NaN₃



Study No.: S240800050

Code in this facility: C12010028

Purpose: Used in the tester strains TA100 and TA1535 in S9-.

Supplier: Amresco Inc., Ohio, USA

Batch No.: 0639

Content or purity: $\geq 99.0\%$

Appearance: White crystalline powder

Storage condition: RT, protected from moisture

Storage site: Stored in the positive cabinet in room A220 of this facility.

Expiry date: Long-term effective

4) Name: 1, 8-Dihydroxyanthraquinone

CAS No.: 117-10-2

Common name/ Abbreviated name: Danthron/ Dorbane

Code in this facility: C12023016

Purpose: Used in the tester strain TA102 in S9+.

Supplier: Shanghai Aladdin Biochemical Technology Co.Ltd.

Batch No.: G2208134

Content or purity(HPLC): 97.01500%

Appearance: Yellow to orange to brown powder, crystals, chunks

Storage conditions: Cold temperature (2~8℃), closed, dry

Storage site: Stored in the refrigerator (P-195) in the test item storage room of this facility.

Expiry date: July 11, 2026 (Date of next testing)

5) Name: 2-Aminoanthracene

CAS No.: 613-13-8

Common name/ Abbreviated name: 2-AA

Code in this facility: C12023014

Purpose: Used in the tester strain TA1535 in S9+.

Supplier: Sigma-Aldrich, LLC.

Batch No.: STBJ3963

Content or purity(HPLC): $> 99.9\%$



Study No.: S240800050

Appearance: Dark yellow-beige powder

Storage conditions: Cool temperature (2–8℃), closed, dry

Storage site: Stored in the refrigerator (P-195) in the test item storage room of this facility.

Expiry date after opening: Feb. 02, 2028.

10. TEST SCHEDULE

Study initiation date: May 27, 2024

Test starting date: May 31, 2024

Test completion date: June 06, 2024

Draft report: June 25, 2024

Final report: July 05, 2024

11. MATERIALS AND METHODS

11.1 PRINCIPAL INSTRUMENTS AND EQUIPMENTS

Name	Brand and type	Instrument code
Electronic balance	Sartorius BS323S	M-15
Electronic balance	Sartorius SQP	Z-43
Constant temperature incubator	Binder BD240	T-9
Biologic safety cabinet	Thermo MSC-Advantage 1.8	R-9
Microtitre plate Shaker	Stuart SSM5	A2019020 and A2019021
Stereomicroscope	OLYMPUS SZX16	P-21
Automatic colony analysis	Shimadzu Icount 30I	Z-72

11.2 MAIN REAGENTS

Name	Purity or Grade
Nutrient broth (powder)	Biochemical reagent (BR)
Agar	BR
NaCl	Analytic reagent (AR)
L-Histidine	98.5–101.0%
D-Biotin	98–100%
NADP-Na2	≥98%
D-Glucose 6-phosphate disodium salt (G-6-P-Na2)	≥95%



Study No.: S240800050

K_2HPO_4	AR
$NaNH_4HPO_4 \cdot 4H_2O$	AR
$C_6H_5O_7 \cdot H_2O$	AR
$MgSO_4 \cdot 7H_2O$	AR
KCl	AR
$MgCl_2 \cdot 6H_2O$	AR

11.3 TESTER SYSTEM

Tester strains: Histidine requiring strains of *Salmonella typhimurium*, including TA97a, TA98, TA100, TA102 and TA1535.

The tester strains supplier: Molecular Toxicology Inc. USA

Receiving date: Sep. 17, 2010

Passage No. (the tester strains purchased): Passage 0

Apperance/color/state (the tester strains purchased): Dry powder, tablet

Passage No. (the frozen stock tester strains used in this study): Passage 2

Genotype verification of the tester strains: In order to provide eligible tester strains for the bacterial reverse mutation test, before being used for Ames test, the genotypes of each batch strains had been tested, including histidine dependence test, Lipopolysaccharide envelope deficiency test (*rfa* mutation), *uvrB* mutation test (sensitivity to ultra-violet light), R factor plasmids test (Ampicillin Resistance), Tetracycline Resistance test and spontaneous mutation frequency test. After the genotype being tested, the qualified strains were stored in ultra-low temperature freezer at approximately -80°C for Ames tests not more than one year. The frozen stock tester strains used in this study were stored in Feb. 19, 2024 and tested from Feb. 19 to 21, 2024. The results showed that all tester strains were qualified and met the requirements of Ames test. The results of genotype verification of Ames tester strains (COA of Ames tester strains) tested by this lab are present in Annex 2.

11.4 TEST CONDITIONS

11.4.1 METABOLIC ACTIVATION SYSTEM

In this study, a cofactor- supplemented post-mitochondrial fraction (S9) prepared from the livers of Sprague Dawley rat treated with enzyme-inducing agents (a combination of phenobarbitone and β - naphthoflavone) was used as the



Study No.: S210800050

metabolic activation system. The information of S9 purchase is showed as below:

Supplier: CHI SCIENTIFIC INC.

Batch No.: 23FS037K

Control date: Jan. 18, 2024

Expiry date: Jan. 17, 2026

Protein concentration: 30 mg/mL

Test of metabolic activation ability: The results showed that the metabolic activation capability of the S9 was in the expected range to meet the test requirements. The COA of this batch of S9 provided by the supplier is presented in Annex 3.

Verification of S9: Before being used, this batch of S9 had been tested in this lab for its sterility (culture after inoculation), its protein content (tested by Hitachi 7180 Full Automatic Biochemical Analyzer in this facility, the protein content was not higher than 40 mg/mL), and its capability to activate known mutagens (2-AF, Danthron and 2-AA) in an Ames test. The S9 batches used in this test were tested from Mar. 06 to 08, 2024. The results indicated that each index of this batch of S9 was in the expected range to meet Ames test requirements. The verification results for S9 (COA of the S9) tested in this lab is presented in Annex 4.

Preparation of 10% S9 mixture (S9 mix): 10% S9 mix was prepared according to the recipe below just before the test and placed in a mixture of water and ice during the test. The remaining S9 mix was discarded.

Ingredient	Formula (10 mL S9 mix)	Actual volume (65 mL S9 mix)
0.2mol/L Phosphate buffer (PBS)	6.0 mL	39.000 mL
NADP-Na ₂ solution (0.025 mol/L)	1.6 mL	10.400 mL
G-6-P-Na ₂ solution (0.05 mol/L)	1.0 mL	6.500 mL
MgCl ₂ (0.4mol/L)-KCl (1.65mol/L) solution	0.2 mL	1.300 mL
Sterile distilled water	0.2 mL	1.300 mL
Liver S9	1.0 mL	6.500 mL

11.4.2 PREPARATION OF CULTURE MEDIA AND SOLUTIONS

The media and solutions used in this test were included:

2.5% (w/v) nutrient broth medium;



Study No.: 5240800050

Bottom-layer agar medium (minimal glucose agar, MGA for short) plates, being made with the mixture of Vogel-Bonner (VB) solution, 20% (w/v) Glucose solution and 1.9% (w/v) agar medium;

Top-layer agar medium with 0.7% (w/v) agar and 0.5% (w/v) sodium chloride;

Histidine-biotin solution containing 7.75 mg L-histidine and 12.2 mg D-biotin per 100mL;

0.5% L- histidine solution containing 5mg L-histidine per 1 mL;

0.01% D- biotin solution containing 0.1mg D-biotin per 1 mL;

0.2 mol/L Phosphate buffer (PBS);

1.65 mol/L KCl+0.4 mol/L MgCl₂ solution;

0.05 mol/L G-6-P-Na₂ solution;

0.025 mol/L NADP-Na₂ solution.

All media and solutions were prepared and sterilized before the start of the test and the details were laid in the preparation records.

11.4.3 PREPARATION OF POSITIVE CONTROLS

The positive control solutions used in this study were prepared by Ames lab technicians in this facility, then these were managed and stored in this facility. All solutions were prepared according to the table below, and stored at 2~8℃ after being dispensed. These positive control solutions were prepared on Feb. 01, 2024, and tested from Feb. 02 to 04, 2024. The test results showed that the prepared solutions of all positive controls met the requirement of validity in Ames test. The validity period is one year.

Name	Dose level (μg/plate)	Preparation Concentration (mg/mL)	Solvent
Dexon	50	0.5	DMSO (AR)
NaN ₃	1.5	0.015	Sterile distilled water
2-AF	20	0.2	DMSO (AR)
Daniltron	50	0.5	DMSO (AR)
2-AA	2.0	0.02	DMSO (AR)

Note: DMSO-dimethyl sulfoxide; AR- analytic reagent

After plate incorporation test, the remaining solutions were kept temporarily in a



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special container in this facility before being transported to external professional environmental protection company for disposal.

11.5 TEST METHODS

11.5.1 PRELIMINARY TEST

The preliminary test (Study No.: S24080005A) for the Bacterial Reverse Mutation Test with L-Ergothioneine had been performed in this lab to evaluate solubility and cytotoxicity of the test item in final treatment mixture, in order to provide a reference basis for the dose (the concentration in final treatment mixture) selection of the test item in Bacterial Reverse Mutation Test.

In the preliminary test, the results of test item solubility test showed that the test item was completely dissolved in H₂O at the concentration of 50 mg/mL, so H₂O was selected as the solvent. The plate incorporation method was performed at 3 doses of the test item, including 5000, 1667 and 556 µg/plate, in five tester strains of TA97a, TA98, TA100, TA102 and TA1535, with and without metabolic activation system. The concurrent solvent control (H₂O) in each tester strain was performed. The dose volume of each group was 0.1 mL/plate, in duplicate. The plates after administration were inverted and incubated for approximately 48 hours at 37±3 °C. The precipitate in two plates of each dose in all tester strains were observed by naked eye before and after the incubation. The number of the revertant colonies in each plate was observed and / or counted, and the background lawn were observed for the plates of 5000µg/mL and all controls by naked eye and microscope after the incubation.

Preliminary test results: No precipitate was observed on the surface of MGA plate at any dose level before and after incubation, with and without metabolic activation system. Moreover, compared to the concurrent solvent control, no cytotoxicity was observed at any dose level for 5 tester strains with and without metabolic activation system.

11.5.2 THE FIRST TEST



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11.5.2.1 Solvent and dose selection

Solvent and dose were selected according to the results of the preliminary test and the related requirements of OECD Guideline for Testing of Chemicals, No.471 (2020).

In the first test, H₂O was used as the solvent, and 6 doses were performed, including 5000, 1500, 500, 150, 50 and 15 µg/plate with and without the metabolic activation system. The concurrent solvent control (H₂O) and positive controls in each tester strain were included. The details are shown as below:

Groups	S9-					S9+				
	TA97a	TA98	TA100	TA102	TA1535	TA97a	TA98	TA100	TA102	TA1535
Dose level (µg/plate)	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000
	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500
	500	500	500	500	500	500	500	500	500	500
	150	150	150	150	150	150	150	150	150	150
	50	50	50	50	50	50	50	50	50	50
	15	15	15	15	15	15	15	15	15	15
Solvent control (µL/plate)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)	H ₂ O (100)
Positive control (µg/plate)	Dexon (50)	Dexon (50)	NaN ₃ (1.5)	Dexon (50)	NaN ₃ (1.5)	2-AF (20)	2-AF (20)	2-AF (20)	Danthron (50)	2-AA (2.0)

11.5.2.2 Preparation of test item

In the test, H₂O was used as the solvent. The preparation concentrations of different dose levels are showed in the table below:

Dose (µg/plate)	Dose volume (µL/plate)	Preparation concentration (mg/mL)
5000	100	50
1500	100	15
500	100	5
150	100	1.5
50	100	0.5
15	100	0.15

Calculation formulas:

Preparation concentration (µg/µL solvent) = Dose (µg/plate) / Dose volume (µL/plate)

Test item weight (g) = Preparation volume (mL) × Preparation concentration (g/mL)



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To prepared the test item, the test item (0.299g and 0.090g) were weighed with the electronic balance (M-15) into a calibrated aseptic tube, respectively, and H₂O was added and mixed with the test item thoroughly to a certain volume (here: 6.000 mL) to obtain 50 and 15 mg/mL test item solutions, respectively. Under aseptic conditions, the test item solutions of the other concentrations were prepared by stepwise dilution. The dilution method is shown in the following table:

Before dilution		Factor	Volume of solvent (μL)	Concentration of the solution after dilution (mg/mL)
Concentration of the solution (mg/mL)	Volume of the solution (μL)			
50	500	10	4500	5
15	500	10	4500	1.5
5	500	10	4500	0.5
1.5	500	10	4500	0.15

The prepared test item solutions were stored in the aseptic test tubes labeled with study No. and preparation concentration, and the test item was wrapped with foil paper to protected from light when the test item was use.

After plate incorporation test, the remaining test item solutions were kept temporarily in a special container in this facility before transported to external professional environmental protection company for disposal.

11.5.2.3 Enrichment culture

One day before the plate-incorporation, the frozen stock tester strains were thawed and 100.0μL of bacterium suspension of each tester strain was inoculated in nutrient broth medium (20mL) and incubated approximately 16 hours at 36.6~37.0 ℃.

11.5.2.4 Viable count

In order to confirm the density of living bacteria in the fresh cultures, a part of the culture of each tester strain was taken out for the viable count. The culture was diluted with sterile stiller water, then 200.0 μL of the dilution of bacteria culture that had been diluted by the factors of 10⁵, 10⁶ or 10⁷ respectively, 100.0 μL of



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histidine solution supplemented with an excess of histidine (0.5% L-histidine solution), 100.0 μL of 0.01% D-biotin solution, 500 μL of PBS and 2000 μL of molten top-layer agar medium kept in a water bath at 43.9~47.6 $^{\circ}\text{C}$ were added to a sterile tube under aseptic conditions. After vortexing, the mixture was rapidly and evenly overlaid on the surface of a MGA plate, and each of the above diluted concentration was performed in triplicate. When solidified, the plates were inverted and incubated for about 69.5 hours at 35.4~37.5 $^{\circ}\text{C}$. Then the number of the colonies in the plates with the 10^6 -fold dilution were counted. The density of living bacteria in the culture was calculated based on the numbers of colonies and dilution factor. The calculation formula is as following:

The density of living bacteria (cells/mL) = Mean number of colonies in three plates $\times 5 \times$ dilution factor

11.5.2.5 Plate-incorporation

1) In the absence of S9 mix: 200.0 μL of histidine-biotin solution, 100.0 μL of fresh cultures of bacteria, 500 μL of PBS, 100.0 μL of the test item formulation or the concurrent solvent control (H_2O) or positive control solution and 2000 μL of molten top-layer agar medium kept in a water bath at 43.9~47.6 $^{\circ}\text{C}$ were added to a sterile test tube under aseptic conditions. After vortexing, the mixture was rapidly and evenly overlaid on the surface of a MGA plate, all dose levels and the controls were performed in triplicate. After the top-layer agar solidified, the plates were inverted and incubated for approximately 69.5 hours at 35.4~37.5 $^{\circ}\text{C}$. The test item precipitate was observed by naked eye in three parallel plates of each dose group of all tester strains before and after incubation. After incubation, the number of revertant colonies in each plate was counted, and the background lawn were observed by naked eye and microscope for the plates of 5000 $\mu\text{g}/\text{plate}$ dose group and all control groups of each tester strain.

2) In the presence of S9 mix: All procedures were the same as those in the absence of S9 mix except PBS was replaced with S9 mix.

11.5.2.6 Data statistics and analysis

In the verified Microsoft Excel (2007) (Bacterial Reverse Mutation Test-the



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statistical results table for the number of colonies 2.0), for three plates of each dose and control group, the mean number and the standard deviation of the number of revertant colonies were calculated. At the same time, the ratio of the mean number of revertant colonies between each group and the concurrent solvent control was calculated.

11.5.3 THE VALIDATION TEST

As there was a negative result in the first test, a validation test was performed. The validation test was conducted at the same dose levels as the first test in the same tester strains, either with or without S9 mix, with the same positive and solvent controls. The test item formulations were prepared using the same way as that in the first test.

As it was good scientific practice to alter one condition in the replicate test, the treatment condition was changed to pre-incubation method in the validation test. In the validation test, bacteria culture was obtained through incubation at 36.4~37.0℃ for about 16 hours. When prepared, the test items (0.302g and 0.090g) were weighed with the electronic balance (M-15) into a calibrated aseptic tube, respectively, and H₂O was added and mixed with the test item thoroughly to a certain volume (here: 6.000mL) to obtain 50 and 15 mg/mL test item solutions. Under aseptic conditions, the test item solutions of the other concentrations were prepared by stepwise dilution. During pre-incubated operation procedure, fresh culture of bacteria, PBS or S9 mix, and the solution of the test item or solvent or positive control were mixed in a sterile test tube and pre-incubated using the shakers (A2019020 and A2019021) at 35.7~36.2℃ for at least 20min, then histidine-biotin solution and top-layer agar were add, and the rest operations were the same as the plate incorporation method. The temperature range of the water bath was 45.0~47.6℃. All tested plates were inverted and incubated for about 48.5 hours at 35.4~37.9℃. The test item precipitate was observed by naked eye in three parallel plates of each dose group of all tester strains before and after incubation. After incubation, the revertant colonies were count and the background lawn were observed by naked eye and microscope for the plates of



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5000 µg/plate dose group and all control groups of each tester strain in two treatment conditions.

11.6 TEST VALIDITY CRITERIA

The test system that meets the conditions below is considered to be valid:

- 1) The density of living bacteria in each fresh tester strain culture should be contain viable bacteria approximately 10^9 cells/mL;
- 2) The mean numbers of revertant colonies in all the concurrent negative control (solvent control) groups are the 95% control limits of historical negative control distribution in this lab, and microscopic examination of the background lawn reveals the presence of densely microcolonies which form a uniform layer;
- 3) The mean numbers of revertant colonies in all positive controls are within the 95% control limits of historical positive control distribution in this lab, and microscopic examination of the background lawn reveals no thin compared with that of the concurrent solvent control observed by microscope.

11.7 EVALUATION CRITERIA OF THE TEST RESULTS

11.7.1 CRITERIA OF CYTOTOXICITY

The test item is considered to be cytotoxicity if one of the below criteria is met.

- 1) Compared with the concurrent solvent control, the number of the revertant colonies has an obvious decrease or is absent;
- 2) Compared with the concurrent solvent control, the signs of the background lawn are thin or clearing.

11.7.2 CRITERIA OF POSITIVE RESULT

- 1) When there is a dose-related increase over the range (equal to or greater than two times of that of the concurrent solvent control in TA97a, TA98, TA100, TA102, and/or equal to or greater than three times of that of the concurrent solvent control in TA1535) in the mean number of revertant colonies in at least one strain with or without the metabolic activation system, the test item is considered clearly positive.



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2) When there is a reproducible increase in the mean number of revertant colonies (equal to or greater than two times of that of the concurrent solvent control in TA97a, TA98, TA100, TA102, and/or equal to or greater than three times of that of the concurrent solvent control in TA1535) at one or more dose levels in the mean number of revertant colonies in at least one strain with or without metabolic activation system, the test item is considered clearly positive.

11.7.3 CRITERIA OF NEGATIVE RESULT

The test item is considered clearly negative if none of the above criteria (See 11.7.2) is met.

11.7.4 CRITERIA OF EQUIVOCAL RESULT

Although most tests give clearly positive or negative results, in rare cases, the generated data prohibit test item from making a definite judgment. Results of this type main remain equivocal or questionable regardless of the number of times the experiment is repeated, the results of this type is reported as equivocal.

12. STUDY DEVIATION

None.

13. RESULTS AND CONCLUSION

13.1 RESULTS

The results of the viable count in two tests showed that the density of living bacteria in the fresh culture of each tester strain was more than 1×10^5 cells/mL, the results met the test requirements. The data of the viable count in the two tests are presented in Table 1.

In the first test and the validation test, the mean numbers of revertant colonies of the concurrent solvent controls and positive controls were within the range of historical control data in this lab. In addition, the background lawn of the concurrent solvent control and positive controls met the test validity criteria the



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presence of densely microcolonies which form a uniform layer observed with microscope. So the sensitivity of the test system and the efficacy of the metabolic activation system were validated. The data of all controls in two tests are presented in Tables 2 and 3. The historical control data of five tester strains in this lab (2023) are presented in Annex 1.

The results of the test item precipitate: In the first test, no precipitate was observed on the surface of MGA plate at any dose level before and after incubation, with and without the metabolic activation system. In the validation test, the same precipitate results were obtained as those in the first test. The observation results of the test-item precipitate are presented in Table 8.

The results of cytotoxicity: In the first test, no cytotoxicity was observed at any dose level of 5 tester strains with and without the metabolic activation system. It showed that the number of revertant colonies was not obvious decrease and the background lawn was not thinner than that of the concurrent solvent control observed by microscope. In the validation test, the similar cytotoxicity results were observed as those in the first test.

The results of colony counting: In the first test, the mean number of revertant colonies at each dose level was less than two times of that of the concurrent solvent controls in TA97a, TA98, TA100, TA102, and less than three times of that of the concurrent solvent controls in TA1535 with and without metabolic activation. In the validation test, the similar results was obtained as those in the first test.

All results of the individual data and cytotoxicity of the first test are presented in Tables 4 and 5. All results of the individual data and cytotoxicity of the validation test are presented in Tables 6 and 7.

13.2 CONCLUSION

The results of the first test and the validation test were negative in this study. Thus, the test item, L-Ergothioneine, was considered to be non-mutagenic in the bacterial reverse mutation test using the histidine requiring tester strains of *Salmonella*



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typhimurium under the conditions of this study.

14. ARCHIVING

The following materials will be maintained in the archives of the facility for a period at least ten years from the date of archiving:

- Study plan;
- Raw data;
- Final report.

However, electronic data will be retained in the NAS system in this facility.

Samples of positive control items for archiving will be archived at least 7 years after the end of the study, or be archived until they cannot be evaluate effectively from the archive day.

If requested, a subsequent retention period and relevant matters will be discussed by the sponsor and test facility.



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15. ATTACHED TABLES

Table 1 Results of the viable count

Test	Tester strain	Dilution multiple	count of colonies per plate (CFU/plate)			Mean±SD	Viable count (cells/mL)
The first test	TA97a	10 ⁶	323	402	382	369±41	1.845×10 ⁹
	TA98	10 ⁶	348	335	362	348±14	1.740×10 ⁹
	TA100	10 ⁷	231	251	265	249±17	1.245×10 ⁹
	TA102	10 ⁶	488	473	493	485±10	2.425×10 ⁹
	TA1535	10 ⁶	405	406	371	394±20	1.970×10 ⁹
The validation test	TA97a	10 ⁶	323	369	298	330±36	1.650×10 ⁹
	TA98	10 ⁶	353	377	311	347±33	1.735×10 ⁹
	TA100	10 ⁶	216	282	200	233±43	1.165×10 ⁹
	TA102	10 ⁶	443	440	469	451±16	2.255×10 ⁹
	TA1535	10 ⁶	447	343	324	371±66	1.855×10 ⁹

Notes: (1) Viable count = Mean number of colonies in three plates × 5 × dilution factor

(2) CFU- Colony forming units

(3) (5) Mean- Mean number

(4) SD- Standard deviation



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Table 2 Results of the controls in the first test

	Control	Plates	Number of the revertant colonies (CFU/plate)				
			TA97a	TA98	TA100	TA102	TA1535
S9-	Solvent control (H ₂ O)	1	132N	22N	158N	247N	17N
		2	160N	18N	139N	282N	13N
		3	152N	27N	152N	264N	14N
		Mean±SD	148±14	22±5	150±10	264±18	15±2
		Ratio	1.00	1.00	1.00	1.00	1.00
	Positive control	Name (dose level)	Dexon (50 µg/plate)	Dexon (50 µg/plate)	NaN ₃ (1.5 µg/plate)	Dexon (50 µg/plate)	NaN ₃ (1.5 µg/plate)
		1	4361G	1889G	1324G	2401G	552G
		2	4120G	1614G	925G	2289G	617G
		3	4412G	1558G	1088G	1953G	524G
		Mean±SD	4298±156	1687±177	1112±201	2214±233	564±48
		Ratio	29.04	76.68	7.41	8.39	37.60
S9+	Solvent control (H ₂ O)	Number of the revertant colonies (CFU/plate)					
			TA97a	TA98	TA100	TA102	TA1535
		1	132N	30N	176N	292N	19N
		2	153N	27N	180N	320N	24N
		3	165N	31N	155N	289N	16N
		Mean±SD	150±17	29±2	170±13	300±17	20±4
		Ratio	1.00	1.00	1.00	1.00	1.00
	Positive control	Name (dose level)	2-AF (20 µg/plate)	2-AF (20 µg/plate)	2-AF (20 µg/plate)	Danthron (50 µg/plate)	2-AA (2.0 µg/plate)
		1	1701G	1975G	1664G	798G	125G
		2	1563G	1691G	1614G	889G	95G
		3	1425G	1779G	1805G	743G	111G
		Mean±SD	1563±138	1815±145	1694±99	810±74	111±14
		Ratio	10.42	62.59	9.96	2.70	3.55

Notes: (1) Ratio – The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control
(2) N- Background lawn revealed the presence of densely microcolonies which form a uniform layer
(3) G- The background lawn was not this compared with that of the concurrent solvent control observed by microscope
(4) CFL- Colony forming units
(5) Mean- Mean number
(6) SD-Standard deviation



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Table 3 Results of the controls in the validation test

	Control	Plates	Number of the revertant colonies (CFU/plate)				
			TA97a	TA98	TA100	TA102	TA1535
S9-	Solvent control (H ₂ O)	1	140N	21N	150N	233N	10N
		2	146N	20N	121N	265N	16N
		3	153N	23N	142N	251N	16N
		Mean±SD	146±7	21±2	138±15	250±16	14±3
		Ratio	1.00	1.00	1.00	1.00	1.00
	Positive control	Name (dose level)	Dexon (50 µg/plate)	Dexon (50 µg/plate)	NaN ₃ (1.5 µg/plate)	Dexon (50 µg/plate)	NaN ₃ (1.5 µg/plate)
		1	4047G	1667G	947G	1780G	493G
		2	3765G	1805G	696G	1936G	553G
		3	3933G	1751G	815G	1606G	506G
		Mean±SD	3915±142	1741±70	819±126	1774±165	517±32
		Ratio	26.82	82.90	5.93	7.10	36.93
S9+	Control	Plates	Number of the revertant colonies (CFU/plate)				
			TA97a	TA98	TA100	TA102	TA1535
	Solvent control (H ₂ O)	1	189N	26N	162N	268N	13N
		2	161N	28N	147N	300N	21N
		3	159N	31N	164N	313N	18N
		Mean±SD	170±17	28±3	158±9	294±23	17±4
		Ratio	1.00	1.00	1.00	1.00	1.00
	Positive control	Name (dose level)	2-AF (20 µg/plate)	2-AF (20 µg/plate)	2-AF (20 µg/plate)	Danthron (50 µg/plate)	2-AA (2.0 µg/plate)
		1	1679G	1591G	1691G	895G	96G
		2	1562G	1824G	1897G	833G	100G
		3	1394G	1726G	1523G	756G	94G
		Mean±SD	1545±143	1714±117	1704±187	828±70	97±3
		Ratio	9.09	61.21	10.78	2.82	5.71

Notes: (1) Ratio = The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control

(2) N- Background lawn revealed the presence of densely microcolonies which form a uniform layer

(3) G- The background lawn was not thin compared with that of the concurrent solvent control observed by microscope

(4) CFU- Colony forming units

(5) Mean- Mean number

(6) SD- Standard deviation



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Table 4 Results of the first test in the absence of S9 mix

Dose level ($\mu\text{g}/\text{plate}$)	Plates	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
5000	1	160G	26G	205G	256G	16G
	2	187G	20G	179G	306G	12G
	3	153G	25G	164G	264G	12G
	Mean $\pm$ SD	167 $\pm$ 18	24 $\pm$ 3	183 $\pm$ 21	275 $\pm$ 27	13 $\pm$ 2
	Ratio	1.13	1.09	1.22	1.04	0.87
1500	1	154	22	141	262	21
	2	131	21	182	277	17
	3	165	24	154	254	13
	Mean $\pm$ SD	150 $\pm$ 17	22 $\pm$ 2	159 $\pm$ 21	264 $\pm$ 12	17 $\pm$ 4
	Ratio	1.01	1.00	1.06	1.00	1.13
500	1	169	26	166	270	18
	2	132	18	167	276	22
	3	173	23	160	291	16
	Mean $\pm$ SD	158 $\pm$ 23	22 $\pm$ 4	164 $\pm$ 4	279 $\pm$ 11	19 $\pm$ 3
	Ratio	1.07	1.00	1.09	1.06	1.27
150	1	142	20	176	259	24
	2	138	23	166	254	16
	3	148	25	150	273	23
	Mean $\pm$ SD	143 $\pm$ 5	23 $\pm$ 3	164 $\pm$ 13	262 $\pm$ 10	21 $\pm$ 4
	Ratio	0.97	1.05	1.09	0.99	1.40
50	1	158	21	153	274	21
	2	143	27	142	241	23
	3	152	19	159	282	15
	Mean $\pm$ SD	151 $\pm$ 8	22 $\pm$ 4	151 $\pm$ 9	266 $\pm$ 22	20 $\pm$ 4
	Ratio	1.02	1.00	1.01	1.01	1.33
15	1	183	22	151	258	17
	2	162	25	152	284	14
	3	145	23	143	278	20
	Mean $\pm$ SD	163 $\pm$ 19	23 $\pm$ 2	149 $\pm$ 5	273 $\pm$ 14	17 $\pm$ 3
	Ratio	1.10	1.05	0.99	1.03	1.13

Notes: (1) Ratio = The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control
(2) G- The background lawn was not illio compared with that of the concurrent solvent control, observed by microscope
(3) CFU- Colony forming units
(4) Mean- Mean number
(5) SD-Standard deviation



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Table 5 Results of the first test in the presence of S9 mix

Dose level (µg/plate)	Plates	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
5000	1	169G	26G	193G	284G	17G
	2	152G	34G	200G	297G	15G
	3	174G	24G	171G	325G	23G
	Mean±SD	165±12	28±5	188±15	302±21	18±4
	Ratio	1.10	0.97	1.11	1.01	0.90
1500	1	133	32	217	285	18
	2	154	25	179	305	21
	3	157	28	170	269	20
	Mean±SD	148±13	28±4	189±25	286±18	20±2
	Ratio	0.99	0.97	1.11	0.95	1.00
500	1	141	26	203	290	18
	2	150	26	163	281	19
	3	162	29	174	277	24
	Mean±SD	151±11	27±2	180±21	283±7	20±3
	Ratio	1.01	0.93	1.06	0.94	1.00
150	1	151	27	193	300	25
	2	139	31	207	335	17
	3	144	23	169	274	20
	Mean±SD	145±6	27±4	190±19	303±31	21±4
	Ratio	0.97	0.93	1.12	1.01	1.05
50	1	183	29	182	357	24
	2	146	31	178	292	20
	3	155	32	152	306	16
	Mean±SD	161±19	31±2	171±16	285±25	20±4
	Ratio	1.07	1.07	1.01	0.95	1.00
15	1	176	28	178	275	21
	2	165	26	176	327	22
	3	170	31	183	297	25
	Mean±SD	170±6	28±3	179±4	300±26	23±2
	Ratio	1.13	0.97	1.05	1.00	1.15

Notes: (1) Ratio= The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control.
(2) G- The background lawn was not thin compared with that of the concurrent solvent control observed by microscope.
(3) CFU- Colony forming units.
(4) Mean- Mean number.
(5) SD-Standard deviation.



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Table 6 Results of the validation test in the absence of S9 mix

Dose level ($\mu\text{g}/\text{plate}$)	Plates	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
5000	1	172G	19G	185G	306G	14G
	2	181G	23G	146G	306G	12G
	3	153G	24G	139G	264G	19G
	Mean $\pm$ SD	169 $\pm$ 14	22 $\pm$ 3	157 $\pm$ 25	292 $\pm$ 24	15 $\pm$ 4
	Ratio	1.16	1.05	1.14	1.17	1.07
1500	1	157	30	187	287	23
	2	157	22	159	254	22
	3	122	23	177	280	15
	Mean $\pm$ SD	145 $\pm$ 20	25 $\pm$ 4	174 $\pm$ 14	274 $\pm$ 17	20 $\pm$ 4
	Ratio	0.99	1.19	1.26	1.10	1.43
500	1	164	22	146	264	13
	2	165	28	148	287	16
	3	181	24	164	262	21
	Mean $\pm$ SD	170 $\pm$ 10	25 $\pm$ 3	153 $\pm$ 10	271 $\pm$ 14	17 $\pm$ 4
	Ratio	1.16	1.19	1.11	1.08	1.21
150	1	155	22	161	307	18
	2	148	20	150	259	23
	3	155	25	157	272	15
	Mean $\pm$ SD	153 $\pm$ 4	22 $\pm$ 3	156 $\pm$ 6	279 $\pm$ 25	19 $\pm$ 4
	Ratio	1.05	1.05	1.13	1.12	1.36
50	1	201	19	143	266	20
	2	182	25	123	296	21
	3	160	28	160	243	16
	Mean $\pm$ SD	181 $\pm$ 21	24 $\pm$ 5	142 $\pm$ 19	268 $\pm$ 27	19 $\pm$ 3
	Ratio	1.24	1.14	1.03	1.07	1.36
15	1	172	22	144	248	18
	2	147	22	149	278	17
	3	143	26	132	270	13
	Mean $\pm$ SD	154 $\pm$ 16	23 $\pm$ 2	142 $\pm$ 9	265 $\pm$ 16	16 $\pm$ 3
	Ratio	1.05	1.10	1.03	1.06	1.14

Notes: (1) Ratio= The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control
(2) G- The background lawn was not this compared with that of the concurrent solvent control observed by microscope
(3) CFU= Colony forming units
(4) Mean= Mean number
(5) SD= Standard deviation



Study No.: S240800050

Table 7 Results of the validation test in the presence of S9 mix

Dose level ($\mu\text{g}/\text{plate}$)	Plates	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
5000	1	183G	31G	151G	306G	18G
	2	189G	32G	135G	306G	21G
	3	162G	27G	174G	275G	16G
	Mean $\pm$ SD	178 $\pm$ 14	30 $\pm$ 3	153 $\pm$ 20	296 $\pm$ 18	18 $\pm$ 3
	Ratio	1.05	1.07	0.97	1.01	1.06
1500	1	167	27	162	287	17
	2	161	24	136	354	19
	3	180	33	170	306	25
	Mean $\pm$ SD	169 $\pm$ 10	28 $\pm$ 5	156 $\pm$ 18	316 $\pm$ 35	20 $\pm$ 4
	Ratio	0.99	1.00	0.99	1.07	1.18
500	1	186	32	157	350	18
	2	152	21	169	272	29
	3	158	28	163	335	22
	Mean $\pm$ SD	165 $\pm$ 18	27 $\pm$ 6	163 $\pm$ 6	319 $\pm$ 41	23 $\pm$ 6
	Ratio	0.97	0.96	1.03	1.09	1.35
150	1	163	29	164	319	15
	2	185	23	187	316	21
	3	154	28	154	294	17
	Mean $\pm$ SD	167 $\pm$ 16	27 $\pm$ 3	168 $\pm$ 17	310 $\pm$ 14	18 $\pm$ 3
	Ratio	0.98	0.96	1.06	1.05	1.06
50	1	188	26	141	306	20
	2	153	35	133	299	22
	3	172	21	175	336	20
	Mean $\pm$ SD	171 $\pm$ 18	27 $\pm$ 7	150 $\pm$ 22	314 $\pm$ 20	21 $\pm$ 1
	Ratio	1.01	0.96	0.95	1.07	1.24
15	1	173	24	153	326	24
	2	176	24	176	318	26
	3	170	27	147	291	21
	Mean $\pm$ SD	173 $\pm$ 3	25 $\pm$ 2	159 $\pm$ 15	312 $\pm$ 18	24 $\pm$ 3
	Ratio	1.02	0.89	1.01	1.06	1.41

Notes: (1) Ratio- The mean number of revertant colonies in each group / the mean number of revertant colonies in the concurrent solvent control
(2) G- The background lawn was not thin compared with that of the concurrent solvent control observed by microscope
(3) CFU- Colony forming units
(4) Mean- Mean number
(5) SD-Standard deviation



Study No.: S240800050

Table 8 Observation result for the test item precipitate

Test	Dose level (µg/plate)	The description of the test item precipitate			
		Before the incubation*		After the incubation*	
		S9-	S9+	S9-	S9+
The first test	5000	None	None	None	None
	1500	None	None	None	None
	500	None	None	None	None
	150	None	None	None	None
	50	None	None	None	None
	15	None	None	None	None
Observed tester strain		All tester strains	All tester strains	All tester strains	All tester strains
The validation test	5000	None	None	None	None
	1500	None	None	None	None
	500	None	None	None	None
	150	None	None	None	None
	50	None	None	None	None
	15	None	None	None	None
Observed tester strain		All tester strains	All tester strains	All tester strains	All tester strains

Note: (1) None-no test item precipitate;

(2) *- Three parallel plates of each dose group were observed.



Study No. S240800050

16. ANNEXES

Annex 1 Historical controls data of Ames test in this lab (2023)

Negative controls (solvent control)

S9-/S9+	Name	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
S9-	Test	45	45	45	45	45
	Data	135	135	135	135	135
	Max	173	31	182	324	20
	Min	102	14	104	215	7
	Mean	132	21	142	264	12
	SD	14	3	14	21	3
95% control limits		103~161	15~27	114~169	223~306	6~18
S9+	Test	45	45	45	45	45
	Data	135	135	135	135	135
	Max	188	38	195	379	25
	Min	128	20	122	251	10
	Mean	154	27	158	311	17
	SD	13	4	15	23	3
95% control limits		128~180	19~35	129~187	265~358	10~24

Note: (1) Solvent control here included H₂O and DMSO; (2) 95% control limits = Mean $\pm$ 2SD;
(3) Mean- Mean number; (4) SD-Standard deviation; (5) CFU- colony forming units;
(6) Test methods: The plate incorporation method and the pre-incubation method;
(7) When calculation, all data were rounded after calculation.

Positive controls

Treatment condition	Name	Number of the revertant colonies (CFU/plate)				
		TA97a	TA98	TA100	TA102	TA1535
S9-	Positive control	Dexon (50μg/plate)	Dexon (50μg/plate)	NaN ₃ (1.5μg/plate)	Dexon (50μg/plate)	NaN ₃ (1.5μg/plate)
	Test	45	45	45	45	45
	Data	135	135	135	135	135
	Max	4592	1956	1363	2509	732
	Min	2934	1415	587	1431	385
	Mean	3861	1708	979	1897	530
	SD	311	116	141	177	63
95% control limits		3329~4483	1477~1939	698~1260	1543~2250	103~857
S9+	Positive control	2-AF (20μg/plate)	2-AF (20μg/plate)	2-AF (20μg/plate)	Danthron (50μg/plate)	2-AA (2.0μg/plate)
	Test	45	45	45	45	45
	Data	135	135	135	135	135
	Max	1771	2319	2124	1207	147
	Min	1011	1432	1344	640	57
	Mean	1355	1784	1747	902	91
	SD	178	170	161	115	15
95% control limits		999~1710	1444~2125	1424~2070	671~1113	61~122

Note: (1) 95% control limits = Mean $\pm$ 2SD (2) Mean- Mean number;
(3) SD-Standard deviation (4) CFU- colony forming units;
(5) Test methods: The plate incorporation method and the pre-incubation method;
(6) When calculation, all data were rounded after calculation.



Study No.: S240808250

Annex 2 Certificate of analysis for genotype of Ames tester strains



CERTIFICATE OF ANALYSIS

Bacterial strains informations

Species: Histidine requiring strains of *Salmonella typhimurium* Strains: TA97a, TA98, TA100, TA102 and TA1535
Supplier: Molecular Toxicology Inc, USA
Passage No. (frozen stock tester strain): Passage 2
Frozen stock date: February 19, 2024 Storage condition: Ultra-low temperature freezer (Appr. -80 °C)
Genotype test date: February 19–21, 2024 Expiry date: February 19, 2025

Tester strain frozen stock culture preparation, genotype marker verification and storage

Before preparation of working tester strains, the frozen cultures of long-term stored strains (passage 1) were quickly thawed in a water bath at approximately 37 °C, and the suspension of each strain was transferred to nutrient broth at a volume ratio of 1:100 (V_{tester}:V_{nutrient broth}). Aseptically, The strain cultures were incubated stationary at 36.6–36.9 °C for approximately 16.5 hours. Part of the fresh culture of bacteria was used for genotype test and the rest was mixed with 96% DMSO (Sigma D-8418) Then the mixtures were dispensed and frozen storage as frozen cultures of the tester strains for Ames test.

Test results

The results table of genotype marker verification for Ames tester strains (2023)

Test			TA97a	TA98	TA100	TA102	TA1535
Histidine dependence test	On the plate with histidine-broth	Criterion	G	G	G	G	G
		Result	P	P	P	P	P
	On the plate with biotin	Criterion	NG	NG	NG	NG	NG
		Result	P	P	P	P	P
Lipopolysaccharide envelope deficiency test (pH Mutation)	Around the filter paper containing crystal violet	Criterion	NG	NG	NG	NG	NG
		Result	P	P	P	P	P
UVB mutation test [Sensitivity to ultra-violet light]		Criterion	NG*	NG*	NG*	G*	NG*
		Result	P	P	P	P	P
Ampicillin resistance test (Ampicillin Resistance)	Around the filter paper containing Ampicillin	Criterion	G	G	G	G	NG
		Result	P	P	P	P	P
Tetracycline Resistance test	Around the filter paper containing Tetracycline	Criterion	NG	NG	NG	G	NG
		Result	P	P	P	P	P
Spontaneous reversion frequency test	Mean ±SD	Historical standard data (2022)	103~151	15~27	114~164	223~366	5~18
		Result	116±11	23±2	141±5	261±12	11±2

Note: NG = No growth; G = Growth; * = On irradiated media; SD = Standard deviation; P = Pass

Conclusion: The test results showed that each tester strain met its genotype criteria. So it is considered that all above tester strains had retained their genetic characteristics.

Wei Min
Study Director

Chen Cong
Executive Manager

Safety Evaluation Center of Shenyang SYRICA Testing Co., Ltd. /
Supervision and Test Center for Pesticide Safety Evaluation and Quality Control

Address: No. 600 Shedian Road, Tiesi District, Shenyang, Liaoning Province, China
Post code: 110141 Tel: 86-24-62353324 Fax: 86-24-62353328



Study No.: S240800050

Annex 3 Certificate of Analysis of S9 provided by the supplier

Q: 010-5779997-6000
F: 010-5779997-2442
E: info@chiscientific.com
W: www.chiscientific.com

CERTIFICATE OF ANALYSIS

Sprague Dawley Rat

INDUCED HEPATIC S9 FRACTIONS, POOL

Catalog number: 6-1202

Batch number: 23FS037K

Origin of Product	Description of Production
SPECIES: Rat	INDUCING AGENTS: <i>β</i> -naphthoflavone and phenobarbital
STRAIN: Sprague Dawley	STORAGE BUFFER: 0.15M KCl, pH 7.4
SEX: Male	WEIGHT: 230 ~ 250 g
AGE: 6-8 weeks	TISSUE: Liver
VOLUME: 1ml	PROTEIN CONCENTRATION: 30 mg/mL
CONTROL DATE: Jan 18, 2024	EXPIRY DATE: Jan 7, 2026

METABOLIC ACTIVITY AFTER THAWING

Substrate	Enzyme	Result (Vmax)
Fluorenone	CYP1A1/2	1210
Propion	CYP2B1	111
Melatonin	CYP1A2	101

Hepatic S9 fractions, at 1 mg/ml are incubated for 30 minutes at 37°C with substrates (n=3) at the following concentrations: Fluorenone: 200 μM, Propion: 100 μM, Melatonin: 50 μM. The reaction is stopped by addition of ice-cold acetonitrile. The metabolite formation is measured by LC-MS/MS. Activities are expressed as pmole/h/mg of protein.

TEST FOR THE PRESENCE OF ADVENTITIOUS AGENTS

10% S9 in the metabolic activation system were tested for their ability to activate 2-aminofluorene(2-AF) (10 μg/plate) to intermediates mutagenic to TA97, TA98 or TA100. Sample of S9 were assayed for the presence of contaminating microflora by plating on overlay Agar (supplemented with 0.05 mM L-histidine and D-biotin) and Minimal agar (Vogel-Bonner E) media. Duplicate plates were read after 48h incubation at 37 °C.

Treatment	TA97		TA98		TA100	
	Control	+S9	Control	+S9	Control	+S9
Reversion	140±78	1140±160	31±4	2542±156	150±36	1584±84

Chinese hamster lung (CHL) cells was carried out in standard chromosomal aberration tests which demonstrated positive results in the metabolic activation system (1.5% S9), using cyclophosphamide (CP, 1.5 μg/L) as positive drug. At least 200 well-spread metaphases have to be scored per test, the experimental unit is the cell, and the percentage of cells with different types of structural chromosome aberration is evaluated and listed.

Test	Concentration (μg/L)	S9	Cell	Types of chromosome aberration						Percentage of aberration	Result
				B	F	P	R	CL	O		
Control	-	+	200	1			1			1%	-
CP	1.5	+	100	14	2	1	1	25		43%	++
MMS	0.5	-	100	13	2	1	1	15		32%	++

B: break; F: dicentric; P: polyploid; R: ring; CL: cross link; O: other

Approved:  Date 2024.1.18

Devin East, Maynard, Massachusetts 01754, USA



Study No.: S240800050

Annex 4 Certificate of Analysis of S9 tested by this lab



CERTIFICATE OF ANALYSIS

S9 informations				
Name:	Sprague Dawley rat induced hepatic S9 fractions (S9), pool			
Supplier:	CHI SCIENTIFIC INC.			
Batch No.:	23F9037K			
Expiry date:	2026-03-17			
Test schedule				
Test start:	2024-03-06			
S9 quality test:	2024-03-06-2024-03-08			
Test end:	2024-03-08			
Certification:	2024-03-18			
Test method				
Before being used, this batch of S9 had been tested in this lab for its sterility (culture after inoculation), its protein content (tested by Hitachi 7180 Full Automatic Biochemical Analyzer in this facility, the protein content was not higher than 40 mg/mL), and its capability to activate known mutagens (2-AP, Dantrolen and 2-AA) in Ames test.				
Result				
Protein Content: 30.2g/L		Sterility: Pass		
The results on the S9 metabolic activation to Mutagens in TA98, TA102 and TA1535 tester strains of <i>Salmonella typhimurium</i> with 10% S9 fraction in the S9 mix are as below:				
Control	Mean number of the revertant colonies (CFU/plate)			
	TA 9 8	TA 1 0 2	TA 1 5 3 5	
Untreated	Mean±SD	27±3	299±18	14±2
	Ratio	1.00	1.00	1.00
Positive	Name (dose level)	2-AP (20µg/plate)	Dantrolen (50µg/plate)	2-AA (2.0µg/plate)
	Mean±SD	1748±65	1026±40	95±9
	Ratio	64.74	3.43	6.79

Conclusion: The results indicated that each index of the S9 was in the expected range to meet the test requirements.

Wei Min
Study Director

Chen Cong
Executive Manager

Safety Evaluation Center of Shenyang SYRCCI Testing Co., Ltd.
(Supervision and Test Center for Pesticide Safety Evaluation and Quality Control)
Address: No. 600 Shenhe Road, Tiesi District, Shenyang, Liaoning Province, China
Post code: 110141 Tel: 86-24-62353324 Fax: 86-24-62353328



Study No.: S240800050

Annex 5 Certificate of Analysis of the test item offered by the Sponsor

Gene III Biotechnology Co., Ltd

East side of 7th Floor, Building 4, No.6 Xuzhuang Road,
Xuesen District, Nanjing, Jiangsu, China
Phone: 400-800-2959

GeneIII 三

Certificate of Analysis

NO.: GS20240207

Product Name: L-Ergothioneine		Batch Number: EGT20240207
Production Date: 2024.02.27		Expiry date: 2026.02.26
Item	Standard	Result
Characteristics	White powder, characteristic smell, no peculiar smell, no visible impurities	White powder, characteristic smell, no peculiar smell, no visible impurities
Identification	HPLC chromatogram matches the standard sample	Conform
Optical rotation	$[\alpha]_D \geq (+)122^\circ$ (C=1, H ₂ O)	+126.0°
Assay of L-Ergothioneine (HPLC)	$\geq 99.5\%$	99.9%
Loss on Drying	$\leq 0.5\%$	0.05%
Total Residual Solvents (Ethanol)	$\leq 1000\text{ppm}$	197ppm
Lead	$\leq 0.5\text{mg/kg}$	<0.05 mg/kg
Arsenic	$\leq 0.5\text{mg/kg}$	<0.05 mg/kg
Mercury	$\leq 0.1\text{mg/kg}$	<0.01mg/kg
Cadmium	$\leq 0.1\text{mg/kg}$	<0.05mg/kg
Total aerobic bacteria count	$\leq 1000\text{ CFU/g}$	<10CFU/g
Molds and Yeasts	$\leq 50\text{ CFU/g}$	<10CFU/g
Escherichia coli	Negative/g	Negative
Salmonella	Negative/25g	Negative
Staphylococcus aureus	Negative/25g	Negative
Conclusion: up to standard		
Report by:	Reviewed by:	Approved by:





Study No.: S240800050

Annex 6 The Endorsement of Compliance with OECD GLP

 entidad responsable de acreditación, s. de. c. registro acreditación en SGA nº de registro: 12590 código de entidad: -001001484361 www.ema.gub.uy	entidad responsable de acreditación, s. de. c. registro acreditación en SGA nº de registro: 12590 código de entidad: -001001484361 www.ema.gub.uy
Shenyang SYRICI Testing Co., Ltd. Safety Evaluation Center. has been Recognized in Good Laboratory Practice in Compliance with the Organization for Economic Co-operation and Development (OECD) Principles of Good Laboratory Practice - (1998). Registry No: BPL-02029 Date of registration: 2023-12-05 Date of issuance: 2023-12-06 Address: No.500 Shenfa Road, Z.P. 110141 Tian District, Shenyang, China Telephone: +86(0)246355450 Email: zhongkuan1@shenchem.com Authorized representative: Zhang Xuan	
Scope of registration: Pharmaceuticals, industrial chemicals, Medical devices	
Areas of expertise in studies:	02 Toxicity testing 03 Mutagenicity testing 04 Environmental toxicology studies on aquatic and terrestrial organisms 05 Studies on behaviour in water, soil and air; bioaccumulation 08 Analytical and clinical chemistry testing 12 Other: Toxicokinetics
Inspection dates: November 10th 2023, and 21st 2023 (2x-EPL007)	Type of inspection: Complete inspection
By the certified accreditation, the accreditation is: Maria Isabel López Martínez General Director	

Página 1 de 1



Study No.: S240800050



entidad mexicana de acreditación s.c.

Granted to the testing facility:

Shenyang SYRICI Testing Co., Ltd.
Safety Evaluation Center

No.600 Shenliao Road, Z.P. 110141
Tiexi District, Shenyang, P.R.China,

**CERTIFICATE OF REGISTRATION OF
CONFORMITY WITH THE OECD
PRINCIPLES OF GOOD LABORATORY
PRACTICE**

The entidad mexicana de acreditación, s.c. (ema) states that according to the requirements of their programme for verifying compliance with Good Laboratory Practice, the testing facility Shenyang SYRICI Testing Co., Ltd. holds the contents of non-clinical studies in the scope of this certificate of registration in compliance with the Organization for Economic Cooperation and Development OECD Principles of good laboratory practices (1998).

Registry No: **BPL-020/23**

Date of registration: **2023/03/08**

Maria Isabel López Martínez
General Director

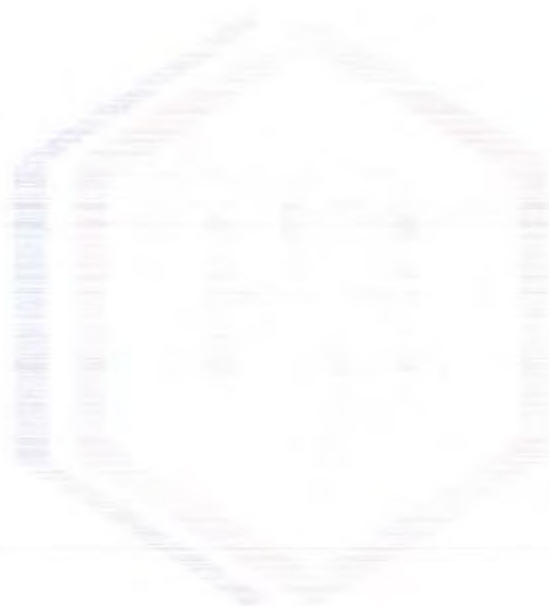
* This document is not valid without its corresponding technical annex: **230P1.00007**

Wanneer het document is gepresenteerd als exemplaar, dient het te worden vergezocht door de technische annex. Het document is niet geldig zonder de technische annex.



Safety Evaluation Center of Shenyang SYRICI Testing Co., Ltd.





DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001270	DATE OF RECEIPT May 7, 2025
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3 Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Ju Hongying		Position or Title Senior Reliability Manager	
	Organization (<i>if applicable</i>) Gene III Biotechnology Co., Ltd.			
	Mailing Address (<i>number and street</i>) East side of 7th Floor, Building 4, No.6 Xuzhuang Road, Xuanwu District			
City Nanjing		State or Province Jiangsu	Zip Code/Postal Code 210023	Country China
Telephone Number +86 18052038505		Fax Number	E-Mail Address juhongying@geneiii.com.cn	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person Stella Si		Position or Title Managing Director	
	Organization (<i>if applicable</i>) Anchor Center for Certification			
	Mailing Address (<i>number and street</i>) No. 1295 Chuan Qiao Road, Building 2, Suite 302			
City Shanghai		State or Province Shanghai	Zip Code/Postal Code 201206	Country China
Telephone Number 02150320118		Fax Number	E-Mail Address stella.si@acclab.cn	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

L-Ergothioneine

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway ☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? *(see 21 CFR 170.225(c)(8))*

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Intended to be used as an ingredient in beverages (coffee, tea, fruit drinks and ades, carbonated soft drinks, sport and energy drinks, nutritional beverages), protein and nutritional powders, bakery and other baked goods (nutrition bars, cereal bars, sweet bakery products) and candy at levels intended to provide 30 - 150 mg of L-Ergothioneine per serving

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☒ Yes ☐ No

Did you include this other information in the list of attachments?

☒ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Gene III Biotechnology Co., Ltd.

(name of notifier)

has concluded that the intended use(s) of L-Ergothioneine

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Gene III Biotechnology Co., Ltd. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

No. 1295 Chuan Qiao Road, Building 2, Suite 302,

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Stella Si

数字签名者: Stella Si
日期: 2025.05.07 17:41:24 +08'00'

Printed Name and Title

Stella Si, Managing Director of ACC Lab

Date (mm/dd/yyyy)

05/06/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	GenellIL-EGTFDAGRASV7.pdf	Submission

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.